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TECHNICAL MEMORANDUM

June 17, 2025 Project# 25608

To: Matt Kelly, Contra Costa Transportation Authority
Shannon McCarthy, Alameda County Transportation Commission

From: Mike Aronson, Sravya Kamalapuram, Kittelson & Associates, Inc.

CC: Rosella Picado, WSP

RE: AlaCC Model Consistency

Introduction

The AlaCC Model is a travel demand forecasting model developed for Alameda and Contra Costa Counties in California. The purpose of this Model Consistency technical memorandum is to provide the deliverables requested by the Metropolitan Transportation Commission (MTC) and required by state legislation for each county's Congestion Management Program (CMP) to establish that the AlaCC Model utilizes land use and travel demand forecasting processes and assumptions consistent with Plan Bay Area (PBA) 2050.

DESCRIPTION OF THE ALACC MODEL

The AlaCC Model is an activity-based travel demand model encompassing ten counties in the San Francisco Bay Area of California, with its origin in MTC Travel Model 1.5.2 (TM 1.5.2). In addition to the nine counties included in MTC TM 1.5.2, the AlaCC Model incorporates San Joaquin County and features more detailed transportation analysis zones (TAZs) and networks in Alameda and Contra Costa Counties. Exhibit 1 lists the main features of the two model systems with the key differences highlighted in blue. This section describes these differences in more detail and explains how they impact the consistency of the AlaCC model outputs relative to TM 1.5.2.

While MTC TM 1.5 includes 1,454 TAZs, the AlaCC Model includes 6,000 TAZs. The AlaCC Model TAZs are generally consistent with the TAZ boundaries of the Alameda Countywide travel model maintained by Alameda CTC[1] for TAZs within Alameda County, the Contra Costa Countywide travel model maintained by CCTA[2] for TAZs within Contra Costa County, and the MTC Travel Model Two[3] TAZs for the other seven Bay

[1] Alameda County Transportation Commission (Alameda CTC).
[2] Contra Costa County Transportation Authority (CCTA).
[3] Travel Model Two is currently under development by MTC to provide improved representation of transportation conditions including more detailed spatial representation and evaluation of the effects of transit crowding.

Area counties. The TAZs within San Joaquin County represent aggregations of TAZs used in the San Joaquin County travel model maintained by SJCOG[4].

The land use and socioeconomic data assumptions are based on MTC PBA 2050 projections (2021 Regional Transportation Plan/Sustainable Communities Strategy) for the nine-county Bay Area. San Joaquin County land use and socioeconomic data assumptions are based on the 2022 San Joaquin COG Regional Transportation Plan.

The validation years for the AlaCC model are 2015 (consistent with TM 1.5) and 2020 (pre-pandemic).

Exhibit 1: Key Features of the AlaCC Model System

Model System Item	AlaCC	TM1.5.2
Input networks	All-streets, version 1	Simple, version PBA 2050
Input zone system	Custom TAZs in Alameda and Contra Costa, TM2 TAZs elsewhere	TAZ1454
Input land use	PBA 2050 incorporating Alameda and Contra Costa local jurisdiction input	PBA 2050
Population synthesizer	PopulationSim	PopulationSim
Resident travel software	Java CT-RAMP1, version TM1.5.2	Java CT-RAMP1, version TM1.5.2
Resident travel specification	TM1.5.2, version PBA 2050	TM1.5.2, version PBA 2050
Non-resident travel software	Cube	Cube
Non-resident travel specification	Refactored legacy Alameda CTC	TM1.0
Assignment and skimming software	Cube	Cube

KEY DIFFERENCES RELATED TO MODEL VALIDATION

The AlaCC model closely follows the MTC TM 1.5.2 procedures for activity, tour and trip estimation, creation of level of service skims and trip assignment. However, there are several differences which are intended to improve travel estimation at the local level in Alameda and Contra Costa counties.

Network and TAZ Detail

The AlaCC model has many more TAZs and smaller TAZs than TM 1.5.2. The TAZ size can affect measures of accessibility, the attractiveness of short trips, and the percentages of “intrazonal” trips that get assigned

[4] San Joaquin Council of Governments (SJCOG).

to the network. In general, smaller TAZs as in the AlaCC model should result in fewer intrazonal trips and more trips assigned to the network even if total activity generation is constant.

The AlaCC model uses a more detailed network representation than TM 1.5.2. The AlaCC model includes most local streets in the Bay Area, like the Travel Model Two (TM2) networks under development by MTC. Local streets and many collector streets were generally excluded from the networks for TM 1.5.2.

The greater spatial detail in AlaCC could affect accessibility measures and zone-to-zone travel distance and travel times, and correspondingly, affect trip lengths and mode shares. However, as there are other differences that affect accessibility and travel times, described below, it is difficult to parse out the contribution of the greater network and zone density from the contribution of other factors.

Unlike TM1.5.2, which classifies freeways into "freeway" and "managed freeways" based on facility type, AlaCC has only one "freeway" facility type which applies to all freeway segments. For reporting VMT and other performance metrics separately for managed and non-managed freeways, AlaCC identifies a road segment as a managed freeway when it meets the conditions listed below:

- The road segment is a freeway or freeway ramp, and
- The road segment is tolled and/or restricted to carpools or trucks.

These conditions are evaluated for each time period; for example, in the AlaCC model a road segment may be a "managed freeway" during peak periods and a "freeway" during off-peak periods if the facility is tolled only during peak periods.

These differences in facility type definitions would affect the VMT, VHT and Average Speed comparison reported in Table 44. Specifically, the differences in VMT for freeways and managed freeways between TM1.5.2 and the AlaCC model would be confounded by the somewhat inconsistent facility type definition. For this reason, these metrics are presented for all freeway segments combined rather than by lane type.

Calibration and Validation Year

Two base year scenarios were created for AlaCC—a 2015 base year for consistency with PBA 2050 and a 2019/2020 (pre-pandemic) scenario to align the model estimates to a recent validation year. The model parameters established as part of the 2019/2020 validation are recommended for future scenarios and were used to generate the Year 2050 estimates reported herein.

The most salient adjustment performed as part of the 2019/2020 calibration was reducing the magnitude of transit constants in the tour mode choice model. The intent of the adjustment was to improve the validation to observed 2019 transit boardings. Exhibit 2 shows the change in transit boardings between 2015 and 2019 for the region's major transit operators as reported in the National Transit Database (NTD). As shown in this exhibit, transit boardings decreased or remained approximately constant between 2015 and 2019 despite the region growing by approximately 2.6% in population[5]. Without the mode choice calibration adjustment, the AlaCC model would overestimate transit boardings in 2019/2020 and it would carry forward the overestimation to future year scenarios.

[5] State of California, Department of Finance, *E-4 Population Estimates for Cities, Counties, and the State, 2011-2020, with 2010 Census Benchmark*. Sacramento, California, May 2022.

Exhibit 2: Change in Average Weekday Transit Ridership for Major Operators, 2015 to 2019

Operator	Technology	2015 Boardings	2019 Boardings	Percent Difference
AC Transit	Bus	181,798	178,702	-1.7%
BART	Heavy Rail	452,126	443,123	-2.0%
Caltrain	Commuter Rail	66,921	64,492	-3.6%
Golden Gate Transit	Bus	12,042	10,723	-11.0%
Golden Gate Transit	Ferry	8,469	8,518	0.6%
SamTrans	Bus	45,162	37,023	-18.0%
SF Muni	Bus	481,607	509,716	5.8%
SF Muni	Light Rail	197,420	200,274	1.4%
VTA	Bus	106,583	89,486	-16.0%
VTA	Light Rail	36,584	27,485	-24.9%
San Joaquin RTD	Bus	15,573	12,782	-17.9%
Total		1,604,285	1,582,324	-1.4%

Source: National Transit Database

Truck Model

The AlaCC model adapts the truck model developed for the Alameda Countywide model in place of the truck model used in TM 1.5.2. The AlaCC truck model estimates higher truck trip generation than TM1.5.2 across all truck types except “very small trucks”, as shown in Exhibit 3. The “very small” truck trips represent two-axle trucks and vans not fully represented in the household activity estimation, such as delivery trucks, service vehicles, construction contractors, etc. The trip generation rates for the small, medium and combination trucks were updated based on the Alameda County CMA truck modeling study completed in early 2010. As a result of the truck trip generation updates, the AlaCC model assigns more vehicles to the network than TM 1.5.2 even when household activity generation is fully consistent, leading to higher VMT.

Exhibit 3: Truck Trip Generation

Truck Type	MTC TM1.5.2	AlaCC
Very Small	1,286,719	2,745,369
Small	205,060	146,393
Medium	17,469	115,650
Combination	38,367	116,200
Total	1,547,615	3,123,612

PBA 2050 Projects and Policies

The AlaCC model's 2050 scenario excludes certain PBA projects and policies. Projects considered highly relevant for regional travel and local travel in Alameda and Contra Costa were retained, as follows:

- Transit: all rail projects in the region, Regional Express Bus projects, ferry projects, BRT and various other transit projects in Alameda and Contra Costa Counties.
- Highway: all Express Lane projects in the region, freeway capacity projects, and various road capacity and interchange improvement projects in Alameda and Contra Costa Counties.
- Non-motorized: Bike/Ped improvements to the San Rafael Bridge and Bay Skyway bike improvements. In addition, all AlaCC scenarios include bike lanes and bike paths in Alameda and Contra Costa Counties, sourced from the Alameda CTC legacy model network.

A complete list of project cards coded for the AlaCC model is available upon request. Appendix A shows the list of PBA 2050 coded for the AlaCC network.

The AlaCC model adopts the TM1.5.2 procedures for representing non-network policies. Exhibit 4 shows the PBA 2050 non-network policies and indicates which ones can be simulated with the AlaCC model. The Complete Streets strategy is not represented TM1.5.2 due to its sparse local road representation. Although AlaCC has the necessary network detail, there is no information readily available about the location or extent of Complete Streets treatments in PBA 2050. Other strategies funded in PBA 2050, such as traffic signal coordination, integrated freeway management, and travel demand management, are not represented in the AlaCC model.

Exhibit 4: Implementation of Non-Network PBA Policies in the AlaCC Model

Policy	Implemented in the AlaCC Model
Parking price updates	Yes
Eliminate free parking eligibility	Yes
Speed overrides for BRT and Bus-on-Shoulder strategies	Yes
Blueprint Regional Transit Fare Policy	Yes
Means-based fare discounts	Yes
Blueprint Vision Zero	Yes
Blueprint per-mile tolling	Yes
Toll rate discounts for income quartiles 1 and 2	Yes
Complete Streets	N/A

Effect on Model Validation and 2050 Consistency

For the validation of the AlaCC model to 2015 and 2019/2020 traffic and transit counts, the model characteristics which result in higher trip estimates than TM 1.5.2 were found to improve the model's representation of observed weekday transportation activity, measured against Caltrans HPMS estimates and traffic counts from Caltrans and other sources. Exhibit 5 compares HPMS VMT estimates by county for Year 2015 to estimates from TM1.5.2 and AlaCC. As shown, whereas TM1.5.2 generates approximately 8% less VMT in 2015 than the HPMS estimate, the AlaCC model estimates 2015 VMT within 1% of HPMS. The model estimates shown in Exhibit 5 include centroid connector VMT and intrazonal trip VMT, for a more consistent comparison to HPMS.

Exhibit 5: Weekday Vehicle Miles Traveled Comparison

County	2015 Caltrans HPMS	MTC TM1.5.2		AlaCC	
		VMT	% Error	VMT	% Error
San Francisco	8,844,00	8,557,000	-3%	9,265,000	5%
San Mateo	19,532,000	16,618,000	-15%	19,417,000	-1%
Santa Clara	42,894,000	41,525,000	-3%	45,350,000	6%
Alameda	41,699,000	37,603,000	-10%	41,514,000	0%
Contra Costa	23,507,000	22,583,000	-4%	24,219,000	3%
Solano	13,154,000	11,674,000	-11%	13,230,000	1%
Napa	3,330,000	3,132,000	-6%	2,984,000	-10%
Sonoma	11,449,000	10,472,000	-9%	10,373,000	-9%
Marin	7,564,000	5,751,000	-24%	6,889,000	-9%
San Joaquin	17,992,000	n/a		16,503,000	-8%
Total, AlaCC model area	189,965,000			189,742,000	0%
Excluding San Joaquin Co.	171,973,000	157,915,000	-8%	173,241,000	1%

The increase in VMT is largely a result of the difference in very small truck trips discussed above. Various model calibration adjustments also contribute to the VMT difference, though in general model calibration adjustments had a more modest effect on VMT than the truck trip generation updates. The increase in VMT is more pronounced in non-peak periods such as early morning and midday, when commercial truck trips make up a higher proportion of overall trips, as shown in Exhibit 6.

Exhibit 6: Trip Shares by Time Period and Vehicle Type

Time Period	MTC TM1.5.2			AlaCC		
	Passenger Vehicles	Very Small Trucks	Other Trucks	Passenger Vehicles	Very Small Trucks	Other Trucks
Early Morning	91%	8%	1%	66%	29%	5%
AM Peak	98%	0%	2%	91%	8%	1%
Midday	84%	14%	2%	80%	16%	5%
PM Peak	94%	5%	1%	83%	14%	3%
Evening	91%	8%	1%	89%	10%	2%

Increased trip-making in off-peak periods results in a better match to traffic counts, on average, in the AlaCC model compared to TM1.5.2. Exhibit 7 shows two measures of aggregate traffic count validation, the slope and r-squared coefficients of a linear regression using observed traffic counts as the independent variable and estimated traffic volume as the dependent variable. Slope coefficients with values lower than 1.0 indicate that the model tends to underestimate traffic counts for the corresponding time period. As shown in Exhibit 7, the AlaCC model exhibits higher slope coefficients than TM1.5.2 in the off-peak periods.

A more detailed validation of the AlaCC model to 2019 traffic counts is available in the Model Calibration and Validation Report (Product 1).

Exhibit 7: Traffic Volume Regression Statistics

Time Period	AlaCC		TM1.5.2	
	R-Squared	Slope Coefficient	R-Squared	Slope Coefficient
Early	0.87	0.87	0.83	0.63
AM Peak	0.95	0.99	0.96	1.00
Midday	0.96	0.88	0.96	0.85
PM Peak	0.95	0.98	0.95	0.98
Evening	0.93	0.86	0.93	0.79
All Day	0.97	0.93	0.96	0.91

As a result of the differences discussed above, the AlaCC model tends to generate higher traffic levels than TM 1.5.2, particularly during off-peak periods. This occurs even when the demographic inputs and household activity generation process are fully consistent with TM 1.5.2. These higher traffic estimates affect consistency measures such as VMT, VHT and Average Speed (Products 15 and 16, Table 1). Furthermore, the higher levels of congestion exhibited by the AlaCC model may result in a more muted model response to strategies like Vision Zero, which are represented by decreasing free-flow speeds. Conversely, strategies that are represented by higher free-flow speeds may have a more pronounced effect in the AlaCC model than in TM1.5.2.

Travel Forecast Model Reporting

The specific checklist of product deliverables was defined by MTC in the 2022 *Guidance for Model Consistency Collaboration and Transparency Annexure I: Scope of Work (Task 4: Travel Forecast Model Requirements).* The required checklist of products is summarized in Table 1.

Item	Item Description	Product	Product Description
A	Base Year(s): Model Development, Calibration, and Validation Report(s) and Model User Guide	Product 1	Model Development, Calibration, & Validation Report(s).
A		Product 2	Model User Guide.
B	Base Year(s): Demographic/ economic/ land use assumptions	Product 2	A statement establishing that the differences between key ABAG land use variables (i.e., population, households, jobs, and employed residents) and those of the CTA fall within a minimum/maximum range for the base year.
B		Product 3	A table comparing the MTC/ABAG land use estimates with the CTA land use estimates by county and superdistrict for population, households, jobs, and employed residents for the base year.
B		Product 4	A table with CTA land use estimates by TAZ for the base year.
C	Forecast Year(s): Demographic/ economic/land use forecasts	Product 5	A statement establishing that the differences between key ABAG land use variables (i.e., population, households, jobs, and employed residents) and those of the CTA fall within plus or minus one percent for the horizon year at both the county and superdistrict level for the subject county. A statement establishing that no differences exist at the TAZ-level outside the county between the MTC/ABAG forecast or the MTC/ABAG/CTA revised forecast.
C		Product 6	A table comparing the MTC/ABAG land use estimates with the CTA land use estimates by county and superdistrict for population, households, jobs, and employed residents for the horizon year.

Item	Item Description	Product	Product Description
C		Product 7	If land use estimates within the CTA's county or superdistricts are modified from MTC/ABAG's projections, agendas, discussion summaries, and action items from each meeting held with cities, MTC, and/or ABAG at which the redistribution was discussed, as well as before/after census-tract-level data summaries and maps.
D	Forecast Year(s): Pricing assumptions	Product 8	A table comparing the assumed automobile operating cost, key transit fares, and roadway/express lane and bridge tolls to MTC's values and tolled extents.
E	Forecast Year(s): Network assumptions	Product 9	Statement establishing satisfaction of the following: (a) use of MTC's regional roadway and transit network assumptions for other Bay Area counties; (b) more detailed network representation in their own counties; (c) only include projects in the fiscally constrained RTP/SCS.
F	Forecast Year(s): Automobile ownership	Product 10	County and superdistrict-level table comparing estimates of households by automobile ownership level (zero, one, two, or more automobiles) to MTC's.
G	Forecast Year(s): Coordinated Daily Activity Pattern model/trip generation	Product 11	Tables comparing estimates of trip frequency by purpose by superdistrict and county of residence to MTC's.
H	Forecast Year(s): Activity/trip location	Product 12	Region-level tables comparing estimates of average trip distance frequency by tour/trip purpose to MTC's.
H		Product 13	County-to-county comparison of journey-to-work or home-based work flow.
I	Forecast Year(s): Travel mode choice	Product 14	County-level (by county of residence) tables comparing travel mode flow estimates by tour/trip purpose to MTC's estimates for the horizon year. The table summaries should be stratified by transit, auto (split into drive alone, shared ride 2, shared ride 3+), walk, and bicycle.
J	Forecast Year(s): Traffic and transit assignment	Product 15	Region-level and by-county, time-period-specific comparison of vehicle miles traveled (VMT) and vehicle hours traveled (VHT) estimates by facility type to MTC's estimates for the horizon year.

Item	Item Description	Product	Product Description
J		Product 16	Region-level and by-county, time-period-specific comparison of estimated average speed on freeways and all other facilities, separately, to MTC's estimates for the horizon year.
J		Product 17	Region-level and by-county, time-period-specific comparison of estimated volumes on bridges and county-lines, separately, to MTC's estimates for the horizon year.
J		Product 18	Region-level and by-county, time-period-specific comparison of estimated transit boardings by operator and technology, separately, to MTC's estimates for the horizon year.

ITEM A: BASE YEAR(S) MODEL DEVELOPMENT, CALIBRATION, AND VALIDATION REPORT(S) AND MODEL USER GUIDE

Product 1: Model Development, Calibration, & Validation Documentation

The Model Development, Calibration and Validation reports for the AlaCC model are available online at https://github.com/Bi-County-AlaCC-Activity-Based-Model/client_actc_ccta_model_doc/wiki/AlaCC-Activity%E2%80%90Based-Model-Documentation.

Product 2: Model User Guide

The Model User Guide and documentation for the AlaCC model are available online at https://github.com/Bi-County-AlaCC-Activity-Based-Model/client_actc_ccta_model_doc/wiki/AlaCC-Activity%E2%80%90Based-Model-Documentation.

ITEM B: BASE YEAR(S) DEMOGRAPHIC/ECONOMIC/LAND USE ASSUMPTIONS

This section presents a comparison of the 2015 "Base Year" demographic/economic/land use assumptions used in the AlaCC Model with MTC PBA 2050.

Product 2: A statement establishing that the differences between key ABAG land use variables (i.e., population, households, jobs, and employed residents) and those of the CTA fall within a minimum/maximum range for the base year.

The AlaCC Model's inputs for population, households, non-institutional group quarters population, jobs, and employed residents, both at the county and superdistrict levels, fall within the minimum/maximum range of the MTC PBA inputs for the nine-county Bay Area region for the 2015 base year.

Product 3: A table comparing the MTC/ABAG land use estimates with the CTA land use estimates by county and superdistrict for population, households, jobs, and employed residents for the base year

The AlaCC model land use estimates are compared with the MTC/ABAG allowable ranges for the 2015 base year for the following categories:

- County-level population (Table 2)
- County-level households (Table 3)
- County-level non-institutional group quarters population (Table 4)
- County-level jobs (Table 5)
- County-level employed residents (Table 6)
- Superdistrict-level population (Table 7)
- Superdistrict-level households (Table 8)
- Superdistrict-level non-institutional group quarters population (Table 9)
- Superdistrict-level jobs (Table 10)
- Superdistrict-level employed residents (Table 11)

The AlaCC model demographics assumptions for the 2015 base year are within the allowable ranges for all categories except two:

- Households in Napa County are higher than the PBA 2050 maximum by 35 households (see Table 3). However, the AlaCC model is within the allowable ranges for households for the two superdistricts in Napa County (see Table 8). Since the AlaCC model is within ranges at the greater level of detail at the superdistrict level, it is assumed that the demographic assumptions are consistent.
- Employed residents in Superdistrict 7 South San Mateo County are slightly higher than the PBA 2050 maximum, by 7 employed persons (see Table 11). All other demographic assumptions for this superdistrict are within acceptable ranges.

Table 2: Demographic/Economic/Land Use Assumptions – County-Level Population – Base Year 2015

County	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
San Francisco	859,200	884,500	883,650	Yes
San Mateo	758,900	774,200	773,676	Yes
Santa Clara	1,889,400	1,927,600	1,915,327	Yes
Alameda	1,593,100	1,625,300	1,594,060	Yes
Contra Costa	1,110,100	1,132,500	1,116,465	Yes
Solano	425,900	434,500	427,981	Yes
Napa	139,200	145,700	142,608	Yes
Sonoma	497,100	511,300	497,968	Yes
Marin	255,900	272,600	262,396	Yes
Bay Area	7,528,800	7,708,200	7,614,131	Yes
San Joaquin	n/a	n/a	708,631	-
AlaCC Model Area	n/a	n/a	8,322,762	-

Table 3: Demographic/Economic/Land Use Assumptions – County-Level Households – Base Year 2015

County	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
San Francisco	353,900	366,200	365,823	Yes
San Mateo	260,900	266,200	265,240	Yes
Santa Clara	620,200	632,700	622,348	Yes
Alameda	551,100	569,600	558,632	Yes
Contra Costa	382,200	390,000	382,266	Yes
Solano	141,700	147,500	141,758	Yes
Napa	49,000	50,400	50,435	No
Sonoma	187,100	190,900	187,973	Yes
Marin	104,700	108,700	106,763	Yes
Bay Area	2,650,800	2,722,200	2,681,238	Yes
San Joaquin	n/a	n/a	223,059	-
AlaCC Model Area	n/a	n/a	2,904,297	-

Table 4: Demographic/Economic/Land Use Assumptions – County-Level Non-Institutional Group Quarters Population – Base Year 2015

County	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
San Francisco	16,100	39,100	18,793	Yes
San Mateo	6,200	45,200	6,461	Yes
Santa Clara	25,900	119,600	27,666	Yes
Alameda	24,700	97,400	25,439	Yes
Contra Costa	5,500	65,400	6,721	Yes
Solano	2,000	26,500	2,525	Yes
Napa	2,400	7,500	2,800	Yes
Sonoma	5,100	26,100	6,151	Yes
Marin	2,900	12,500	3,860	Yes
Bay Area	90,800	439,300	100,416	Yes
San Joaquin	n/a	n/a	n/a	-
AlaCC Model Area	n/a	n/a	100,416	-

Table 5: Demographic/Economic/Land Use Assumptions – County-Level Jobs – Base Year 2015

County	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
San Francisco	636,600	682,600	682,001	Yes
San Mateo	390,100	398,000	392,033	Yes
Santa Clara	988,700	1,100,500	1,098,064	Yes
Alameda	801,800	867,600	867,532	Yes
Contra Costa	403,100	416,300	403,648	Yes
Solano	131,900	157,700	132,019	Yes
Napa	71,400	86,100	71,530	Yes
Sonoma	220,700	241,100	220,904	Yes
Marin	134,800	137,500	136,701	Yes
Bay Area	3,779,100	4,087,400	4,004,432	Yes
San Joaquin	n/a	n/a	234,826	-
AlaCC Model Area	n/a	n/a	4,239,258	-

Table 6: Demographic/Economic/Land Use Assumptions – County-Level Employed Residents – Base Year 2015

County	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
San Francisco	502,600	512,700	508,241	Yes
San Mateo	398,800	406,900	405,525	Yes
Santa Clara	952,400	971,600	953,042	Yes
Alameda	826,900	851,300	849,388	Yes
Contra Costa	535,700	594,000	555,684	Yes
Solano	200,300	232,900	232,654	Yes
Napa	71,100	80,600	80,498	Yes
Sonoma	249,700	281,600	270,437	Yes
Marin	130,600	136,000	135,847	Yes
Bay Area	3,868,100	4,067,600	3,991,316	Yes
San Joaquin	n/a	n/a	296,608	-
AlaCC Model Area	n/a	n/a	4,287,924	-

Table 7: Demographic/Economic/Land Use Assumptions – Superdistrict-Level Population – Base Year 2015

MTC Superdistrict	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
1-4. San Francisco County (Combined)	859,200	884,500	883,650	Yes
5. North San Mateo County	301,000	307,000	303,156	Yes
6. Central San Mateo County	221,900	233,000	232,824	Yes
7. South San Mateo County	232,300	237,900	237,696	Yes
8. Northwest Santa Clara County	188,400	195,700	195,526	Yes
9. North Santa Clara County	280,200	289,400	289,089	Yes
10. West Santa Clara County	329,200	340,800	340,427	Yes
11. Central Santa Clara County	321,800	328,500	322,132	Yes
12. East Santa Clara County	415,600	424,000	417,350	Yes
13. Central South Santa Clara County	229,800	234,500	232,279	Yes
14. South Santa Clara County	118,400	120,800	118,524	Yes
15. East Alameda County	211,100	226,900	213,430	Yes
16. South Alameda County	342,800	349,700	345,555	Yes
17. Central Alameda County	376,000	389,500	376,466	Yes
18. North Alameda County	475,100	484,700	476,969	Yes
19. Northwest Alameda County	179,500	183,100	181,640	Yes
20. West Contra Costa County	261,500	266,800	261,522	Yes
21. North Contra Costa County	235,000	241,500	235,382	Yes
22. Central Contra Costa County	148,100	151,000	150,622	Yes
23. South Contra Costa County	152,200	162,000	158,581	Yes
24. East Contra Costa County	309,100	315,400	310,358	Yes
25. South Solano County	148,800	154,300	151,428	Yes
26. North Solano County	275,900	281,500	276,553	Yes
27. South Napa County	102,000	104,000	103,840	Yes
28. North Napa County	36,900	42,000	38,768	Yes
29. South Sonoma County	170,800	175,300	171,394	Yes
30. Central Sonoma County	241,000	245,900	241,039	Yes
31. North Sonoma County	81,800	93,600	85,535	Yes
32. North Marin County	61,500	62,700	61,517	Yes
33. Central Marin County	107,000	113,900	109,193	Yes
34. South Marin County	86,600	96,800	91,686	Yes
Bay Area	7,500,500	7,736,700	7,614,131	Yes
San Joaquin	n/a	n/a	708,631	-
AlaCC Model Area	n/a	n/a	8,322,762	-

Table 8: Demographic/Economic/Land Use Assumptions – Superdistrict-Level Households – Base Year 2015

MTC Superdistrict	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
1-4. San Francisco County (Combined)	353,900	366,200	365,823	Yes
5. North San Mateo County	97,400	99,400	97,581	Yes
6. Central San Mateo County	83,200	87,300	87,197	Yes
7. South San Mateo County	79,100	80,700	80,462	Yes
8. Northwest Santa Clara County	71,100	73,900	75,564	Yes
9. North Santa Clara County	104,600	107,100	106,973	Yes
10. West Santa Clara County	118,700	121,100	120,736	Yes
11. Central Santa Clara County	104,900	108,500	105,013	Yes
12. East Santa Clara County	108,000	114,700	108,163	Yes
13. Central South Santa Clara County	73,100	75,000	73,151	Yes
14. South Santa Clara County	34,700	37,400	34,681	Yes
15. East Alameda County	72,300	79,900	72,354	Yes
16. South Alameda County	104,900	107,800	105,075	Yes
17. Central Alameda County	120,300	128,100	120,719	Yes
18. North Alameda County	179,200	182,800	180,163	Yes
19. Northwest Alameda County	72,000	73,400	75,501	Yes
20. West Contra Costa County	88,700	90,500	88,911	Yes
21. North Contra Costa County	84,700	89,500	84,769	Yes
22. Central Contra Costa County	60,100	61,300	60,526	Yes
23. South Contra Costa County	52,200	54,600	54,310	Yes
24. East Contra Costa County	93,700	96,900	93,750	Yes
25. South Solano County	52,300	53,400	52,529	Yes
26. North Solano County	89,200	94,200	89,229	Yes
27. South Napa County	34,100	34,800	34,500	Yes
28. North Napa County	14,600	16,000	15,935	Yes
29. South Sonoma County	64,100	65,400	64,228	Yes
30. Central Sonoma County	88,000	92,700	88,090	Yes
31. North Sonoma County	32,100	35,700	35,655	Yes
32. North Marin County	23,100	24,600	23,140	Yes
33. Central Marin County	42,400	44,100	43,967	Yes
34. South Marin County	37,800	41,500	41,512	Yes
Bay Area	2,634,500	2,738,500	2,681,238	Yes
San Joaquin	n/a	n/a	223,059	-
AlaCC Model Area	n/a	n/a	2,903,266	-

Table 9: Demographic/Economic/Land Use Assumptions – Superdistrict-Level Non-Institutional Group Quarters Population – Base Year 2015

MTC Superdistrict	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
1-4. San Francisco County (Combined)	16,100	39,100	18,793	Yes
5. North San Mateo County	1,700	15,900	1,855	Yes
6. Central San Mateo County	1,900	13,700	1,942	Yes
7. South San Mateo County	2,600	15,600	2,664	Yes
8. Northwest Santa Clara County	8,600	9,100	8,713	Yes
9. North Santa Clara County	900	22,000	1,006	Yes
10. West Santa Clara County	700	18,000	928	Yes
11. Central Santa Clara County	11,300	21,800	12,040	Yes
12. East Santa Clara County	2,200	25,100	2,318	Yes
13. Central South Santa Clara County	900	15,500	1,232	Yes
14. South Santa Clara County	1,200	8,000	1,429	Yes
15. East Alameda County	800	13,800	1,199	Yes
16. South Alameda County	1,500	21,500	1,965	Yes
17. Central Alameda County	3,000	24,300	3,498	Yes
18. North Alameda County	5,900	29,300	6,258	Yes
19. Northwest Alameda County	8,500	13,400	12,519	Yes
20. West Contra Costa County	1,100	15,100	1,304	Yes
21. North Contra Costa County	1,100	15,100	1,426	Yes
22. Central Contra Costa County	2,000	6,800	2,170	Yes
23. South Contra Costa County	300	8,400	617	Yes
24. East Contra Costa County	1,000	20,000	1,204	Yes
25. South Solano County	1,000	9,400	1,194	Yes
26. North Solano County	1,100	17,100	1,331	Yes
27. South Napa County	900	6,200	1,200	Yes
28. North Napa County	1,300	1,600	1,565	Yes
29. South Sonoma County	2,700	9,000	3,277	Yes
30. Central Sonoma County	1,800	13,700	2,129	Yes
31. North Sonoma County	600	3,300	745	Yes
32. North Marin County	500	3,100	859	Yes
33. Central Marin County	1,800	5,600	2,134	Yes
34. South Marin County	600	3,800	867	Yes
Bay Area	85,600	444,300	100,381	Yes
San Joaquin	n/a	n/a	n/a	-
AlaCC Model Area	n/a	n/a	100,381	-

Table 10: Demographic/Economic/Land Use Assumptions – Superdistrict-Level Jobs – Base Year 2015

MTC Superdistrict	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
1-4. San Francisco County (Combined)	636,600	682,600	682,001	Yes
5. North San Mateo County	130,200	134,800	130,309	Yes
6. Central San Mateo County	110,100	118,300	111,709	Yes
7. South San Mateo County	142,100	152,600	150,015	Yes
8. Northwest Santa Clara County	126,500	180,400	178,855	Yes
9. North Santa Clara County	332,800	370,100	369,700	Yes
10. West Santa Clara County	124,400	145,000	144,817	Yes
11. Central Santa Clara County	176,600	180,200	177,535	Yes
12. East Santa Clara County	116,000	121,000	120,911	Yes
13. Central South Santa Clara County	56,700	61,500	56,789	Yes
14. South Santa Clara County	48,500	49,500	49,457	Yes
15. East Alameda County	136,100	138,800	138,800	Yes
16. South Alameda County	141,500	160,700	141,617	Yes
17. Central Alameda County	156,400	163,400	156,917	Yes
18. North Alameda County	223,000	275,500	274,801	Yes
19. Northwest Alameda County	118,600	155,400	155,397	Yes
20. West Contra Costa County	74,800	79,300	79,193	Yes
21. North Contra Costa County	119,000	121,600	121,446	Yes
22. Central Contra Costa County	80,800	82,400	80,871	Yes
23. South Contra Costa County	66,200	71,800	66,635	Yes
24. East Contra Costa County	55,500	68,000	55,503	Yes
25. South Solano County	45,300	49,900	45,322	Yes
26. North Solano County	86,600	107,800	86,697	Yes
27. South Napa County	47,500	54,100	47,610	Yes
28. North Napa County	23,900	31,900	23,920	Yes
29. South Sonoma County	72,100	81,300	72,175	Yes
30. Central Sonoma County	117,400	127,300	117,516	Yes
31. North Sonoma County	31,200	32,500	31,213	Yes
32. North Marin County	27,600	28,900	28,866	Yes
33. Central Marin County	61,000	63,100	63,054	Yes
34. South Marin County	44,000	47,600	44,781	Yes
Bay Area	3,729,000	4,137,300	4,004,432	Yes
San Joaquin	n/a	n/a	234,826	-
AlaCC Model Area	n/a	n/a	4,239,258	-

Table 11: Demographic/Economic/Land Use Assumptions – Superdistrict-Level Employed Residents – Base Year 2015

MTC Superdistrict	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
1-4. San Francisco County (Combined)	502,600	512,700	508,241	Yes
5. North San Mateo County	155,400	166,700	162,271	Yes
6. Central San Mateo County	118,000	125,000	121,747	Yes
7. South San Mateo County	119,100	121,500	121,507	No
8. Northwest Santa Clara County	94,200	104,100	99,205	Yes
9. North Santa Clara County	153,100	156,200	154,037	Yes
10. West Santa Clara County	162,300	170,800	166,714	Yes
11. Central Santa Clara County	156,300	172,100	159,345	Yes
12. East Santa Clara County	193,500	209,000	196,805	Yes
13. Central South Santa Clara County	116,500	120,400	119,231	Yes
14. South Santa Clara County	56,200	59,500	57,705	Yes
15. East Alameda County	114,900	117,600	115,587	Yes
16. South Alameda County	175,100	190,000	189,181	Yes
17. Central Alameda County	191,700	195,600	191,863	Yes
18. North Alameda County	242,000	250,100	249,275	Yes
19. Northwest Alameda County	97,300	103,800	103,482	Yes
20. West Contra Costa County	130,000	133,900	132,799	Yes
21. North Contra Costa County	123,600	135,800	130,843	Yes
22. Central Contra Costa County	69,600	84,100	72,676	Yes
23. South Contra Costa County	71,900	94,500	78,086	Yes
24. East Contra Costa County	140,600	145,700	141,280	Yes
25. South Solano County	69,400	85,600	85,532	Yes
26. North Solano County	130,900	147,300	147,122	Yes
27. South Napa County	52,800	54,000	53,938	Yes
28. North Napa County	18,300	26,600	26,560	Yes
29. South Sonoma County	87,900	101,500	91,568	Yes
30. Central Sonoma County	121,200	132,200	130,948	Yes
31. North Sonoma County	40,800	48,000	47,921	Yes
32. North Marin County	22,800	31,900	22,843	Yes
33. Central Marin County	54,700	55,800	54,956	Yes
34. South Marin County	43,200	58,100	58,048	Yes
Bay Area	3,825,900	4,110,100	3,991,316	Yes
San Joaquin	n/a	n/a	296,608	-
AlaCC Model Area	n/a	n/a	4,287,924	-

Product 4: A table with CTA land use estimates by TAZ for the base year. Where differences exist between MTC/ABAG estimates, documentation of the CTA data sources should be included.

AlaCC model base year land uses by TAZ are available upon request. Allocation factors to map MTC RTAZs and MAZs to AlaCC TAZs were developed using geographic correspondence tables, master address files and a mapping of Census Blocks to AlaCC TAZs. Using these factors, PBA 2050 MAZ and TAZ data (households and employment) were allocated to AlaCC TAZs. Local jurisdictions reviewed, and in some cases, commented on specific allocations to AlaCC TAZs. These comments were incorporated to the extent possible while maintaining the PBA 2050 control totals at the superdistrict level. Documentation of the local jurisdiction review comments and their disposition is available upon request.

ITEM C: FORECAST YEAR(S) DEMOGRAPHIC/ECONOMIC/ LAND USE FORECASTS

This section presents a comparison of the 2050 "Forecast Year" demographic/economic/land use assumptions used in the AlaCC Model compared to MTC's PBA 2050.

Product 5: A statement establishing that the differences between key ABAG land use variables (i.e., population, households, jobs, and employed residents) and those of the CTA fall within +/- 1% for the horizon year at both the county and superdistrict level for the subject county. A statement establishing that no differences exist at the TAZ-level outside the county between the MTC/ABAG forecast or the MTC/ABAG/CTA revised forecast.

The AlaCC Model's estimates for population, households, non-institutional group quarters population, jobs, and employed residents for the 2050 forecast year, both at the county and superdistrict levels, fall within the plus or minus one percent of the MTC PBA inputs for the nine-county Bay Area region. In addition, no differences exist at the TAZ-level outside Alameda and Contra Costa Counties.

Product 6: A table comparing the MTC/ABAG land use estimates with the CTA land use estimates by county and superdistrict for population, households, jobs, and employed residents for the horizon year.

The AlaCC model land use estimates are compared with the MTC/ABAG allowable ranges for the 2050 forecast year for the following categories:

- County-level population (Table 12)
- County-level households (Table 13)
- County-level non-institutional group quarters population (Table 14)
- County-level jobs (Table 15)

- County-level employed residents (Table 16)
- Superdistrict-level population (Table 17)
- Superdistrict-level households (Table 18)
- Superdistrict-level non-institutional group quarters population (Table 19)
- Superdistrict-level jobs (Table 20)
- Superdistrict-level employed residents (Table 21)

The AlaCC model demographics assumptions for the 2050 forecast year are within the allowable ranges for all categories at both the county and superdistrict levels.

Table 12: Demographic/Economic/Land Use Assumptions – County-Level Population – Forecast Year 2050

County	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
San Francisco	1,260,100	1,285,500	1,272,099	Yes
San Mateo	1,004,400	1,024,700	1,015,062	Yes
Santa Clara	2,862,500	2,920,300	2,891,555	Yes
Alameda	2,154,100	2,197,600	2,175,231	Yes
Contra Costa	1,440,400	1,469,500	1,447,330	Yes
Solano	485,000	494,800	490,273	Yes
Napa	145,400	148,300	147,135	Yes
Sonoma	535,100	546,000	540,533	Yes
Marin	335,100	341,900	338,383	Yes
Bay Area	10,222,100	10,428,600	10,317,601	Yes
San Joaquin	n/a	n/a	1,014,184	-
AlaCC Model Area	n/a	n/a	11,331,785	-

Table 13: Demographic/Economic/Land Use Assumptions – County-Level Households – Forecast Year 2050

County	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
San Francisco	572,600	584,200	578,394	Yes
San Mateo	389,800	397,700	393,748	Yes
Santa Clara	1,064,400	1,085,900	1,075,202	Yes
Alameda	838,300	855,200	847,415	Yes
Contra Costa	545,900	556,900	551,276	Yes
Solano	174,800	178,300	176,474	Yes
Napa	55,000	56,200	55,659	Yes
Sonoma	217,700	222,100	219,937	Yes
Marin	144,300	147,200	145,754	Yes
Bay Area	4,002,800	4,083,700	4,043,859	Yes
San Joaquin	n/a	n/a	312,322	-
AlaCC Model Area	n/a	n/a	4,356,181	-

Table 14: Demographic/Economic/Land Use Assumptions – County-Level Non-Institutional Group Quarters Population – Forecast Year 2050

County	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
San Francisco	31,000	31,600	31,070	Yes
San Mateo	11,200	11,400	11,233	Yes
Santa Clara	37,200	37,900	37,633	Yes
Alameda	43,700	44,600	44,195	Yes
Contra Costa	12,400	12,700	12,517	Yes
Solano	16,200	16,600	16,486	Yes
Napa	4,800	4,900	4,854	Yes
Sonoma	12,500	12,700	12,608	Yes
Marin	5,400	5,500	5,500	Yes
Bay Area	174,400	177,900	176,096	Yes
San Joaquin	n/a	n/a	n/a	-
AlaCC Model Area	n/a	n/a	176,096	-

Table 15: Demographic/Economic/Land Use Assumptions – County-Level Jobs – Forecast Year 2050

County	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
San Francisco	909,000	927,400	918,225	Yes
San Mateo	502,200	512,400	507,320	Yes
Santa Clara	1,594,400	1,626,600	1,611,763	Yes
Alameda	1,170,200	1,193,800	1,181,986	Yes
Contra Costa	528,400	539,100	534,711	Yes
Solano	199,400	203,500	201,429	Yes
Napa	85,800	87,600	86,733	Yes
Sonoma	248,800	253,900	251,355	Yes
Marin	116,000	118,300	117,034	Yes
Bay Area	5,354,200	5,462,600	5,410,556	Yes
San Joaquin	n/a	n/a	318,655	-
AlaCC Model Area	n/a	n/a	5,729,211	-

Table 16: Demographic/Economic/Land Use Assumptions – County-Level Employed Residents – Forecast Year 2050

County	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
San Francisco	701,800	716,000	708,628	Yes
San Mateo	515,800	526,300	521,488	Yes
Santa Clara	1,462,900	1,492,500	1,479,028	Yes
Alameda	1,139,400	1,162,400	1,149,616	Yes
Contra Costa	758,600	773,900	763,809	Yes
Solano	254,500	259,700	257,453	Yes
Napa	77,800	79,400	78,676	Yes
Sonoma	291,100	297,000	293,931	Yes
Marin	163,200	166,500	164,734	Yes
Bay Area	5,365,100	5,473,700	5,417,363	Yes
San Joaquin	n/a	n/a	357,535	-
AlaCC Model Area	n/a	n/a	5,774,898	-

Table 17: Demographic/Economic/Land Use Assumptions – Superdistrict-Level Population – Forecast Year 2050

MTC Superdistrict	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
1-4. San Francisco County (Combined)	1,260,100	1,285,500	1,272,099	Yes
5. North San Mateo County	442,100	451,100	446,608	Yes
6. Central San Mateo County	285,400	291,100	288,320	Yes
7. South San Mateo County	276,900	282,500	280,134	Yes
8. Northwest Santa Clara County	235,600	240,400	238,311	Yes
9. North Santa Clara County	747,500	762,600	754,817	Yes
10. West Santa Clara County	428,400	437,100	432,714	Yes
11. Central Santa Clara County	438,200	447,100	442,802	Yes
12. East Santa Clara County	624,700	637,300	631,105	Yes
13. Central South Santa Clara County	255,000	260,200	257,479	Yes
14. South Santa Clara County	133,100	135,800	134,327	Yes
15. East Alameda County	343,600	350,600	349,708	Yes
16. South Alameda County	440,000	448,900	441,794	Yes
17. Central Alameda County	444,900	453,900	447,801	Yes
18. North Alameda County	680,000	693,800	685,591	Yes
19. Northwest Alameda County	245,500	250,500	250,337	Yes
20. West Contra Costa County	325,900	332,400	327,023	Yes
21. North Contra Costa County	339,600	346,500	341,477	Yes
22. Central Contra Costa County	192,700	196,600	193,504	Yes
23. South Contra Costa County	181,200	184,900	182,828	Yes
24. East Contra Costa County	401,000	409,100	402,498	Yes
25. South Solano County	149,800	152,900	150,377	Yes
26. North Solano County	335,100	341,900	339,896	Yes
27. South Napa County	106,800	108,900	108,100	Yes
28. North Napa County	38,600	39,400	39,035	Yes
29. South Sonoma County	201,200	205,300	203,218	Yes
30. Central Sonoma County	241,400	246,300	243,816	Yes
31. North Sonoma County	92,500	94,400	93,499	Yes
32. North Marin County	71,000	72,400	71,832	Yes
33. Central Marin County	159,000	162,200	160,443	Yes
34. South Marin County	105,100	107,300	106,108	Yes
Bay Area	10,221,900	10,428,900	10,317,601	Yes
San Joaquin	n/a	n/a	1,014,184	
AlaCC Model Area	n/a	n/a	11,331,785	

Table 18: Demographic/Economic/Land Use Assumptions – Superdistrict-Level Households – Forecast Year 2050

MTC Superdistrict	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
1-4. San Francisco County (Combined)	572,600	584,200	578,394	Yes
5. North San Mateo County	164,700	168,000	166,351	Yes
6. Central San Mateo County	120,100	122,500	121,249	Yes
7. South San Mateo County	105,000	107,200	106,148	Yes
8. Northwest Santa Clara County	101,000	103,000	102,004	Yes
9. North Santa Clara County	316,300	322,700	319,502	Yes
10. West Santa Clara County	170,000	173,400	171,670	Yes
11. Central Santa Clara County	166,100	169,500	167,784	Yes
12. East Santa Clara County	178,300	181,900	180,138	Yes
13. Central South Santa Clara County	90,300	92,100	91,212	Yes
14. South Santa Clara County	42,500	43,300	42,892	Yes
15. East Alameda County	130,500	133,200	133,027	Yes
16. South Alameda County	150,600	153,700	151,005	Yes
17. Central Alameda County	158,900	162,100	159,375	Yes
18. North Alameda County	284,200	290,000	287,790	Yes
19. Northwest Alameda County	114,100	116,400	116,218	Yes
20. West Contra Costa County	121,500	123,900	122,602	Yes
21. North Contra Costa County	132,600	135,200	133,554	Yes
22. Central Contra Costa County	88,000	89,800	88,668	Yes
23. South Contra Costa County	69,200	70,500	69,428	Yes
24. East Contra Costa County	134,700	137,400	137,024	Yes
25. South Solano County	56,500	57,600	56,679	Yes
26. North Solano County	118,300	120,700	119,795	Yes
27. South Napa County	39,200	40,000	39,635	Yes
28. North Napa County	15,900	16,200	16,024	Yes
29. South Sonoma County	82,500	84,100	83,294	Yes
30. Central Sonoma County	96,700	98,600	97,646	Yes
31. North Sonoma County	38,600	39,400	38,997	Yes
32. North Marin County	29,300	29,900	29,689	Yes
33. Central Marin County	65,200	66,500	65,789	Yes
34. South Marin County	49,700	50,800	50,276	Yes
Bay Area	4,003,100	4,083,800	4,043,859	Yes
San Joaquin	n/a	n/a	312,322	
AlaCC Model Area	n/a	n/a	4,356,181	

Table 19: Demographic/Economic/Land Use Assumptions – Superdistrict-Level Non-Institutional Group Quarters Population – Forecast Year 2050

MTC Superdistrict	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
1-4. San Francisco County (Combined)	31,000	31,600	31,070	Yes
5. North San Mateo County	3,700	3,800	3,766	Yes
6. Central San Mateo County	3,400	3,500	3,451	Yes
7. South San Mateo County	4,000	4,100	4,016	Yes
8. Northwest Santa Clara County	10,200	10,400	10,355	Yes
9. North Santa Clara County	2,000	2,000	2,000	Yes
10. West Santa Clara County	3,100	3,200	3,100	Yes
11. Central Santa Clara County	14,400	14,700	14,618	Yes
12. East Santa Clara County	4,200	4,200	4,200	Yes
13. Central South Santa Clara County	1,300	1,300	1,300	Yes
14. South Santa Clara County	2,000	2,100	2,060	Yes
15. East Alameda County	2,400	2,500	2,448	Yes
16. South Alameda County	3,200	3,300	3,292	Yes
17. Central Alameda County	7,800	8,000	7,808	Yes
18. North Alameda County	11,400	11,600	11,594	Yes
19. Northwest Alameda County	18,900	19,200	19,053	Yes
20. West Contra Costa County	2,300	2,300	2,300	Yes
21. North Contra Costa County	3,100	3,200	3,165	Yes
22. Central Contra Costa County	4,200	4,200	4,200	Yes
23. South Contra Costa County	700	800	715	Yes
24. East Contra Costa County	2,100	2,200	2,137	Yes
25. South Solano County	2,400	2,400	2,400	Yes
26. North Solano County	13,800	14,100	14,086	Yes
27. South Napa County	1,500	1,500	1,500	Yes
28. North Napa County	3,300	3,400	3,354	Yes
29. South Sonoma County	6,600	6,800	6,646	Yes
30. Central Sonoma County	4,600	4,700	4,670	Yes
31. North Sonoma County	1,200	1,300	1,292	Yes
32. North Marin County	900	900	900	Yes
33. Central Marin County	3,400	3,400	3,400	Yes
34. South Marin County	1,200	1,200	1,200	Yes
Bay Area	174,300	177,900	176,096	Yes
San Joaquin	n/a	n/a	n/a	
AlaCC Model Area	n/a	n/a	176,096	

Table 20: Demographic/Economic/Land Use Assumptions – Superdistrict-Level Jobs – Forecast Year 2050

MTC Superdistrict	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
1-4. San Francisco County (Combined)	909,000	927,400	918,225	Yes
5. North San Mateo County	186,300	190,100	188,200	Yes
6. Central San Mateo County	121,800	124,300	122,616	Yes
7. South San Mateo County	194,100	198,000	196,504	Yes
8. Northwest Santa Clara County	204,800	209,000	208,282	Yes
9. North Santa Clara County	622,500	635,100	628,745	Yes
10. West Santa Clara County	194,800	198,700	196,739	Yes
11. Central Santa Clara County	260,800	266,100	263,471	Yes
12. East Santa Clara County	168,000	171,400	169,660	Yes
13. Central South Santa Clara County	76,500	78,100	77,300	Yes
14. South Santa Clara County	66,900	68,200	67,566	Yes
15. East Alameda County	154,400	157,500	157,372	Yes
16. South Alameda County	218,700	223,100	222,394	Yes
17. Central Alameda County	282,000	287,700	282,017	Yes
18. North Alameda County	354,500	361,700	357,771	Yes
19. Northwest Alameda County	160,600	163,900	162,432	Yes
20. West Contra Costa County	130,400	133,000	130,945	Yes
21. North Contra Costa County	182,300	186,000	185,780	Yes
22. Central Contra Costa County	73,000	74,500	74,474	Yes
23. South Contra Costa County	59,900	61,100	60,046	Yes
24. East Contra Costa County	82,900	84,500	83,466	Yes
25. South Solano County	61,600	62,900	62,220	Yes
26. North Solano County	137,800	140,600	139,209	Yes
27. South Napa County	65,600	66,900	66,243	Yes
28. North Napa County	20,300	20,700	20,490	Yes
29. South Sonoma County	79,600	81,200	80,385	Yes
30. Central Sonoma County	129,800	132,400	131,137	Yes
31. North Sonoma County	39,400	40,200	39,833	Yes
32. North Marin County	28,500	29,100	28,827	Yes
33. Central Marin County	48,200	49,100	48,791	Yes
34. South Marin County	39,300	40,100	39,416	Yes
Bay Area	5,354,300	5,462,600	5,410,556	Yes
San Joaquin	n/a	n/a	318,655	
AlaCC Model Area	n/a	n/a	5,729,211	

Table 21: Demographic/Economic/Land Use Assumptions – Superdistrict-Level Employed Residents –Forecast Year 2050

MTC Superdistrict	PBA 2050 Minimum	PBA 2050 Maximum	AlaCC Model	Within Range
1-4. San Francisco County (Combined)	701,800	716,000	708,628	Yes
5. North San Mateo County	224,500	229,000	226,733	Yes
6. Central San Mateo County	151,800	154,800	153,272	Yes
7. South San Mateo County	139,600	142,400	141,483	NO
8. Northwest Santa Clara County	132,700	135,400	134,401	Yes
9. North Santa Clara County	418,000	426,400	422,958	Yes
10. West Santa Clara County	214,600	218,900	216,774	Yes
11. Central Santa Clara County	214,500	218,900	216,885	Yes
12. East Santa Clara County	285,900	291,700	288,907	Yes
13. Central South Santa Clara County	133,700	136,400	134,994	Yes
14. South Santa Clara County	63,500	64,800	64,109	Yes
15. East Alameda County	196,700	200,600	199,167	Yes
16. South Alameda County	240,000	244,800	240,643	Yes
17. Central Alameda County	217,700	222,100	218,624	Yes
18. North Alameda County	348,400	355,500	351,862	Yes
19. Northwest Alameda County	136,600	139,400	139,320	Yes
20. West Contra Costa County	163,500	166,800	164,677	Yes
21. North Contra Costa County	186,100	189,800	188,204	Yes
22. Central Contra Costa County	113,100	115,400	113,227	Yes
23. South Contra Costa County	112,500	114,800	113,436	Yes
24. East Contra Costa County	183,400	187,100	184,265	Yes
25. South Solano County	80,600	82,200	81,029	Yes
26. North Solano County	174,000	177,500	176,424	Yes
27. South Napa County	53,500	54,600	54,133	Yes
28. North Napa County	24,300	24,800	24,543	Yes
29. South Sonoma County	115,200	117,600	116,365	Yes
30. Central Sonoma County	129,500	132,100	130,685	Yes
31. North Sonoma County	46,400	47,400	46,881	Yes
32. North Marin County	26,600	27,200	26,947	Yes
33. Central Marin County	72,600	74,100	73,282	Yes
34. South Marin County	64,000	65,300	64,505	Yes
Bay Area	5,365,300	5,473,800	5,417,363	Yes
San Joaquin	n/a	n/a	357,535	
AlaCC Model Area	n/a	n/a	5,774,898	

Product 7: If land use estimates within the CTA's county or superdistricts are modified from MTC/ABAG's projections, agendas, discussion summaries, and action items from each meeting held with cities, MTC, and/or ABAG at which the redistribution was discussed, as well as before/after census-tract-level data summaries and maps.

Local jurisdictions reviewed and, in some cases, commented on the TAZ level allocations of PBA 2050 land use. These comments were incorporated to the extent possible while maintaining the PBA 2050 control totals at the superdistrict level. Appendix B includes census-tract-level maps of the difference in households and total employment for the 2050 Forecast Year between PBA 2050 and AlaCC.

ALAMEDA COUNTY LAND USE REVIEW

Exhibit 8 summarizes the principal comments received from Alameda County jurisdictions as well as their disposition. A more detailed list of local jurisdiction comments is available upon request.

Exhibit 8: Alameda County Local Jurisdiction Land Use Review

Jurisdiction	Jurisdiction's Comments	Comment Disposition
City of Dublin	Proposed increasing the 2015 household counts, based on their own surveys.	Revised household counts incorporated in AlaCC land use input dataset.
	Proposed increasing 2015 and 2050 jobs counts.	Partially accepted. Additional job counts were restricted to amounts that fit within the PBA allowable min/max range.
City of Fremont	Restricted household growth in areas zoned for commercial and industrial land uses.	Incorporated.
	Restricted job growth in areas zoned for residential land uses.	Incorporated.
City of Hayward	Restricted housing growth in areas with topographical challenges, including zones located over the Hayward Fault. The city also reduced housing growth in areas restricted by FAA regulations and areas with approved commercial/industrial developments.	Incorporated. Households removed from these zones were re-allocated to areas within the city without these restrictions.
City of Livermore	Proposed increasing future household allocation by approximately 20%, compared to the PBA proposed growth of 5.1%.	Partially incorporated. Household growth of 5.5% was accepted. Additional growth would exceed the PBA maximum allowance for the Livermore Sphere of Influence.

City of Piedmont	Requested a 12% increase in their household allowance to comply with their RNHA allocation.	Incorporated.
Union City	Proposed increasing the 2050 jobs counts.	Partially incorporated. Additional job growth was accepted up to the maximum allowed for the South Alameda County Superdistrict.

CONTRA COSTA COUNTY LAND USE REVIEW

As part of the 2023 Subregional Action Plan Updates, Contra Costa jurisdictions were asked to review the Plan Bay Area 2050 land use allocations to the CCTA Countywide Model 3,120 TAZ system. The jurisdictions of Clayton, Oakley, Pleasant Hill, Richmond, and Contra Costa County provided comments to CCTA. Comments included suggested shifts in the number of households or employees from one CCTA-TAZ to another. CCTA then shifted units when the change could be accommodated under the Countywide +/-1% tolerance, which was the metric at the time. Upon commencing the AlaCC model update, the CCTA-TAZ land use was re-allocated to the AlaCC model TAZ system, which required additional shifts in jurisdictional allocations based on the +/-1% tolerance at the MTC Superdistrict level.

ITEM D: FORECAST YEAR(S) PRICING ASSUMPTIONS

This section presents a comparison of the 2050 Forecast Year' automobile operating costs, transit fares and roadway/express lane and bridge tolls assumptions used in the AlaCC Model against the MTC PBA assumptions.

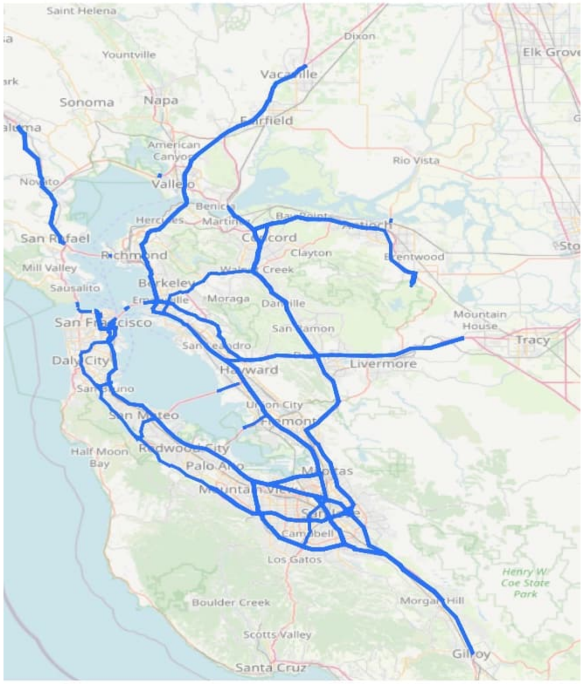
Product 8: Table comparing the assumed automobile operating cost, key transit fares, and roadway/express lane and bridge tolls to MTC's values for the horizon year. For express lanes and roadway tolls, documentation comparing the tolled extents should be included.

Table 22 lists the MTC PBA 2050 pricing assumptions for the 2050 forecast year, as obtained from the TM1.5.2 input files. Peak period tolls are reported whenever tolls vary by time of day. The AlaCC model uses the same pricing assumptions. The tolled extents are shown in Exhibit 9.

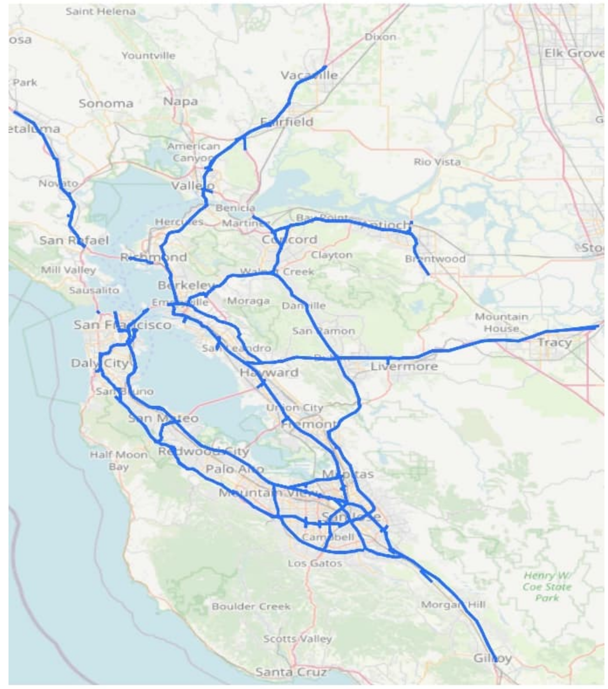
Table 22: Pricing Assumptions - Forecast Year 2050 PBA 2050 and AlaCC Model

Pricing Assumption	2050 Value in 2000 dollars		2050 Value in 2010 dollars	
Automobile Operating Cost (per mi.)		$0.174		$0.220
Transit Fares				
Muni Local Bus		$1.52		$1.92
AC Transit Local Bus		$1.35		$1.70
VTA Local Bus		$1.35		$1.70
SamTrans Local Bus		$1.35		$1.70
Bridge Tolls	Single Occ.	Carpool	Single Occ.	Carpool
San Francisco/Oakland Bay Bridge	$4.83	$3.76	$6.08	$2.71
Antioch Bridge	$4.29	$2.15	$5.41	$2.71
Benicia/Martinez Bridge	$4.29	$2.15	$5.41	$2.71
Carquinez Bridge	$4.29	$2.15	$5.41	$2.71
Dumbarton Bridge	$4.29	$2.15	$5.41	$2.71
Richmond/San Rafael Bridge	$4.29	$2.15	$5.41	$2.71
San Mateo Bridge	$4.29	$2.15	$5.41	$2.71
Golden Gate Bridge	$4.70	$3.62	$5.92	$4.56
Express Lane (Dynamic Tolling) Cost (per mile)				
Minimum	$0.030	$0.000	$0.038	$0.000
Maximum	$0.910	$0.410	$1.147	$0.517
Strategy T5: Per-Mile Roadway Toll Cost	$0.093	$0.031	$0.117	$0.039
SR-37 WB Per-Mile Roadway Toll Cost	$0.622	$0.311	$0.784	$0.392

Exhibit 9: Comparison of Roadway Network Tolled Extents for the 2050 Scenario



PBA 2050



AlaCC 2050

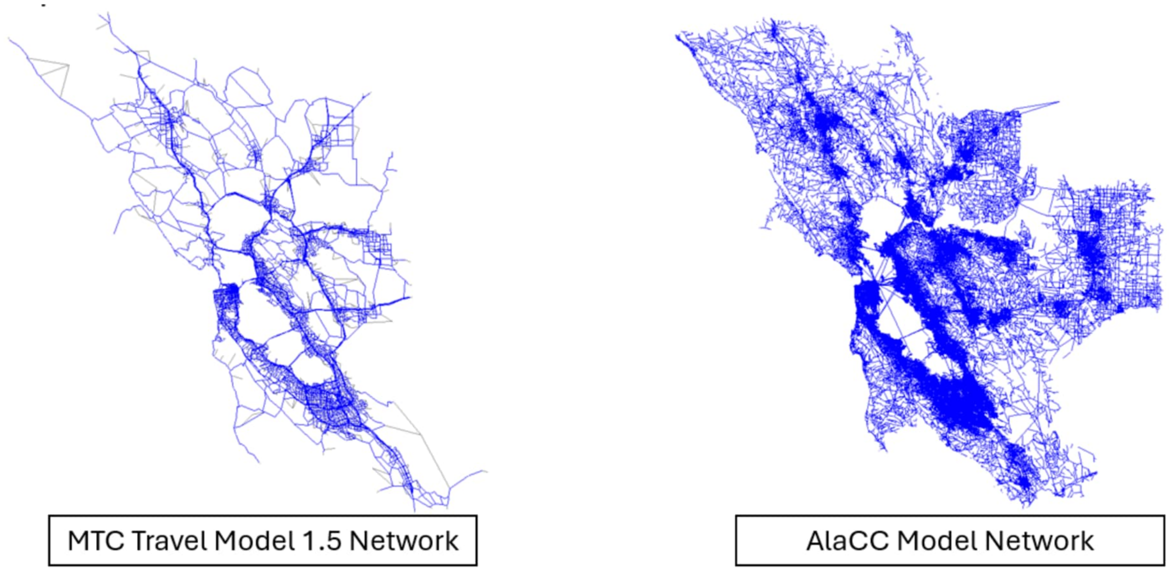
ITEM E: FORECAST YEAR(S) NETWORK ASSUMPTIONS

Product 9: Statement of network compatibility.

The AlaCC Model's roadway network is based on the road network for MTC Travel Model Two which was in turn based on the Open Streets Map (OSM) network as of December 2018. In contrast to MTC TM 1.5.2 which excludes most local streets, the AlaCC Model features enhanced network details across the nine Bay Area counties as well as San Joaquin County (shown in Exhibit 10 for 2015 Base Year).

The transit network for the 2015 base year was sourced from GTFS from years 2015 to 2017 (depending on the transit operator) and is routed over the OSM-based roadway network. Additionally, the AlaCC Model incorporates pedestrian, and bicycle networks derived from the December 2018 OSM data.

Exhibit 10: MTC TM 1.5.2 Network vs AlaCC Model Network - 2015 “Base Year”



The AlaCC Model includes high priority transportation projects included in the PBA 2050 Regional Transportation Plan in Alameda and Contra Costa counties and regionally significant projects in the other Bay Area counties and San Joaquin County. The projects for the forecast years—which strictly speaking include the 2019/2020 AlaCC scenario—are coded as changes from the 2015 base year networks using the MTC Network Wrangler project card format.

A list of the PBA 2050 projects coded for AlaCC is available in Appendix A.

ITEM F: FORECAST YEAR(S) AUTOMOBILE OWNERSHIP

This section presents a comparison of the 2050 forecast year automobile ownership estimates from the AlaCC Model with the MTC PBA estimates.

Product 10: County and superdistrict-level table comparing estimates of households by automobile ownership level (zero, one, two, or more automobiles) to MTC’s estimates for the horizon year.

Households by number of automobiles for the 2050 forecast year are listed by county of residence for PBA 2050 (Table 23), the AlaCC model (Table 24) and a comparison between the AlaCC model and PBA 2050 (Table 25).

Table 23: Households by Number of Automobiles, by County of Residence – Forecast Year 2050 - PBA 2050

County	Zero Autos	One Auto	Two Autos	Three Autos	Four-Plus Autos	Total
San Francisco	203,426	264,524	103,068	28,020	7,818	606,856
San Mateo	32,844	135,748	146,502	59,168	27,742	402,004
Santa Clara	99,870	409,794	375,292	146,776	76,806	1,108,538
Alameda	149,614	286,458	280,622	117,656	53,276	887,626
Contra Costa	28,484	164,838	221,668	103,344	41,374	559,708
Solano	7,840	44,266	68,948	38,674	20,090	179,818
Napa	3,128	16,298	22,856	11,280	5,666	59,228
Sonoma	15,872	62,742	88,418	42,844	19,884	229,760
Marin	7,780	46,312	60,890	25,136	9,426	149,544
Bay Area	548,858	1,430,980	1,368,264	572,898	262,082	4,183,082
Percent of Households	13%	34%	33%	14%	6%	100%

Table 24: Households by Number of Automobiles, by County of Residence – Forecast Year 2050- AlaCC Model

County	Zero Autos	One Auto	Two Autos	Three Autos	Four-Plus Autos	Total
San Francisco	169,707	255,127	132,833	42,300	8,227	608,193
San Mateo	27,093	128,433	155,507	72,013	25,013	408,060
Santa Clara	70,907	410,920	394,867	174,787	66,093	1,117,573
Alameda	134,927	278,673	288,133	137,360	48,267	887,360
Contra Costa	40,127	162,960	210,280	110,533	32,547	556,447
Solano	7,780	49,140	74,133	46,713	15,193	192,960
Napa	3,053	16,233	23,327	13,027	4,633	60,273
Sonoma	11,587	62,333	93,347	50,520	15,667	233,453
Marin	7,167	43,367	64,673	28,620	7,153	150,980
Bay Area	472,347	1,407,187	1,437,100	675,873	222,793	4,215,300
Percent of Households	11%	33%	34%	16%	5%	100%
San Joaquin	17,727	71,427	129,073	65,047	24,533	307,807
AlaCC Model Area	490,073	1,478,613	1,566,173	740,920	247,327	4,523,107

Table 25: Households by Number of Automobiles, by County of Residence – Forecast Year 2050- AlaCC Model Compared to PBA 2050

County	Zero Autos	One Auto	Two Autos	Three Autos	Four-Plus Autos	Total
San Francisco	-17%	-4%	29%	51%	5%	0%
San Mateo	-18%	-5%	6%	22%	-10%	2%
Santa Clara	-29%	0%	5%	19%	-14%	1%
Alameda	-10%	-3%	3%	17%	-9%	0%
Contra Costa	41%	-1%	-5%	7%	-21%	-1%
Solano	-1%	11%	8%	21%	-24%	7%
Napa	-2%	0%	2%	15%	-18%	2%
Sonoma	-27%	-1%	6%	18%	-21%	2%
Marin	-8%	-6%	6%	14%	-24%	1%
Bay Area	-14%	-2%	5%	18%	-15%	1%

The AlaCC model allocation of households by number of automobiles is very close to PBA 2050 for the highest number categories of one auto, two autos and total households. Compared to PBA 2050, the AlaCC model allocates fewer households to the zero-auto and four-plus auto categories, and higher numbers of households to the three-auto category. However, when comparing the share of households by number of automobiles, the AlaCC model is within one to two percentage points of PBA 2050 for all categories.

Households by number of automobiles for the 2050 forecast year are listed by superdistrict of residence for PBA 2050 (Table 26), the AlaCC model (Table 27) and the percentage differences between the AlaCC model and PBA 2050 (Table 28). As with the county-level comparison, the AlaCC model estimates similar numbers of households as PBA 2050 for the larger categories of one-auto and two-auto households.

Table 26: Households by Number of Automobiles, by Superdistrict of Residence – Forecast Year 2050 - PBA 2050

Superdistrict	Zero Autos	One Auto	Two Autos	Three Autos	Four-Plus Autos	Total
San Francisco County (Combined)	203,426	264,524	103,068	28,020	7,818	606,856
North San Mateo Co.	15,944	55,188	58,644	25,858	13,462	169,096
Central San Mateo Co.	9,780	44,056	46,032	17,106	6,724	123,698
South San Mateo Co.	7,120	36,504	41,826	16,204	7,556	109,210
Northwest Santa Clara Co.	11,840	44,910	38,260	12,800	4,852	112,662
North Santa Clara Co.	34,412	156,742	92,938	25,286	12,036	321,414
West Santa Clara Co.	8,246	56,708	70,042	26,504	11,248	172,748
Central Santa Clara Co.	30,752	71,078	51,100	18,900	10,702	182,532
East Santa Clara County	9,994	44,822	67,428	36,508	24,224	182,976
Central South Santa Clara Co.	3,276	24,736	38,030	17,428	8,510	91,980
South Santa Clara Co.	1,350	10,798	17,494	9,350	5,234	44,226
East Alameda County	4,562	40,074	56,452	22,998	8,812	132,898
South Alameda County	11,856	38,480	61,504	28,994	14,474	155,308
Central Alameda Co.	16,584	46,960	57,802	29,634	14,790	165,770
North Alameda County	70,538	109,444	76,756	28,220	12,282	297,240
Northwest Alameda Co.	46,074	51,500	28,108	7,810	2,918	136,410
West Contra Costa Co.	11,480	38,758	44,816	20,614	9,324	124,992
North Contra Costa Co.	5,954	43,782	52,446	23,378	9,828	135,388
Central Contra Costa Co.	6,288	33,338	34,120	13,798	3,984	91,528
South Contra Costa Co.	986	16,534	32,874	14,596	5,070	70,060
East Contra Costa Co.	3,776	32,426	57,412	30,958	13,168	137,740
South Solano County	3,632	16,272	21,868	11,206	5,640	58,618
North Solano County	4,208	27,994	47,080	27,468	14,450	121,200
South Napa County	2,130	10,740	15,842	8,030	4,274	41,016
North Napa County	998	5,558	7,014	3,250	1,392	18,212
South Sonoma County	5,790	25,856	33,810	16,392	7,370	89,218
Central Sonoma County	8,648	25,878	38,000	18,624	9,066	100,216
North Sonoma County	1,434	11,008	16,608	7,828	3,448	40,326
North Marin County	1,372	8,240	12,650	5,754	2,220	30,236
Central Marin County	3,658	22,104	27,016	11,210	4,438	68,426
South Marin County	2,750	15,968	21,224	8,172	2,768	50,882
Bay Area	548,858	1,430,980	1,368,264	572,898	262,082	4,183,082

Table 27: Households by Number of Automobiles, by Superdistrict of Residence – Forecast Year 2050 - AlaCC Model

Superdistrict	Zero Autos	One Auto	Two Autos	Three Autos	Four-Plus Autos	Total
San Francisco County (Combined)	169,707	255,127	132,833	42,300	8,227	608,193
North San Mateo Co.	13,660	55,033	61,047	30,080	11,420	171,240
Central San Mateo Co.	7,813	40,300	50,460	21,233	6,413	126,220
South San Mateo Co.	5,620	33,100	44,000	20,700	7,180	110,600
Northwest Santa Clara Co.	9,487	41,600	41,293	15,973	4,373	112,727
North Santa Clara Co.	21,187	165,100	96,553	30,847	10,720	324,407
West Santa Clara Co.	7,033	55,340	73,993	29,353	10,007	175,727
Central Santa Clara Co.	22,027	70,853	55,787	23,973	9,720	182,360
East Santa Clara County	7,373	44,660	70,393	42,540	19,433	184,400
Central South Santa Clara Co.	2,840	22,753	38,407	21,073	7,533	92,607
South Santa Clara Co.	960	10,613	18,440	11,027	4,307	45,347
East Alameda County	6,260	39,593	54,213	27,960	8,260	136,287
South Alameda County	7,493	41,060	61,480	32,073	11,480	153,587
Central Alameda Co.	13,420	46,427	59,867	32,633	12,893	165,240
North Alameda County	67,853	103,227	80,607	33,987	12,047	297,720
Northwest Alameda Co.	39,900	48,367	31,967	10,707	3,587	134,527
West Contra Costa Co.	12,573	39,027	43,193	19,873	7,620	122,287
North Contra Costa Co.	14,433	41,113	48,847	23,467	6,813	134,673
Central Contra Costa Co.	4,960	29,600	33,987	18,153	4,860	91,560
South Contra Costa Co.	2,740	16,333	30,113	16,847	4,367	70,400
East Contra Costa Co.	5,420	36,887	54,140	32,193	8,887	137,527
South Solano County	2,607	14,020	22,887	14,367	4,313	58,193
North Solano County	5,173	35,120	51,247	32,347	10,880	134,767
South Napa County	1,853	10,807	16,447	8,433	3,240	40,780
North Napa County	1,200	5,427	6,880	4,593	1,393	19,493
South Sonoma County	4,893	26,773	35,960	18,233	5,007	90,867
Central Sonoma County	5,487	26,427	40,027	22,353	7,447	101,740
North Sonoma County	1,207	9,133	17,360	9,933	3,213	40,847
North Marin County	1,233	8,500	13,167	6,313	1,293	30,507
Central Marin County	4,007	21,633	28,933	11,420	2,893	68,887
South Marin County	1,927	13,233	22,573	10,887	2,967	51,587
Bay Area	472,347	1,407,187	1,437,100	675,873	222,793	4,215,300
San Joaquin County	17,727	71,427	129,073	65,047	24,533	307,807
AlaCC Model Area	490,073	1,478,613	1,566,173	740,920	247,327	4,523,107

Table 28: Households by Number of Automobiles, by Superdistrict of Residence – Forecast Year 2050 - AlaCC Model Compared to PBA 2050

Superdistrict	Zero Autos	One Auto	Two Autos	Three Autos	Four-Plus Autos	Total
San Francisco County (Combined)	-17%	-4%	29%	51%	5%	0%
North San Mateo Co.	-14%	0%	4%	16%	-15%	1%
Central San Mateo Co.	-20%	-9%	10%	24%	-5%	2%
South San Mateo Co.	-21%	-9%	5%	28%	-5%	1%
Northwest Santa Clara Co.	-20%	-7%	8%	25%	-10%	0%
North Santa Clara Co.	-38%	5%	4%	22%	-11%	1%
West Santa Clara Co.	-15%	-2%	6%	11%	-11%	2%
Central Santa Clara Co.	-28%	0%	9%	27%	-9%	0%
East Santa Clara County	-26%	0%	4%	17%	-20%	1%
Central South Santa Clara Co.	-13%	-8%	1%	21%	-11%	1%
South Santa Clara Co.	-29%	-2%	5%	18%	-18%	3%
East Alameda County	37%	-1%	-4%	22%	-6%	3%
South Alameda County	-37%	7%	0%	11%	-21%	-1%
Central Alameda Co.	-19%	-1%	4%	10%	-13%	0%
North Alameda County	-4%	-6%	5%	20%	-2%	0%
Northwest Alameda Co.	-13%	-6%	14%	37%	23%	-1%
West Contra Costa Co.	10%	1%	-4%	-4%	-18%	-2%
North Contra Costa Co.	142%	-6%	-7%	0%	-31%	-1%
Central Contra Costa Co.	-21%	-11%	0%	32%	22%	0%
South Contra Costa Co.	178%	-1%	-8%	15%	-14%	0%
East Contra Costa Co.	44%	14%	-6%	4%	-33%	0%
South Solano County	-28%	-14%	5%	28%	-24%	-1%
North Solano County	23%	25%	9%	18%	-25%	11%
South Napa County	-13%	1%	4%	5%	-24%	-1%
North Napa County	20%	-2%	-2%	41%	0%	7%
South Sonoma County	-15%	4%	6%	11%	-32%	2%
Central Sonoma County	-37%	2%	5%	20%	-18%	2%
North Sonoma County	-16%	-17%	5%	27%	-7%	1%
North Marin County	-10%	3%	4%	10%	-42%	1%
Central Marin County	10%	-2%	7%	2%	-35%	1%
South Marin County	-30%	-17%	6%	33%	7%	1%
Bay Area	-14%	-2%	5%	18%	-15%	1%

ITEM G: FORECAST YEAR(S) COORDINATED DAILY ACTIVITY PATTERN MODEL/TRIP GENERATION

This section presents a comparison of the 2050 Forecast Year estimates of trip frequency by purpose from the AlaCC Model versus the MTC PBA estimates.

Product 11: Tables comparing estimates of trip frequency by purpose by superdistrict and county of residence to MTC's estimates for the horizon year.

The AlaCC model estimates of trip frequency by purpose are compared with the MTC PBA 2050 estimates for the 2050 forecast year in the following tables:

- Trip frequency by purpose, county-level, PBA 2050 (Table 29
- Trip frequency by purpose, county-level, AlaCC model (Table 30)
- Trip frequency by purpose, county-level, AlaCC model compared to PBA 2050 (Table 31)
- Trip frequency by purpose, superdistrict-level, PBA 2050 (Table 32)
- Trip frequency by purpose, superdistrict -level, AlaCC model (Table 33)
- Trip frequency by purpose, superdistrict -level, AlaCC model compared to PBA 2050 (Table 34)

The AlaCC Model estimates a higher number of Work and School trips and lower number of Eat Out, Shopping, Social and Other trips compared to the MTC PBA 2050 estimates. Total trips are within four percent of the PBA 2050 totals for the Bay Area.

Table 29: Trip Frequency by Purpose, by County of Residence – Forecast Year 2050 – PBA 2050

County	Work	University	School	At-Work	Eat Out	Escort	Shopping	Social	Other	All Purposes
San Francisco	1,439,406	145,202	210,458	307,770	264,462	206,162	674,056	168,380	932,928	4,348,824
San Mateo	1,023,706	51,232	298,854	233,368	202,940	250,402	597,494	144,974	878,710	3,681,680
Santa Clara	2,824,866	259,134	904,426	604,000	575,896	799,038	1,685,098	432,676	2,511,598	10,596,732
Alameda	1,994,558	258,762	647,198	435,472	432,830	556,718	1,287,452	333,814	1,872,686	7,819,490
Contra Costa	1,339,380	74,704	493,098	319,046	285,474	408,692	912,304	219,434	1,328,242	5,380,374
Solano	465,242	24,722	169,378	106,668	89,592	146,044	297,624	73,644	439,600	1,812,514
Napa	151,396	7,288	46,322	33,140	27,084	39,298	89,328	21,472	131,302	546,630
Sonoma	538,424	33,652	170,224	112,730	101,816	144,622	338,710	81,304	486,154	2,007,636
Marin	331,150	17,264	92,210	69,362	70,586	86,488	227,236	50,506	314,716	1,259,518
Bay Area	10,108,128	871,960	3,032,168	2,221,556	2,050,680	2,637,464	6,109,302	1,526,204	8,895,936	37,453,398

Table 30: Trip Frequency by Purpose, by County of Residence – Forecast Year 2050 – AlaCC Model

County	Work	University	School	At-Work	Eat Out	Escort	Shopping	Social	Other	All Purposes
San Francisco	1,779,813	115,393	328,547	464,900	242,053	243,380	643,160	150,060	875,753	4,843,060
San Mateo	1,038,393	42,540	357,467	213,560	154,773	249,493	489,107	100,687	619,320	3,265,340
Santa Clara	3,074,147	225,727	1,147,127	707,420	493,120	868,647	1,441,880	333,487	1,969,680	10,261,233
Alameda	2,227,820	204,140	852,567	506,740	357,393	622,120	1,061,780	259,973	1,422,120	7,514,653
Contra Costa	1,269,080	72,553	563,020	251,160	226,407	417,380	747,080	164,007	964,120	4,674,807
Solano	502,220	25,980	213,640	98,473	71,493	144,673	237,767	53,327	310,520	1,658,093
Napa	169,113	8,180	53,607	37,440	22,053	37,140	69,200	14,940	91,400	503,073
Sonoma	580,347	35,313	209,167	114,880	87,707	154,433	289,907	61,393	383,393	1,916,540
Marin	315,847	14,000	117,313	56,733	54,093	88,427	180,180	36,613	225,213	1,088,420
Bay Area	10,956,780	743,827	3,842,453	2,451,307	1,709,093	2,825,693	5,160,060	1,174,487	6,861,520	35,725,220
San Joaquin	683,087	52,700	477,273	149,547	114,653	330,620	466,373	108,393	604,660	2,987,307
AlaCC Model Area	11,639,867	796,527	4,319,727	2,600,853	1,823,747	3,156,313	5,626,433	1,282,880	7,466,180	38,712,527

Table 31: Trip Frequency by Purpose, by County of Residence – Forecast Year 2050 – AlaCC Model Compared to PBA 2050

County	Work	University	School	At-Work	Eat Out	Escort	Shopping	Social	Other	All Purposes
San Francisco	24%	-21%	56%	51%	-8%	18%	-5%	-11%	-6%	11%
San Mateo	1%	-17%	20%	-8%	-24%	0%	-18%	-31%	-30%	-11%
Santa Clara	9%	-13%	27%	17%	-14%	9%	-14%	-23%	-22%	-3%
Alameda	12%	-21%	32%	16%	-17%	12%	-18%	-22%	-24%	-4%
Contra Costa	-5%	-3%	14%	-21%	-21%	2%	-18%	-25%	-27%	-13%
Solano	8%	5%	26%	-8%	-20%	-1%	-20%	-28%	-29%	-9%
Napa	12%	12%	16%	13%	-19%	-5%	-23%	-30%	-30%	-8%
Sonoma	8%	5%	23%	2%	-14%	7%	-14%	-24%	-21%	-5%
Marin	-5%	-19%	27%	-18%	-23%	2%	-21%	-28%	-28%	-14%
Bay Area	8%	-15%	27%	10%	-17%	7%	-16%	-23%	-23%	-5%

Table 32: Trip Frequency by Purpose, by MTC Superdistrict – Forecast Year 2050 – PBA 2050

Superdistrict	Work	University	School	At-Work	Eat Out	Escort	Shopping	Social	Other	All Purposes
San Francisco Co. (Combined)	1,439,406	145,202	210,458	307,770	264,462	206,162	674,056	168,380	932,928	4,348,824
North San Mateo County	464,992	24,590	124,886	106,590	85,572	101,146	249,154	62,904	362,602	1,582,436
Central San Mateo County	292,234	13,666	81,586	67,822	60,962	70,408	182,122	41,620	261,488	1,071,908
South San Mateo County	266,480	12,976	92,382	58,956	56,406	78,848	166,218	40,450	254,620	1,027,336
NW Santa Clara County	241,928	47,240	56,986	53,286	56,168	60,834	156,618	34,894	218,670	926,624
North Santa Clara County	803,332	57,238	212,930	166,862	152,798	203,288	411,500	103,226	615,250	2,726,424
West Santa Clara County	415,690	36,406	146,292	93,832	92,912	123,854	281,970	65,822	408,978	1,665,756
Central Santa Clara County	454,170	52,424	124,324	87,882	86,820	110,746	248,486	67,850	372,252	1,604,954
East Santa Clara County	546,644	41,356	213,616	123,364	111,572	180,144	360,166	99,230	551,458	2,227,550
Central South Santa Clara Co.	247,704	16,608	94,878	54,498	52,018	78,502	154,868	41,590	230,584	971,250
South Santa Clara County	115,398	7,862	55,400	24,276	23,608	41,670	71,490	20,064	114,406	474,174
East Alameda County	353,206	14,094	130,626	81,572	73,658	102,574	194,718	51,600	290,564	1,292,612
South Alameda County	384,114	30,490	154,514	91,812	87,224	141,444	271,268	72,602	408,662	1,642,130
Central Alameda Co.	394,224	28,882	134,344	90,148	86,884	117,758	278,658	69,934	415,404	1,616,236
North Alameda County	608,666	56,536	208,208	125,260	126,972	156,258	396,388	105,142	570,920	2,354,350
NW Alameda County	254,348	128,760	19,506	46,680	58,092	38,684	146,420	34,536	187,136	914,162

Superdistrict	Work	University	School	At-Work	Eat Out	Escort	Shopping	Social	Other	All Purposes
West Contra Costa County	290,418	18,998	104,832	65,304	63,836	84,880	202,768	49,864	294,814	1,175,714
North Contra Costa County	340,862	15,968	109,480	80,406	68,766	90,734	209,660	49,538	302,352	1,267,766
Central Contra Costa County	186,694	10,368	47,332	46,396	41,626	46,502	136,610	31,204	192,950	739,682
South Contra Costa County	174,660	7,504	70,248	43,780	37,372	60,184	117,240	28,128	173,866	712,982
East Contra Costa County	346,746	21,866	161,206	83,160	73,874	126,392	246,026	60,700	364,260	1,484,230
South Solano County	141,412	8,254	50,366	33,506	28,826	41,522	94,252	22,512	133,582	554,232
North Solano County	323,830	16,468	119,012	73,162	60,766	104,522	203,372	51,132	306,018	1,258,282
South Napa County	113,102	4,414	37,238	25,098	20,230	30,824	64,744	15,814	98,548	410,012
North Napa County	38,294	2,874	9,084	8,042	6,854	8,474	24,584	5,658	32,754	136,618
South Sonoma County	213,844	18,330	61,724	45,018	38,510	50,976	123,704	29,894	177,682	759,682
Central Sonoma County	240,442	12,156	77,872	50,048	46,066	67,734	153,816	37,022	222,010	907,166
North Sonoma County	84,138	3,166	30,628	17,664	17,240	25,912	61,190	14,388	86,462	340,788
North Marin County	71,112	2,574	20,934	14,048	14,912	18,502	47,142	10,864	64,466	264,554
Central Marin County	156,396	8,734	43,844	32,230	32,678	39,918	102,236	23,166	146,568	585,770
South Marin County	103,642	5,956	27,432	23,084	22,996	28,068	77,858	16,476	103,682	409,194
Bay Area	10,108,128	871,960	3,032,168	2,221,556	2,050,680	2,637,464	6,109,302	1,526,204	8,895,936	37,453,398

Table 33: Trip Frequency by Purpose, by MTC Superdistrict – Forecast Year 2050 – AlaCC 2050

Superdistrict	Work	University	School	At-Work	Eat Out	Escort	Shopping	Social	Other	All Purposes
San Francisco Co. (Combined)	1,779,813	115,393	328,547	464,900	242,053	243,380	643,160	150,060	875,753	4,843,060
North San Mateo County	428,413	18,813	148,093	77,807	65,540	101,260	208,280	43,053	254,100	1,345,360
Central San Mateo County	283,467	12,720	99,920	56,340	47,893	72,367	155,360	30,107	189,507	947,680
South San Mateo County	326,513	11,007	109,453	79,413	41,340	75,867	125,467	27,527	175,713	972,300
NW Santa Clara County	333,853	44,173	89,427	104,207	53,653	72,147	151,640	32,727	196,633	1,078,460
North Santa Clara County	1,036,313	34,273	246,567	269,013	140,067	205,113	373,080	88,240	542,100	2,934,767
West Santa Clara County	423,060	35,493	182,427	91,987	79,860	135,020	243,333	52,353	321,893	1,565,427
Central Santa Clara County	515,333	69,527	175,333	115,607	85,627	132,360	234,480	61,567	345,927	1,735,760
East Santa Clara County	443,907	27,660	268,780	70,140	79,280	197,620	263,040	58,253	336,520	1,745,200
Central South Santa Clara Co.	201,367	7,627	117,653	32,147	38,000	80,467	121,027	27,253	152,360	777,900
South Santa Clara County	120,313	6,973	66,940	24,320	16,633	45,920	55,280	13,093	74,247	423,720
East Alameda County	346,540	17,373	174,947	70,820	62,287	130,447	191,347	41,473	242,053	1,277,287
South Alameda County	410,547	26,080	182,800	92,867	63,467	138,793	196,607	48,240	258,760	1,418,160
Central Alameda Co.	484,667	38,267	170,100	123,453	70,840	125,447	224,047	52,293	293,667	1,582,780
North Alameda County	683,880	43,240	263,007	146,727	105,753	173,993	287,860	82,700	442,633	2,229,793
NW Alameda County	302,187	79,180	61,713	72,873	55,047	53,440	161,920	35,267	185,007	1,006,633

Superdistrict	Work	University	School	At-Work	Eat Out	Escort	Shopping	Social	Other	All Purposes
West Contra Costa County	283,907	15,093	111,660	58,427	46,467	85,507	155,727	35,420	198,807	991,013
North Contra Costa County	374,407	25,027	126,840	95,140	60,113	99,020	185,693	46,427	266,280	1,278,947
Central Contra Costa County	190,267	10,953	59,807	34,833	34,200	46,433	118,233	22,253	133,140	650,120
South Contra Costa County	142,540	5,447	78,600	29,593	33,387	65,327	114,360	21,140	144,727	635,120
East Contra Costa County	277,960	16,033	186,113	33,167	52,240	121,093	173,067	38,767	221,167	1,119,607
South Solano County	152,887	6,493	63,773	31,307	21,907	43,960	73,967	15,993	97,873	508,160
North Solano County	349,333	19,487	149,867	67,167	49,587	100,713	163,800	37,333	212,647	1,149,933
South Napa County	130,193	4,960	43,060	30,727	16,667	28,440	52,027	10,873	69,233	386,180
North Napa County	38,920	3,220	10,547	6,713	5,387	8,700	17,173	4,067	22,167	116,893
South Sonoma County	200,733	15,207	76,107	34,420	32,847	55,300	105,887	20,987	137,053	678,540
Central Sonoma County	291,333	18,160	96,053	65,040	42,560	74,560	144,360	31,187	192,813	956,067
North Sonoma County	88,280	1,947	37,007	15,420	12,300	24,573	39,660	9,220	53,527	281,933
North Marin County	69,700	1,507	24,700	14,400	10,407	16,567	36,020	7,513	42,620	223,433
Central Marin County	138,453	6,860	57,413	23,600	24,707	41,433	81,220	15,693	103,933	493,313
South Marin County	107,693	5,633	35,200	18,733	18,980	30,427	62,940	13,407	78,660	371,673
Bay Area	10,956,780	743,827	3,842,453	2,451,307	1,709,093	2,825,693	5,160,060	1,174,487	6,861,520	35,725,220
San Joaquin Co.	683,087	52,700	477,273	149,547	114,653	330,620	466,373	108,393	604,660	2,987,307
AlaCC Model	11,639,867	796,527	4,319,727	2,600,853	1,823,747	3,156,313	5,626,433	1,282,880	7,466,180	38,712,527

Table 34: Trip Frequency by Purpose, by MTC Superdistrict – Forecast Year 2050 – AlaCC Model Compared to PBA 2050

Superdistrict	Work	University	School	At-Work	Eat Out	Escort	Shopping	Social	Other	All Purposes
San Francisco Co. (Combined)	24%	-21%	56%	51%	-8%	18%	-5%	-11%	-6%	11%
North San Mateo County	-8%	-23%	19%	-27%	-23%	0%	-16%	-32%	-30%	-15%
Central San Mateo County	-3%	-7%	22%	-17%	-21%	3%	-15%	-28%	-28%	-12%
South San Mateo County	23%	-15%	18%	35%	-27%	-4%	-25%	-32%	-31%	-5%
NW Santa Clara County	38%	-6%	57%	96%	-4%	19%	-3%	-6%	-10%	16%
North Santa Clara County	29%	-40%	16%	61%	-8%	1%	-9%	-15%	-12%	8%
West Santa Clara County	2%	-3%	25%	-2%	-14%	9%	-14%	-20%	-21%	-6%
Central Santa Clara County	13%	33%	41%	32%	-1%	20%	-6%	-9%	-7%	8%
East Santa Clara County	-19%	-33%	26%	-43%	-29%	10%	-27%	-41%	-39%	-22%
Central South Santa Clara Co.	-19%	-54%	24%	-41%	-27%	3%	-22%	-34%	-34%	-20%
South Santa Clara County	4%	-11%	21%	0%	-30%	10%	-23%	-35%	-35%	-11%
East Alameda County	-2%	23%	34%	-13%	-15%	27%	-2%	-20%	-17%	-1%
South Alameda County	7%	-14%	18%	1%	-27%	-2%	-28%	-34%	-37%	-14%
Central Alameda Co.	23%	32%	27%	37%	-18%	7%	-20%	-25%	-29%	-2%
North Alameda County	12%	-24%	26%	17%	-17%	11%	-27%	-21%	-22%	-5%
NW Alameda County	19%	-39%	216%	56%	-5%	38%	11%	2%	-1%	10%

Superdistrict	Work	University	School	At-Work	Eat Out	Escort	Shopping	Social	Other	All Purposes
West Contra Costa County	-2%	-21%	7%	-11%	-27%	1%	-23%	-29%	-33%	-16%
North Contra Costa County	10%	57%	16%	18%	-13%	9%	-11%	-6%	-12%	1%
Central Contra Costa County	2%	6%	26%	-25%	-18%	0%	-13%	-29%	-31%	-12%
South Contra Costa County	-18%	-27%	12%	-32%	-11%	9%	-2%	-25%	-17%	-11%
East Contra Costa County	-20%	-27%	15%	-60%	-29%	-4%	-30%	-36%	-39%	-25%
South Solano County	8%	-21%	27%	-7%	-24%	6%	-22%	-29%	-27%	-8%
North Solano County	8%	18%	26%	-8%	-18%	-4%	-19%	-27%	-31%	-9%
South Napa County	15%	12%	16%	22%	-18%	-8%	-20%	-31%	-30%	-6%
North Napa County	2%	12%	16%	-17%	-21%	3%	-30%	-28%	-32%	-14%
South Sonoma County	-6%	-17%	23%	-24%	-15%	8%	-14%	-30%	-23%	-11%
Central Sonoma County	21%	49%	23%	30%	-8%	10%	-6%	-16%	-13%	5%
North Sonoma County	5%	-39%	21%	-13%	-29%	-5%	-35%	-36%	-38%	-17%
North Marin County	-2%	-41%	18%	3%	-30%	-10%	-24%	-31%	-34%	-16%
Central Marin County	-11%	-21%	31%	-27%	-24%	4%	-21%	-32%	-29%	-16%
South Marin County	4%	-5%	28%	-19%	-17%	8%	-19%	-19%	-24%	-9%
Bay Area	8%	-15%	27%	10%	-17%	7%	-16%	-23%	-23%	-5%

ITEM H: FORECAST YEAR(S) ACTIVITY/TRIP LOCATION

This section presents a comparison of the 2050 Forecast Year estimates of activity/trip location estimated in the AlaCC Model versus the MTC PBA.

Product 12: Region-level tables comparing estimates of average trip distance frequency distribution by tour/trip purpose to MTC's estimates for the horizon year.

Average tip distances by trip purpose are compared between PBA 2050 and the AlaCC model for the 2050 forecast year (Table 35). The AlaCC model estimates five percent longer average trip distances than TM1.5.2. Average trip distance for all purposes is within 10% of the average distance estimated by TM1.5 except for university trips.

Table 35: Average Trip Distance by Trip Purpose - Forecast Year 2050

Tour Purpose	PBA 2050 Average Trip Distance (miles)	AlaCC Model Average Trip Distance (miles)	Difference
Work	9.62	10.05	4%
University	5.40	6.97	29%
School	2.86	2.66	-7%
At-Work	3.37	3.71	10%
Eat Out	5.59	5.23	-6%
Escort	3.61	3.82	6%
Shopping	4.24	4.62	9%
Social	5.39	5.34	-1%
Other	4.95	5.20	5%
All Purposes	5.89	6.15	5%

Product 13: County-to-county comparison of journey-to-work or home-based work flow estimates to MTC's estimates for the horizon year.

Journey to work flows by county of residence are listed for the 2050 forecast year for PBA 2050 (Table 36), the AlaCC model (Table 37), and the percentages of work flows to each county (Table 38 and Table 39). These tables show the count of workers excluding workers that do not travel on the simulation day due to telecommuting, sick day, vacation day, etc.

Table 36: Journey-to-Work Flows – 2050 Forecast Year – PBA 2050

County of Residence	County of Work									
	San Francisco	San Mateo	Santa Clara	Alameda	Contra Costa	Solano	Napa	Sonoma	Marin	Bay Area
San Francisco	377,800	59,100	57,900	43,000	7,300	500	200	200	5,100	551,100
San Mateo	102,700	178,200	69,600	21,400	2,200	100	0	0	1,500	375,700
Santa Clara	26,300	57,300	859,200	74,100	1,400	0	0	0	100	1,018,400
Alameda	90,700	31,300	73,400	484,100	45,600	3,000	1,000	100	2,200	731,400
Contra Costa	50,100	9,800	14,200	119,700	264,400	15,300	3,200	400	4,000	481,100
Solano	6,500	400	200	9,700	23,300	108,700	13,000	1,300	1,800	164,900
Napa	1,500	100	0	2,100	3,900	7,500	32,900	4,100	1,100	53,200
Sonoma	4,200	300	0	1,500	1,800	2,000	5,400	161,500	12,400	189,100
Marin	35,300	3,100	200	10,200	8,900	1,500	1,000	4,200	54,200	118,600
Bay Area	695,100	339,600	1,074,700	765,800	358,800	138,600	56,700	171,800	82,400	3,683,500

Note: Values represent work tours, excluding workers that telecommute on the simulation day.

Table 37: Journey-to-Work Flows – 2050 Forecast Year – AlaCC Model

County of Residence	County of Work									
	San Francisco	San Mateo	Santa Clara	Alameda	Contra Costa	Solano	Napa	Sonoma	Marin	Bay Area
San Francisco	440,180	45,960	54,580	39,987	4,253	307	173	200	6,173	591,813
San Mateo	106,127	217,407	66,333	15,120	1,893	127	53	80	1,253	408,393
Santa Clara	32,100	54,913	934,767	61,907	1,387	27	7	0	73	1,085,180
Alameda	93,173	32,920	62,360	558,320	43,480	1,227	407	260	2,007	794,153
Contra Costa	43,160	10,373	11,940	99,180	315,853	12,920	3,087	953	5,340	502,807
Solano	7,007	1,060	400	14,493	14,360	133,527	14,707	2,220	2,633	190,407
Napa	2,073	113	20	2,080	2,833	6,013	40,687	4,320	833	58,973
Sonoma	7,000	587	53	2,900	2,200	2,073	6,033	180,087	13,840	214,773
Marin	33,320	4,373	540	12,507	11,507	1,240	853	7,467	61,887	133,693
Bay Area	764,140	367,707	1,130,993	806,493	397,767	157,460	66,007	195,587	94,040	3,980,193

Note: Values represent work tours, excluding workers that telecommute on the simulation day.

Table 38: Journey-to-Work Percentages -2050 Forecast Year – PBA 2050

County of Residence	County of Work									
	San Francisco	San Mateo	Santa Clara	Alameda	Contra Costa	Solano	Napa	Sonoma	Marin	Bay Area
San Francisco	68.6%	10.7%	10.5%	7.8%	1.3%	0.1%	0.0%	0.0%	0.9%	100.0%
San Mateo	27.3%	47.4%	18.5%	5.7%	0.6%	0.0%	0.0%	0.0%	0.4%	100.0%
Santa Clara	2.6%	5.6%	84.4%	7.3%	0.1%	0.0%	0.0%	0.0%	0.0%	100.0%
Alameda	12.4%	4.3%	10.0%	66.2%	6.2%	0.4%	0.1%	0.0%	0.3%	100.0%
Contra Costa	10.4%	2.0%	3.0%	24.9%	55.0%	3.2%	0.7%	0.1%	0.8%	100.0%
Solano	3.9%	0.2%	0.1%	5.9%	14.1%	65.9%	7.9%	0.8%	1.1%	100.0%
Napa	2.8%	0.2%	0.0%	3.9%	7.3%	14.1%	61.8%	7.7%	2.1%	100.0%
Sonoma	2.2%	0.2%	0.0%	0.8%	1.0%	1.1%	2.9%	85.4%	6.6%	100.0%
Marin	29.8%	2.6%	0.2%	8.6%	7.5%	1.3%	0.8%	3.5%	45.7%	100.0%
Bay Area	18.9%	9.2%	29.2%	20.8%	9.7%	3.8%	1.5%	4.7%	2.2%	100.0%

Table 39: Journey-to-Work Percentages -2050 Forecast Year – AlaCC Model

County of Residence	County of Work									
	San Francisco	San Mateo	Santa Clara	Alameda	Contra Costa	Solano	Napa	Sonoma	Marin	Bay Area
San Francisco	74.4%	7.8%	9.2%	6.8%	0.7%	0.1%	0.0%	0.0%	1.0%	100.0%
San Mateo	26.0%	53.2%	16.2%	3.7%	0.5%	0.0%	0.0%	0.0%	0.3%	100.0%
Santa Clara	3.0%	5.1%	86.1%	5.7%	0.1%	0.0%	0.0%	0.0%	0.0%	100.0%
Alameda	11.7%	4.1%	7.9%	70.3%	5.5%	0.2%	0.1%	0.0%	0.3%	100.0%
Contra Costa	8.6%	2.1%	2.4%	19.7%	62.8%	2.6%	0.6%	0.2%	1.1%	100.0%
Solano	3.7%	0.6%	0.2%	7.6%	7.5%	70.1%	7.7%	1.2%	1.4%	100.0%
Napa	3.5%	0.2%	0.0%	3.5%	4.8%	10.2%	69.0%	7.3%	1.4%	100.0%
Sonoma	3.3%	0.3%	0.0%	1.4%	1.0%	1.0%	2.8%	83.8%	6.4%	100.0%
Marin	24.9%	3.3%	0.4%	9.4%	8.6%	0.9%	0.6%	5.6%	46.3%	100.0%
Bay Area	19.2%	9.2%	28.4%	20.3%	10.0%	4.0%	1.7%	4.9%	2.4%	100.0%

The AlaCC Model estimates approximately eight percent more workers traveling to work compared to the MTC PBA 2050 estimates for the nine counties in the Bay Area region. Comparing the percentages of work flows to destination counties, the AlaCC model is typically within five percent of the PBA 2050 percentages. The AlaCC model estimates higher percentages of worker flows staying within Alameda County and Contra Costa County, which may be due to the higher level of network and TAZ detail within those counties in the AlaCC model.

ITEM I: FORECAST YEAR(S) TRAVEL MODE CHOICE

This section presents a comparison of the 2050 Forecast Year estimates of travel mode choice estimated by the AlaCC Model versus the MTC PBA.

Product 14: County-level (by county of residence) tables comparing travel mode flow estimates by tour purpose to MTC's estimates for the horizon year. The table summaries should be stratified by transit, auto (split into drive alone, shared ride 2, shared ride 3+), walk, and bicycle.

Mode shares by county of residence and tour purpose for the 2050 forecast year are listed for PBA 2050 (Table 40) and the AlaCC model (Table 41). For the total model areas, the AlaCC model estimates higher drive alone percentages (40.1% versus 36.7%) but lower overall auto percentages (70.1% versus 72.1%). The AlaCC model estimates lower transit percentages (6.4% versus 8.6%) and higher percentages of trips by active transportation (15.4% versus 18.2% for walking and 8.1% versus 6.4% for bicycling). These trends also apply to trips made by residents of Alameda and Contra Costa counties. The higher percentages of active transportation modes may be due to the more detailed local networks included in the AlaCC model for all nine Bay Area counties.

Table 40: Table 40: Trip Mode Choice by Tour Purpose and County of Residence - Forecast Year 2050 - PBA 2050

County of Residence	Trip Mode									
	Tour Purpose	Drive Alone	Shared Ride 2	Shared Ride 3+	Taxi & Ride-Hailing	All Auto	Transit	Walk	Bicycle	All Modes
San Francisco	Work	20.8%	5.5%	2.1%	3.2%	31.5%	48.7%	12.1%	7.7%	100.0%
	University	2.9%	6.5%	0.9%	3.7%	14.0%	53.2%	26.6%	6.2%	100.0%
	School	6.7%	15.4%	21.5%	0.9%	44.5%	9.1%	28.3%	18.0%	100.0%
	At-Work	16.7%	10.5%	3.7%	2.9%	33.9%	5.1%	53.2%	7.8%	100.0%
	Eat Out	11.0%	17.6%	6.9%	6.2%	41.7%	25.6%	23.9%	8.7%	100.0%
	Escort	12.8%	17.7%	11.3%	5.2%	46.9%	14.9%	30.2%	8.0%	100.0%
	Shopping	25.1%	16.0%	7.7%	6.6%	55.3%	19.5%	21.1%	4.1%	100.0%
	Social	11.9%	15.3%	8.8%	6.5%	42.5%	24.5%	25.2%	7.8%	100.0%
	Other	18.1%	15.7%	9.2%	7.6%	50.6%	20.7%	21.8%	6.9%	100.0%
	Total	18.0%	11.9%	6.5%	4.9%	41.3%	29.4%	21.8%	7.5%	100.0%
San Mateo	Work	54.7%	10.2%	3.9%	1.4%	70.2%	19.4%	3.1%	7.3%	100.0%
	University	26.6%	22.4%	3.5%	9.5%	62.0%	23.7%	4.3%	10.0%	100.0%
	School	10.7%	23.7%	33.0%	1.0%	68.4%	2.5%	15.1%	14.1%	100.0%
	At-Work	37.1%	11.7%	5.2%	1.2%	55.2%	1.6%	37.2%	5.9%	100.0%
	Eat Out	28.4%	21.8%	12.4%	5.9%	68.5%	7.2%	16.7%	7.6%	100.0%
	Escort	21.9%	29.5%	17.4%	3.4%	72.1%	2.6%	18.9%	6.3%	100.0%
	Shopping	44.4%	21.2%	12.5%	6.0%	84.0%	3.3%	9.4%	3.3%	100.0%
	Social	27.9%	21.2%	15.0%	7.4%	71.5%	6.7%	14.9%	7.0%	100.0%
	Other	33.5%	20.7%	16.6%	8.3%	79.0%	4.9%	11.0%	5.0%	100.0%
	Total	38.1%	18.2%	12.6%	4.5%	73.4%	8.6%	11.5%	6.5%	100.0%

County of Residence	Trip Mode									
	Tour Purpose	Drive Alone	Shared Ride 2	Shared Ride 3+	Taxi & Ride-Hailing	All Auto	Transit	Walk	Bicycle	All Modes
Santa Clara	Work	60.1%	10.8%	3.9%	1.7%	76.5%	11.5%	3.5%	8.6%	100.0%
	University	27.2%	21.7%	3.5%	8.6%	61.1%	21.4%	8.5%	9.0%	100.0%
	School	10.0%	26.4%	35.8%	1.1%	73.4%	2.2%	13.5%	10.9%	100.0%
	At-Work	45.3%	13.1%	5.6%	1.2%	65.2%	0.9%	26.5%	7.5%	100.0%
	Eat Out	29.7%	21.8%	13.7%	6.8%	71.9%	3.8%	17.1%	7.2%	100.0%
	Escort	22.3%	30.3%	18.1%	3.6%	74.2%	2.1%	18.5%	5.2%	100.0%
	Shopping	43.9%	21.6%	14.2%	6.7%	86.4%	2.7%	8.3%	2.6%	100.0%
	Social	29.3%	21.3%	16.5%	8.4%	75.4%	3.4%	15.1%	6.2%	100.0%
	Other	33.6%	21.1%	18.4%	9.5%	82.7%	2.6%	10.2%	4.5%	100.0%
	Total	39.6%	19.2%	13.9%	5.1%	77.8%	5.4%	10.5%	6.4%	100.0%
Alameda	Work	52.9%	10.0%	3.9%	1.6%	68.4%	18.5%	4.6%	8.5%	100.0%
	University	18.6%	16.0%	2.5%	6.4%	43.5%	24.1%	19.2%	13.2%	100.0%
	School	10.0%	25.1%	34.2%	1.0%	70.2%	3.2%	15.0%	11.6%	100.0%
	At-Work	38.4%	11.6%	5.0%	1.3%	56.4%	1.4%	35.2%	7.0%	100.0%
	Eat Out	27.0%	21.3%	12.2%	6.3%	66.7%	6.6%	19.6%	7.1%	100.0%
	Escort	22.1%	29.7%	17.6%	3.3%	72.6%	2.1%	19.8%	5.4%	100.0%
	Shopping	41.5%	21.3%	13.4%	6.1%	82.3%	4.8%	10.2%	2.7%	100.0%
	Social	25.4%	20.3%	15.7%	7.9%	69.2%	6.1%	18.3%	6.4%	100.0%
	Other	31.4%	20.5%	17.1%	8.5%	77.5%	5.0%	12.7%	4.8%	100.0%
	Total	35.6%	18.4%	13.1%	4.7%	71.8%	8.6%	13.0%	6.6%	100.0%

County of Residence	Trip Mode									
	Tour Purpose	Drive Alone	Shared Ride 2	Shared Ride 3+	Taxi & Ride-Hailing	All Auto	Transit	Walk	Bicycle	All Modes
Contra Costa	Work	62.5%	10.9%	4.5%	0.8%	78.7%	12.9%	2.8%	5.6%	100.0%
	University	34.3%	29.0%	4.4%	7.1%	74.8%	13.3%	3.5%	8.4%	100.0%
	School	11.9%	24.9%	34.3%	0.8%	71.8%	1.4%	12.2%	14.5%	100.0%
	At-Work	42.1%	11.4%	5.3%	0.9%	59.7%	0.8%	34.9%	4.5%	100.0%
	Eat Out	29.3%	21.7%	13.6%	5.5%	70.1%	3.4%	19.6%	7.0%	100.0%
	Escort	21.4%	28.7%	16.9%	2.7%	69.7%	0.9%	23.3%	6.1%	100.0%
	Shopping	44.3%	21.7%	13.7%	5.7%	85.3%	1.3%	10.4%	3.0%	100.0%
	Social	28.1%	22.4%	15.8%	7.1%	73.3%	3.1%	17.3%	6.3%	100.0%
	Other	33.3%	21.2%	16.9%	8.1%	79.6%	2.0%	13.5%	4.9%	100.0%
	Total	39.7%	19.2%	13.8%	4.2%	76.9%	4.6%	12.6%	5.9%	100.0%
Solano	Work	70.2%	11.1%	4.3%	0.7%	86.3%	5.1%	2.3%	6.3%	100.0%
	University	39.6%	34.9%	6.1%	5.7%	86.2%	9.5%	1.4%	2.9%	100.0%
	School	10.8%	26.0%	35.5%	0.9%	73.1%	0.6%	12.9%	13.5%	100.0%
	At-Work	48.6%	11.4%	5.3%	0.8%	66.1%	0.3%	28.4%	5.2%	100.0%
	Eat Out	27.6%	22.4%	12.8%	5.5%	68.2%	1.1%	23.0%	7.7%	100.0%
	Escort	21.3%	28.0%	17.2%	2.8%	69.3%	0.4%	24.9%	5.4%	100.0%
	Shopping	43.5%	22.4%	13.3%	6.1%	85.4%	0.4%	11.4%	2.8%	100.0%
	Social	26.7%	21.9%	16.0%	8.6%	73.2%	0.9%	19.2%	6.7%	100.0%
	Other	30.8%	21.2%	17.6%	9.8%	79.3%	0.4%	15.5%	4.8%	100.0%
	Total	41.2%	19.5%	14.0%	4.6%	79.3%	1.8%	13.0%	5.9%	100.0%

County of Residence	Trip Mode									
	Tour Purpose	Drive Alone	Shared Ride 2	Shared Ride 3+	Taxi & Ride-Hailing	All Auto	Transit	Walk	Bicycle	All Modes
Napa	Work	71.1%	10.9%	3.8%	0.7%	86.6%	3.8%	2.9%	6.7%	100.0%
	University	37.3%	32.3%	5.2%	11.7%	86.6%	3.6%	1.4%	8.4%	100.0%
	School	11.1%	27.9%	39.1%	0.9%	79.0%	0.8%	9.5%	10.7%	100.0%
	At-Work	49.7%	11.2%	5.1%	0.6%	66.7%	0.3%	27.4%	5.6%	100.0%
	Eat Out	34.0%	20.7%	12.8%	4.5%	72.1%	1.2%	19.5%	7.2%	100.0%
	Escort	24.0%	31.1%	17.7%	3.0%	75.7%	0.6%	18.7%	5.0%	100.0%
	Shopping	46.4%	22.8%	13.0%	5.0%	87.1%	0.7%	9.7%	2.4%	100.0%
	Social	32.4%	23.6%	14.2%	5.8%	76.0%	0.9%	16.6%	6.4%	100.0%
	Other	36.0%	21.3%	15.9%	9.6%	82.8%	0.6%	12.0%	4.7%	100.0%
	Total	45.1%	19.5%	13.2%	4.2%	82.0%	1.6%	10.7%	5.7%	100.0%
Sonoma	Work	73.1%	10.8%	3.5%	1.0%	88.4%	3.1%	2.0%	6.6%	100.0%
	University	35.6%	24.0%	3.9%	9.4%	72.8%	8.0%	11.9%	7.3%	100.0%
	School	11.6%	29.7%	40.1%	0.8%	82.1%	0.7%	7.0%	10.2%	100.0%
	At-Work	54.4%	12.0%	5.1%	0.8%	72.3%	0.3%	21.8%	5.6%	100.0%
	Eat Out	35.6%	23.7%	11.2%	4.8%	75.4%	1.0%	16.3%	7.3%	100.0%
	Escort	25.1%	32.5%	18.6%	2.8%	79.1%	0.6%	15.1%	5.3%	100.0%
	Shopping	49.2%	23.4%	11.6%	4.7%	88.9%	0.8%	7.9%	2.4%	100.0%
	Social	34.4%	24.0%	14.1%	6.6%	79.1%	0.9%	13.2%	6.8%	100.0%
	Other	38.2%	23.2%	15.2%	7.3%	84.0%	0.6%	10.5%	4.9%	100.0%
	Total	46.8%	20.6%	12.8%	3.8%	84.0%	1.4%	8.9%	5.7%	100.0%

County of Residence	Trip Mode									
	Tour Purpose	Drive Alone	Shared Ride 2	Shared Ride 3+	Taxi & Ride-Hailing	All Auto	Transit	Walk	Bicycle	All Modes
Marin	Work	60.9%	11.5%	5.3%	0.7%	78.4%	14.1%	2.2%	5.3%	100.0%
	University	31.2%	23.0%	3.9%	6.5%	64.6%	15.9%	6.1%	13.4%	100.0%
	School	11.3%	24.1%	33.6%	0.8%	69.7%	1.4%	12.7%	16.2%	100.0%
	At-Work	40.5%	11.6%	5.3%	0.9%	58.4%	1.2%	36.6%	3.7%	100.0%
	Eat Out	30.9%	21.9%	12.2%	4.9%	70.0%	2.7%	19.2%	8.1%	100.0%
	Escort	21.9%	27.6%	17.0%	2.4%	68.9%	0.8%	24.4%	6.0%	100.0%
	Shopping	46.5%	21.7%	11.9%	4.4%	84.6%	1.1%	11.2%	3.2%	100.0%
	Social	31.2%	21.5%	14.5%	6.0%	73.2%	2.2%	16.9%	7.7%	100.0%
	Other	36.2%	22.3%	14.8%	5.7%	79.1%	1.6%	13.6%	5.6%	100.0%
	Total	41.4%	19.2%	12.5%	3.3%	76.4%	5.0%	12.5%	6.1%	100.0%
Bay Area	Total	36.7%	18.2%	12.6%	4.6%	72.1%	8.6%	12.8%	6.4%	100.0%

Table 41: Trip Mode Choice by Tour Purpose and County of Residence - Forecast Year 2050 – AlaCC Model

County of Residence	Trip Mode									
	Tour Purpose	Drive Alone	Shared Ride 2	Shared Ride 3+	Taxi & Ride-Hailing	All Auto	Transit	Walk	Bicycle	All Modes
San Francisco	Work	23.4%	6.1%	2.6%	2.8%	34.9%	44.9%	11.8%	8.5%	100%
	University	8.2%	10.9%	1.5%	4.1%	24.8%	44.1%	23.5%	7.7%	100%
	School	7.4%	15.2%	22.0%	0.8%	45.4%	6.7%	30.0%	18.0%	100%
	At-Work	16.7%	9.8%	4.1%	1.8%	32.3%	4.0%	56.6%	7.1%	100%
	Eat Out	17.9%	14.2%	4.8%	3.0%	40.0%	19.9%	29.0%	11.1%	100%
	Escort	15.2%	19.6%	12.3%	4.1%	51.2%	9.0%	32.0%	7.8%	100%
	Shopping	34.2%	14.5%	7.4%	4.2%	60.5%	14.1%	21.1%	4.3%	100%
	Social	21.3%	12.4%	5.1%	2.9%	41.8%	19.1%	29.2%	10.0%	100%
	Other	27.7%	14.3%	7.1%	4.0%	53.1%	15.7%	23.3%	7.8%	100%
	Total	22.8%	11.1%	6.2%	3.1%	43.1%	25.1%	23.4%	8.4%	100%
San Mateo	Work	59.3%	11.0%	4.3%	1.2%	75.8%	11.7%	3.7%	8.8%	100%
	University	28.1%	26.2%	3.9%	10.3%	68.6%	14.1%	3.7%	13.7%	100%
	School	11.3%	22.0%	29.9%	0.9%	64.0%	1.7%	18.9%	15.4%	100%
	At-Work	41.2%	12.3%	5.0%	1.1%	59.6%	1.0%	32.5%	7.0%	100%
	Eat Out	32.5%	17.7%	8.1%	2.8%	61.1%	3.8%	24.5%	10.6%	100%
	Escort	21.2%	27.2%	16.8%	3.1%	68.4%	1.4%	24.0%	6.2%	100%
	Shopping	50.0%	19.1%	9.6%	3.2%	81.8%	2.2%	12.1%	3.9%	100%
	Social	34.8%	16.4%	8.6%	3.0%	62.9%	3.6%	23.9%	9.6%	100%
	Other	42.3%	18.9%	10.2%	2.8%	74.3%	3.0%	15.6%	7.1%	100%
	Total	42.9%	16.9%	10.3%	2.2%	72.3%	5.4%	13.9%	8.3%	100%

County of Residence	Trip Mode									
	Tour Purpose	Drive Alone	Shared Ride 2	Shared Ride 3+	Taxi & Ride-Hailing	All Auto	Transit	Walk	Bicycle	All Modes
Santa Clara	Work	60.9%	11.4%	4.5%	1.4%	78.2%	7.8%	4.4%	9.6%	100%
	University	28.3%	23.6%	4.0%	8.1%	64.0%	15.5%	9.2%	11.3%	100%
	School	10.4%	25.2%	35.3%	1.0%	71.9%	1.3%	15.8%	11.1%	100%
	At-Work	44.2%	13.2%	5.6%	1.0%	64.0%	0.4%	27.7%	7.9%	100%
	Eat Out	36.3%	18.2%	8.6%	2.8%	65.9%	2.3%	23.1%	8.8%	100%
	Escort	22.6%	30.1%	18.2%	3.1%	73.9%	1.0%	20.2%	4.9%	100%
	Shopping	51.7%	20.1%	10.9%	3.1%	85.8%	1.7%	9.9%	2.7%	100%
	Social	37.6%	18.6%	9.2%	3.0%	68.5%	2.2%	21.1%	8.2%	100%
	Other	44.3%	20.0%	11.4%	3.6%	79.3%	1.9%	13.1%	5.6%	100%
	Total	43.7%	18.3%	11.7%	2.4%	76.2%	3.7%	12.6%	7.5%	100%
Alameda	Work	55.9%	11.0%	4.5%	1.5%	72.8%	12.2%	5.2%	9.7%	100%
	University	23.6%	21.4%	3.6%	6.7%	55.2%	18.1%	14.3%	12.3%	100%
	School	9.5%	23.0%	31.3%	0.8%	64.6%	2.1%	20.7%	12.6%	100%
	At-Work	41.1%	11.7%	5.3%	1.2%	59.3%	0.7%	32.2%	7.8%	100%
	Eat Out	32.5%	17.6%	7.8%	3.3%	61.1%	3.5%	26.2%	9.1%	100%
	Escort	21.2%	28.6%	17.1%	3.4%	70.2%	1.3%	23.2%	5.3%	100%
	Shopping	47.4%	19.1%	10.5%	4.2%	81.3%	3.4%	12.1%	3.3%	100%
	Social	32.8%	17.2%	8.4%	3.5%	61.9%	3.8%	25.7%	8.6%	100%
	Other	39.5%	19.0%	10.9%	4.5%	74.0%	3.3%	16.4%	6.3%	100%
	Total	39.7%	17.3%	11.0%	2.8%	70.8%	5.9%	15.3%	8.0%	100%

County of Residence	Trip Mode									
	Tour Purpose	Drive Alone	Shared Ride 2	Shared Ride 3+	Taxi & Ride-Hailing	All Auto	Transit	Walk	Bicycle	All Modes
Contra Costa	Work	62.6%	11.5%	4.5%	1.1%	79.7%	7.5%	4.2%	8.6%	100%
	University	33.7%	27.9%	4.9%	8.0%	74.5%	11.6%	3.2%	10.8%	100%
	School	11.2%	21.6%	29.5%	0.7%	63.1%	1.0%	20.1%	15.9%	100%
	At-Work	36.0%	8.8%	4.3%	3.3%	52.4%	1.0%	37.9%	8.6%	100%
	Eat Out	30.5%	15.6%	7.7%	5.5%	59.3%	2.9%	24.5%	13.3%	100%
	Escort	18.8%	24.6%	14.8%	5.6%	63.8%	1.2%	27.0%	8.0%	100%
	Shopping	43.0%	16.8%	8.8%	9.4%	78.1%	1.6%	14.4%	5.9%	100%
	Social	31.1%	15.8%	7.4%	6.2%	60.5%	3.2%	23.6%	12.6%	100%
	Other	35.9%	17.1%	9.2%	8.5%	70.7%	2.0%	18.1%	9.1%	100%
	Total	39.3%	16.3%	10.4%	4.9%	71.0%	3.4%	16.1%	9.5%	100%
Solano	Work	72.1%	11.7%	4.2%	0.7%	88.8%	2.4%	2.5%	6.4%	100%
	University	33.3%	41.3%	7.2%	5.9%	87.8%	5.8%	2.6%	3.8%	100%
	School	12.3%	26.1%	34.5%	0.7%	73.6%	0.4%	14.0%	12.0%	100%
	At-Work	49.5%	10.3%	4.4%	0.7%	64.9%	0.0%	29.5%	5.6%	100%
	Eat Out	38.6%	17.5%	7.6%	2.3%	66.1%	0.4%	24.3%	9.2%	100%
	Escort	23.3%	29.5%	16.9%	2.5%	72.3%	0.2%	22.0%	5.5%	100%
	Shopping	52.6%	20.3%	10.4%	2.9%	86.2%	0.3%	10.3%	3.3%	100%
	Social	39.0%	19.1%	8.9%	2.8%	69.8%	0.4%	21.0%	8.8%	100%
	Other	42.9%	21.0%	10.8%	3.4%	78.1%	0.3%	15.3%	6.4%	100%
	Total	47.4%	19.0%	11.7%	1.9%	80.0%	1.0%	12.3%	6.7%	100%

County of Residence	Trip Mode									
	Tour Purpose	Drive Alone	Shared Ride 2	Shared Ride 3+	Taxi & Ride-Hailing	All Auto	Transit	Walk	Bicycle	All Modes
Napa	Work	72.7%	10.6%	3.4%	0.7%	87.5%	1.5%	4.0%	7.0%	100%
	University	38.5%	40.5%	7.6%	6.5%	93.1%	0.6%	2.4%	3.9%	100%
	School	12.6%	25.0%	33.2%	1.1%	71.9%	0.5%	15.1%	12.4%	100%
	At-Work	52.2%	11.0%	5.3%	0.8%	69.3%	0.0%	25.6%	5.1%	100%
	Eat Out	35.8%	17.1%	6.1%	1.9%	60.9%	0.7%	29.2%	9.2%	100%
	Escort	22.3%	26.2%	16.6%	2.8%	67.9%	0.3%	26.4%	5.4%	100%
	Shopping	51.6%	20.0%	9.0%	2.8%	83.4%	0.3%	13.0%	3.2%	100%
	Social	37.7%	19.6%	7.3%	2.6%	67.1%	0.3%	24.6%	7.9%	100%
	Other	43.5%	19.2%	9.0%	2.8%	74.5%	0.3%	19.2%	6.1%	100%
	Total	49.6%	17.2%	9.8%	1.8%	78.4%	0.7%	14.1%	6.7%	100%
Sonoma	Work	70.5%	11.7%	3.6%	0.9%	86.8%	2.4%	3.4%	7.4%	100%
	University	33.8%	24.9%	3.6%	10.4%	72.8%	6.5%	10.3%	10.5%	100%
	School	11.8%	24.9%	34.8%	0.8%	72.4%	0.4%	14.7%	12.4%	100%
	At-Work	48.1%	10.5%	4.7%	0.6%	63.8%	0.1%	28.9%	7.2%	100%
	Eat Out	37.2%	17.8%	7.4%	2.0%	64.4%	0.6%	26.5%	8.5%	100%
	Escort	23.4%	29.3%	17.0%	2.6%	72.3%	0.4%	22.3%	5.0%	100%
	Shopping	53.9%	20.2%	9.2%	2.3%	85.6%	0.5%	11.2%	2.7%	100%
	Social	38.3%	18.0%	8.2%	2.6%	67.1%	0.5%	24.0%	8.4%	100%
	Other	45.2%	20.3%	9.1%	2.8%	77.4%	0.5%	16.3%	5.8%	100%
	Total	48.2%	18.2%	10.4%	1.9%	78.7%	1.1%	13.3%	6.8%	100%

County of Residence	Trip Mode									
	Tour Purpose	Drive Alone	Shared Ride 2	Shared Ride 3+	Taxi & Ride-Hailing	All Auto	Transit	Walk	Bicycle	All Modes
Marin	Work	62.1%	13.4%	5.0%	0.7%	81.2%	8.0%	3.3%	7.5%	100%
	University	28.8%	27.9%	4.9%	5.6%	67.1%	17.4%	5.4%	10.1%	100%
	School	13.0%	21.0%	27.9%	0.6%	62.5%	1.2%	19.0%	17.4%	100%
	At-Work	42.6%	10.7%	5.2%	0.8%	59.4%	0.1%	33.1%	7.4%	100%
	Eat Out	33.0%	17.4%	6.7%	1.7%	58.9%	1.3%	29.4%	10.4%	100%
	Escort	20.5%	26.1%	15.3%	2.5%	64.3%	0.7%	28.6%	6.4%	100%
	Shopping	49.4%	19.3%	9.7%	2.4%	80.8%	1.0%	14.4%	3.8%	100%
	Social	32.1%	18.0%	7.5%	2.7%	60.2%	1.3%	28.6%	9.9%	100%
	Other	42.0%	19.1%	9.0%	2.2%	72.3%	0.9%	19.0%	7.9%	100%
	Total	43.3%	17.8%	10.1%	1.6%	72.8%	3.2%	15.8%	8.2%	100%
Bay Area	Total	40.0%	16.7%	10.3%	2.8%	69.8%	6.9%	15.2%	8.1%	100%
San Joaquin	Work	73.1%	10.3%	3.3%	0.8%	87.4%	0.7%	3.0%	8.9%	100%
	University	40.9%	27.8%	4.3%	6.0%	79.1%	6.6%	4.3%	10.0%	100%
	School	9.2%	22.8%	30.0%	0.6%	62.6%	0.4%	17.1%	20.0%	100%
	At-Work	50.7%	10.0%	4.4%	0.5%	65.6%	0.0%	27.2%	7.1%	100%
	Eat Out	31.1%	18.7%	7.7%	2.4%	59.9%	0.1%	28.0%	12.0%	100%
	Escort	20.2%	27.0%	15.4%	2.5%	65.2%	0.1%	26.6%	8.1%	100%
	Shopping	48.1%	20.0%	11.3%	3.3%	82.7%	0.3%	12.6%	4.4%	100%
	Social	32.8%	16.9%	8.9%	3.5%	62.2%	0.1%	26.0%	11.8%	100%
	Other	37.3%	18.8%	12.3%	4.4%	72.8%	0.1%	19.0%	8.0%	100%
	Total	41.1%	18.2%	12.4%	2.3%	74.1%	0.4%	15.6%	9.9%	100%
AlaCC Model Area	Total	40.0%	16.9%	10.5%	2.8%	70.1%	6.4%	15.4%	8.1%	100%

ITEM J: FORECAST YEAR(S) TRAFFIC AND TRANSIT ASSIGNMENT

This section presents a comparison of the 2050 Forecast Year traffic and transit assignment estimates from AlaCC Model versus the MTC PBA estimates.

Product 15: Region-level and by county, time-period-specific comparison of vehicle miles traveled (VMT) and vehicle hours traveled (VHT) estimates by facility type to MTC's estimates for the horizon year.

Product 16: Region-level, time-period-specific comparison of estimated average speed on freeways and all other facilities, separately, to MTC's estimates for the horizon year.

Region level time-period specific listings of vehicle-miles of travel (VMT), vehicle hours of travel (VHT) and average speeds by facility type for the 2050 forecast year are shown for PBA 2050 (Table 42), the AlaCC model (Table 43) and a comparison between the AlaCC model and PBA 2050 (Table 43). Many of the differences are related to the adjustments made in the AlaCC model as described in the Model Overview section.

The AlaCC model estimates total VMT 18 percent higher than PBA 2050. The AlaCC model estimates similar amounts of total freeway VMT (86.2 million versus 75.7 million, about 16 percent difference). The AlaCC model estimates VMT on arterial roads, collector roads and local roads approximately 20 percent higher than PBA 2050. In addition, the AlaCC model shows a proportionally higher share of VMT on collectors and local roads, compared to PBA 2050. This is most likely due to the higher density of local street network included in the AlaCC model network; the PBA 2050 model network does not include most local streets.

As noted in the Model Overview, the AlaCC model provides good estimates of VMT for the 2015 base year validation, as compared to HPMS, while TM1.5.2 estimates approximately eight percent lower VMT than HPMS. In addition, as shown in the Model Development Report, the AlaCC model estimates VMT within four percent of the HMPS estimate for 2019. This indicates that the higher VMT estimated by the AlaCC model for 2050 is partly due to base year calibration differences with respect to TM1.5.2. A second contributing factor is the differences in transit projects; as noted in the Model Overview, only a subset of the PBA transit projects is represented in the AlaCC 2050 scenario.

The AlaCC model estimates 57 percent higher VHT than PBA 2050. The estimates of VHT for total freeways and major arterials are similar for the two models though higher in the AlaCC model, reflecting the increase in vehicular demand (cars and trucks) described above. The AlaCC model estimates significantly higher VHT on collector roads and other (local) streets, accounting for most of the difference in total VHT. As with the VMT estimates, the higher VHT on the local street network is due to the higher density of representation of local streets in the AlaCC model, providing more opportunities to assign traffic and report congestion on local streets.

Correspondingly, average speeds are similar between the AlaCC model and PBA 2050 on freeways, expressways and arterials, but significantly lower on collectors and local roads.

Table 42: VMT, VHT and Average Speed by Facility Type and Time Period – Forecast Year 2050 – PBA 2050

		Facility Type					
Category	Time Period	Freeways	Expressways	Major Arterials	Collectors	Others	All Facilities
Vehicle Miles Travelled (VMT)	Early AM (3-6 AM)	4,215,000	798,000	1,879,000	526,000	600,000	8,018,000
	AM Peak (6-10 AM)	19,866,000	3,839,000	13,582,000	4,223,000	5,017,000	46,527,000
	Midday (10 AM-3 PM)	18,032,000	4,043,000	15,390,000	4,430,000	6,336,000	48,231,000
	PM Peak (3-7 PM)	19,937,000	4,209,000	15,756,000	4,929,000	6,253,000	51,084,000
	Evening (7 PM-3 AM)	13,703,000	2,860,000	9,913,000	2,758,000	3,650,000	32,884,000
	All Time Periods	75,753,000	15,749,000	56,520,000	16,866,000	21,856,000	186,744,000
Vehicle Hours Travelled (VHT)	Early AM (3-6 AM)	77,000	16,000	63,000	20,000	30,000	206,000
	AM Peak (6-10 AM)	424,000	95,000	549,000	207,000	268,000	1,543,000
	Midday (10 AM-3 PM)	336,000	91,000	588,000	202,000	326,000	1,543,000
	PM Peak (3-7 PM)	410,000	109,000	664,000	254,000	328,000	1,765,000
	Evening (7 PM-3 AM)	250,000	60,000	335,000	111,000	184,000	940,000
	All Time Periods	1,497,000	371,000	2,199,000	794,000	1,136,000	5,997,000
Average Speed (miles per hour)	Early AM (3-6 AM)	54.7	48.4	30.0	25.7	19.8	38.8
	AM Peak (6-10 AM)	46.9	40.4	24.7	20.4	18.7	30.1
	Midday (10 AM-3 PM)	53.7	44.3	26.2	22.0	19.4	31.3
	PM Peak (3-7 PM)	48.6	38.6	23.7	19.4	19.1	28.9
	Evening (7 PM-3 AM)	54.8	47.6	29.5	24.8	19.8	34.9
	All Time Periods	50.6	42.4	25.7	21.2	19.2	31.1

Table 43: VMT, VHT and Average Speed by Facility Type and Time Period – Forecast Year 2050 – AlaCC Model

		Facility Type					
Category	Time Period	Freeways	Expressways	Major Arterials	Collectors	Others	All Facilities
Vehicle Miles Travelled (VMT)	Early AM (3-6 AM)	4,969,866	1,318,288	2,387,133	1,722,684	1,369,455	11,767,426
	AM Peak (6-10 AM)	22,417,549	4,456,923	11,197,822	7,416,978	8,479,683	53,968,955
	Midday (10 AM-3 PM)	20,064,656	5,234,949	13,306,981	8,126,551	10,035,012	56,768,150
	PM Peak (3-7 PM)	22,206,634	4,895,265	13,424,478	8,578,483	10,661,442	59,766,302
	Evening (7 PM-3 AM)	16,556,324	3,476,712	7,734,334	4,944,135	5,073,292	37,784,798
	All Time Periods	86,215,029	19,382,137	48,050,749	30,788,831	35,618,885	220,055,630
Vehicle Hours Travelled (VHT)	Early AM (3-6 AM)	94,794	27,108	94,095	95,878	69,322	381,197
	AM Peak (6-10 AM)	539,290	125,932	551,685	589,614	499,676	2,306,197
	Midday (10 AM-3 PM)	414,408	126,037	626,425	661,920	699,374	2,528,164
	PM Peak (3-7 PM)	529,118	152,104	719,410	883,937	670,011	2,954,579
	Evening (7 PM-3 AM)	318,496	72,574	308,403	197,474	328,355	1,225,302
	All Time Periods	1,896,107	503,755	2,300,017	2,428,823	2,266,737	9,395,439
Average Speed (miles per hour)	Early AM (3-6 AM)	52.4	48.6	25.4	18.0	19.8	30.9
	AM Peak (6-10 AM)	41.6	35.4	20.3	12.6	17.0	23.4
	Midday (10 AM-3 PM)	48.4	41.5	21.2	12.3	14.3	22.5
	PM Peak (3-7 PM)	42.0	32.2	18.7	9.7	15.9	20.2
	Evening (7 PM-3 AM)	52.0	47.9	25.1	25.0	15.5	30.8
	All Time Periods	45.5	38.5	20.9	12.7	15.7	23.4

Table 44: VMT, VHT and Average Speed by Facility Type and Time Period – Forecast Year 2050 – AlaCC Model Compared to PBA 2050

		Facility Type					
Category	Time Period	Freeways	Expressways	Major Arterials	Collectors	Others	All Facilities
Vehicle Miles Travelled (VMT)	Early AM (3-6 AM)	18%	65%	27%	228%	128%	47%
	AM Peak (6-10 AM)	13%	16%	-18%	76%	69%	16%
	Midday (10 AM-3 PM)	11%	29%	-14%	83%	58%	18%
	PM Peak (3-7 PM)	11%	16%	-15%	74%	71%	17%
	Evening (7 PM-3 AM)	21%	22%	-22%	79%	39%	15%
	All Time Periods	14%	23%	-15%	83%	63%	18%
Vehicle Hours Travelled (VHT)	Early AM (3-6 AM)	23%	69%	49%	379%	131%	85%
	AM Peak (6-10 AM)	27%	33%	0%	185%	86%	49%
	Midday (10 AM-3 PM)	23%	39%	7%	228%	115%	64%
	PM Peak (3-7 PM)	29%	40%	8%	248%	104%	67%
	Evening (7 PM-3 AM)	27%	21%	-8%	78%	78%	30%
	All Time Periods	27%	36%	5%	206%	100%	57%
Average Speed (miles per hour)	Early AM (3-6 AM)	-4%	0%	-16%	-30%	0%	-20%
	AM Peak (6-10 AM)	-11%	-12%	-18%	-38%	-9%	-22%
	Midday (10 AM-3 PM)	-10%	-6%	-19%	-44%	-26%	-28%
	PM Peak (3-7 PM)	-14%	-17%	-21%	-50%	-17%	-30%
	Evening (7 PM-3 AM)	-5%	1%	-15%	1%	-22%	-12%
	All Time Periods	-10%	-9%	-19%	-40%	-18%	-25%

Product 17: Region-level, time-period-specific comparison of estimated volumes on bridges and county-lines, separately, to MTC's estimates for the horizon year.

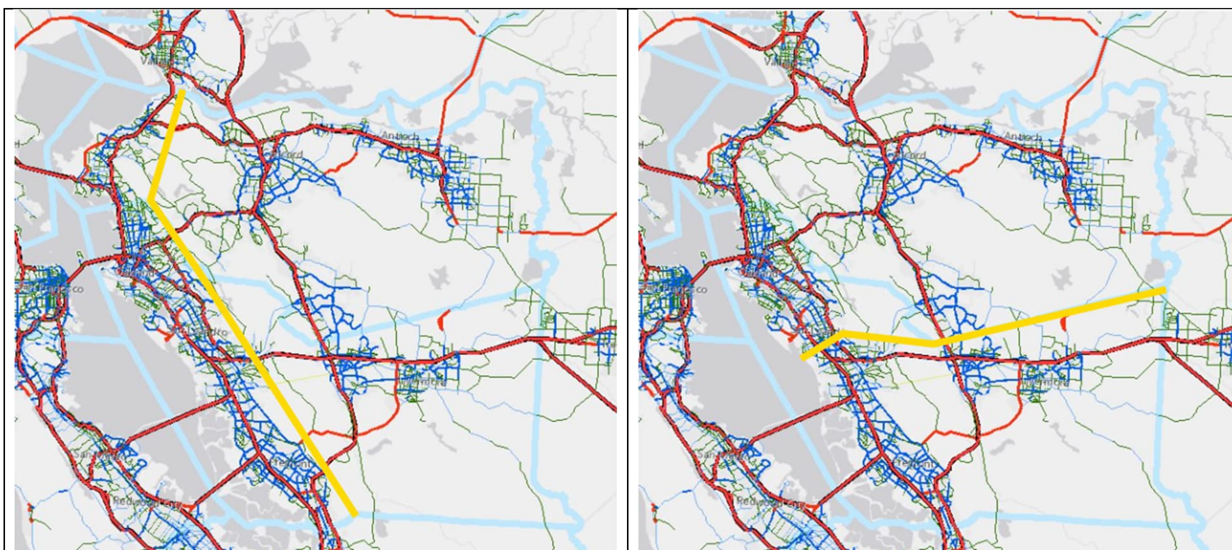
Bridge and screenline traffic volumes are reported for the 2050 forecast year for PBA 2050 (Table 45), the AlaCC model (Table 46) and a comparison between the AlaCC model and PBA 2050 (Table 47). Traffic is reported at seven screenline locations, listed below.

- San Francisco Bay Bridges Screenline
- Contra Costa – Solano County Line
- Contra Costa – Alameda County Line
- Alameda – Santa Clara County Line
- San Joaquin County Line
- Alameda / Contra Costa East-West Screenline
- Alameda / Contra Costa North-South Screenline

Exhibit 11 shows the location of the East-West and North-South screenlines. In these tables, freeway traffic is not reported by lane type (i.e., managed vs non-managed). Traffic on roadways not represented in TM 1.5.2 is reported as Not Available (N/A). The San Joaquin County screenline location in AlaCC follows the county line whereas in TM 1.5.2 it represents locations west of the county line.

The AlaCC model estimates higher volumes on the Bay Bridge compared to PBA 2050, particularly during off-peak periods. These differences are related to the adjustments made in the AlaCC model as described above. It is worth noting that TM1.5.2 reports a lower 2050 traffic volume in the PBA 2050 scenario than in 2015, whereas AlaCC reports increasing volume approximately commensurate with the increase in population. In general, AlaCC reports higher volumes than TM 1.5.2 in the east-west direction and lower traffic in the north-south direction.

Exhibit 11: Alameda / Contra Costa Screenlines



East-West Screenline | North-South Screenline

Table 45: Forecasted Bridge and Screenline Volumes by Time Period – Forecast Year 2050 – PBA 2050

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
San Francisco Bay Screenline		26,999	162,865	123,097	141,938	121,550	576,449
I 580 W	Richmond-San Rafael Bridge (CC to Marin)	1,629	10,002	12,197	15,556	12,199	51,583
I 580 E	Richmond-San Rafael Bridge (Marin to CC)	2,894	18,992	13,095	11,975	7,493	54,449
I 80 W	Bay Bridge (Alameda to San Francisco)	6,822	36,889	20,018	22,597	21,951	108,277
I 80 E	Bay Bridge (San Francisco to Alameda)	5,054	30,523	20,791	28,127	33,787	118,282
SR 92 W	San Mateo Bridge (Alameda to San Mateo)	3,527	19,265	14,706	15,734	11,385	64,617
SR 92 E	San Mateo Bridge (San Mateo to Alameda)	2,002	13,677	13,027	15,443	11,223	55,372
SR 84 W	Dumbarton Bridge (Alameda to San Mateo)	2,577	17,652	12,866	9,853	6,974	49,922
SR 84 E	Dumbarton Bridge (San Mateo to Alameda)	2,494	15,865	16,397	22,653	16,538	73,947
Contra Costa – Solano County Line		17,981	73,441	66,149	71,608	49,841	279,020
I 80 W	Carquinez Bridge (Contra Costa to Solano)	2,877	13,921	14,365	18,545	13,855	63,563
I 80 E	Carquinez Bridge (Solano to Contra Costa)	5,031	22,177	14,982	14,448	9,861	66,499
I 680 W	Benicia Bridge (Contra Costa to Solano)	3,612	16,994	16,482	19,159	12,294	68,541
I 680 E	Benicia Bridge (Solano to Contra Costa)	4,448	14,831	15,014	13,650	10,190	58,133
SR 160 W	Antioch Bridge (Contra Costa to Solano)	1,149	3,142	2,711	2,755	1,536	11,293
SR 160 E	Antioch Bridge (Solano to Contra Costa)	864	2,376	2,595	3,051	2,105	10,991

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Contra Costa – Alameda County Line		15,221	90,570	96,244	94,892	70,030	366,957
I 580 W	Interstate 580 W	1,965	12,996	16,731	16,681	13,397	61,770
I 580 E	Interstate 580 E	2,919	18,218	18,256	16,434	9,596	65,423
I 80 W	Interstate 80 W	5,150	27,448	20,573	17,681	11,482	82,334
I 80 E	Interstate 80 E	2,860	14,674	20,490	24,816	20,712	83,552
N	San Pablo Avenue NB	366	2,694	4,135	3,927	3,809	14,931
S	San Pablo Avenue SB	734	4,138	4,580	3,937	2,772	16,161
N	Ashbury Avenue NB	13	254	376	513	144	1,300
S	Ashbury Avenue SB	11	505	482	468	145	1,611
N	Colusa Ave NB	33	214	357	943	284	1,831
S	Colusa Ave SB	38	1,156	1,350	707	321	3,572
N	Arlington Ave NB	338	2,153	3,231	3,129	3,570	12,421
S	Arlington Ave SB	547	3,380	3,577	3,028	2,310	12,842
N	Grizzly Peak Blvd NB	131	1,155	1,264	1,479	690	4,719
S	Grizzly Peak Blvd SB	116	1,585	842	1,149	798	4,490
N	All other roads	193	1,878	2,396	2,707	1,523	8,697
S	All other roads	210	2,714	2,035	2,209	1,335	8,503

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Alameda – Santa Clara County Line		29,000	148,855	115,916	138,956	96,190	528,917
N	Fremont Blvd NB	1,047	6,595	4,275	4,595	3,342	19,854
S	Fremont Blvd SB	879	4,985	4,182	6,545	4,008	20,599
I 880 N	Interstate 880 N	7,862	34,493	31,533	33,991	31,886	139,765
I 880 S	Interstate 880 N	11,091	39,491	36,306	34,207	28,911	150,006
N	Milmont Dr NB	905	3,145	3,613	3,375	3,680	14,718
S	Milmont Dr SB	1,135	2,735	3,669	3,300	3,216	14,055
N	Warm Springs Blvd NB	1,232	8,456	8,518	8,418	6,571	33,195
S	Warm Springs Blvd SB	1,464	8,653	8,991	8,546	5,675	33,329
I 680 NB	Interstate 680 NB	1,364	11,071	5,483	13,261	4,195	35,374
I 680 SB	Interstate 680 SB	1,947	20,818	6,748	15,126	4,103	48,742
N	N Park Victoria Dr NB	23	3,227	981	3,613	358	8,202
S	N Park Victoria Dr SB	51	5,186	1,617	3,979	245	11,078
N	Calaveras Rd NB	N/A	N/A	N/A	N/A	N/A	N/A
S	Calaveras Rd SB	N/A	N/A	N/A	N/A	N/A	N/A
N	Mines Rd NB	N/A	N/A	N/A	N/A	N/A	N/A
S	Mines Rd SB	N/A	N/A	N/A	N/A	N/A	N/A

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
San Joaquin County Line		20,898	44,263	50,940	50,095	28,490	194,686
SR 4 W	State Highway Route 4 W	850	1,509	1,535	1,331	710	5,935
SR 4 E	State Highway Route 4 E	442	1,101	1,593	1,715	1,089	5,940
W	Byron Highway WB	900	1,594	1,616	1,398	743	6,251
E	Byron Highway EB	461	1,155	1,676	1,809	1,149	6,250
W	W Grant Line Rd WB	N/A	N/A	N/A	N/A	N/A	N/A
E	W Grant Line Rd EB	N/A	N/A	N/A	N/A	N/A	N/A
I 205 W	Interstate 205 WB [*]	N/A	N/A	N/A	N/A	N/A	N/A
I 205 E	Interstate 205 EB [*]	N/A	N/A	N/A	N/A	N/A	N/A
I 580 W	Interstate 580 WB	12,300	23,526	21,766	18,611	9,419	85,622
I 580 E	Interstate 580 EB	5,945	15,378	22,754	25,231	15,380	84,688
W	Patterson Pass Rd WB	N/A	N/A	N/A	N/A	N/A	N/A
E	Patterson Pass Rd EB	N/A	N/A	N/A	N/A	N/A	N/A
W	Tesla Rd WB	N/A	N/A	N/A	N/A	N/A	N/A
E	Tesla Rd EB	N/A	N/A	N/A	N/A	N/A	N/A
[*]	The external station node is located west of the junction with I-580						

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Alameda / Contra Costa East-West Screenline		44,676	188,110	158,602	192,958	145,001	729,347
W	Cummings Skyway WB	N/A	N/A	N/A	N/A	N/A	N/A
E	Cummings Skyway EB	N/A	N/A	N/A	N/A	N/A	N/A
SR 4 WB	State Highway Route 4 WB	1,092	8,330	5,426	6,661	4,271	25,780
SR 4 EB	State Highway Route 4 EB	819	6,824	5,257	7,492	4,246	24,638
W	Alhambra Valley Rd WB	58	702	663	745	411	2,579
E	Alhambra Valley Rd EB	69	636	599	802	542	2,648
W	San Pablo Dam Rd WB	272	4,075	2,292	2,667	1,204	10,510
E	San Pablo Dam Rd EB	165	1,660	2,106	3,814	2,094	9,839
SR 24 W	State Highway Route 24 WB	2,771	21,326	10,396	9,003	5,463	48,959
SR 24 E	State Highway Route 24 EB	1,526	9,560	11,888	18,156	16,800	57,930
W	Canyon Rd WB	1,503	4,939	4,543	3,870	3,589	18,444
E	Canyon Rd EB	661	3,712	4,242	4,378	6,414	19,407
W	Crow Canyon Rd WB	1,105	3,785	4,522	4,567	6,661	20,640
E	Crow Canyon Rd EB	1,105	3,785	4,522	4,567	6,661	20,640
W	Jensen Rd WB	N/A	N/A	N/A	N/A	N/A	N/A
E	Jensen Rd EB	N/A	N/A	N/A	N/A	N/A	N/A
W	E Castro Valley Blvd WB	371	4,194	4,691	4,311	2,651	16,218
E	E Castro Valley Blvd EB	305	2,930	4,369	5,004	2,936	15,544
I 580 W	Interstate 580 WB	4,274	21,313	10,683	12,064	6,203	54,537
I 580 E	Interstate 580 EB	1,981	11,992	11,366	20,687	10,588	56,614

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
W	Dublin Canyon Rd WB	N/A	N/A	N/A	N/A	N/A	N/A
E	Dublin Canyon Rd EB	N/A	N/A	N/A	N/A	N/A	N/A
SR 84 W	State Highway Route 84 WB	5,221	9,197	9,384	7,992	7,723	39,517
SR 84 E	State Highway Route 84 EB	3,522	11,538	12,909	15,144	14,327	57,440
I 680 W	Interstate 680 WB	11,837	33,623	24,153	24,558	15,858	110,029
I 680 E	Interstate 680 EB	6,019	23,989	24,591	36,476	26,359	117,434

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Alameda / Contra Costa North-South Screenline		39,524	222,277	224,556	243,137	167,300	896,794
N	Wicks Blvd NB	95	1,022	2,699	3,308	906	8,030
S	Wicks Blvd SB	110	2,076	2,218	2,800	768	7,972
I 880 N	Interstate 880 NB	4,591	26,884	23,226	24,067	13,125	91,893
I 880 S	Interstate 880 SB	3,456	21,999	21,609	25,999	18,817	91,880
N	Washington Ave NB	995	8,093	9,417	7,943	5,076	31,524
S	Washington Ave SB	404	6,295	8,545	8,089	4,989	28,322
N	Hesperian Blvd NB	948	8,734	11,762	11,598	7,699	40,741
S	Hesperian Blvd SB	1,311	11,167	13,102	11,197	8,472	45,249
N	E 14th St NB	1,137	4,077	5,024	3,955	5,805	19,998
S	E 14th St SB	733	3,814	4,898	4,048	5,446	18,939
I 580 W	Interstate 580 WB	1,539	7,993	7,095	9,487	4,336	30,450
I 580 E	Interstate 580 EB	1,656	10,626	9,918	12,354	9,482	44,036
N	Redwood Rd NB	175	1,175	940	1,115	1,452	4,857
S	Redwood Rd SB	365	1,380	916	1,179	869	4,709
N	Crow Cayon Rd NB	1,097	3,772	4,241	4,482	5,839	19,431
S	Crow Cayon Rd SB	1,845	4,766	4,179	3,892	4,540	19,222
N	San Ramon Rd NB	658	5,560	7,139	9,228	4,900	27,485
S	San Ramon Rd SB	519	7,767	4,729	5,620	2,105	20,740
I 680 W	Interstate 680 WB	3,781	22,134	21,227	29,012	17,695	93,849
I 680 E	Interstate 680 EB	4,806	26,348	21,105	22,794	12,650	87,703

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
N	Village Pkwy NB	N/A	N/A	N/A	N/A	N/A	N/A
S	Village Pkwy SB	N/A	N/A	N/A	N/A	N/A	N/A
N	Dougherty Rd NB	304	1,238	1,348	944	640	4,474
S	Dougherty Rd SB	135	895	1,725	2,123	1,822	6,700
N	Tassajara Rd NB	483	4,259	7,856	8,366	6,175	27,139
S	Tassajara Rd SB	1,155	8,181	8,329	6,927	3,635	28,227
N	Collier Canyon Rd NB	-	-	3	20	1	24
S	Collier Canyon Rd SB	2	69	-	2	-	73
N	N Livermore Ave NB	140	1,147	1,218	1,760	1,096	5,361
S	N Livermore Ave SB	178	1,789	1,335	1,401	773	5,476
N	Vasco Rd NB	2,295	8,411	9,383	10,644	11,903	42,636
S	Vasco Rd SB	4,611	10,606	9,370	8,783	6,284	39,654

Table 46: Forecasted Bridge and Screenline Volumes by Time Period – Forecast Year 2050 – AlaCC Model

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
San Francisco Bay Screenline		38,609	188,698	149,851	174,543	132,316	684,017
I 580 W	Richmond-San Rafael Bridge (CC to Marin)	2,164	14,888	9,332	17,963	11,614	55,961
I 580 E	Richmond-San Rafael Bridge (Marin to CC)	4,254	23,274	13,681	15,938	9,355	66,501
I 80 W	Bay Bridge (Alameda to San Francisco)	12,623	43,623	37,155	30,309	29,152	152,861
I 80 E	Bay Bridge (San Francisco to Alameda)	7,248	36,664	34,647	41,045	41,296	160,899
SR 92 W	San Mateo Bridge (Alameda to San Mateo)	3,572	20,369	12,658	15,644	7,864	60,106
SR 92 E	San Mateo Bridge (San Mateo to Alameda)	2,422	12,088	11,919	16,535	10,232	53,196
SR 84 W	Dumbarton Bridge (Alameda to San Mateo)	3,241	22,133	12,884	13,207	6,128	57,593
SR 84 E	Dumbarton Bridge (San Mateo to Alameda)	3,085	15,660	17,575	23,903	16,676	76,899
Contra Costa – Solano County Line		24,514	82,912	73,963	82,583	55,719	319,691
I 80 W	Carquinez Bridge (Contra Costa to Solano)	4,105	17,438	7,192	9,046	7,605	45,387
I 80 E	Carquinez Bridge (Solano to Contra Costa)	1,934	10,488	6,780	13,567	11,778	44,547
I 680 W	Benicia Bridge (Contra Costa to Solano)	5,922	20,400	22,035	22,762	17,430	88,549
I 680 E	Benicia Bridge (Solano to Contra Costa)	8,301	23,945	23,460	23,473	12,645	91,825
SR 160 W	Antioch Bridge (Contra Costa to Solano)	3,068	6,600	7,479	6,698	2,961	26,806
SR 160 E	Antioch Bridge (Solano to Contra Costa)	1,183	4,041	7,016	7,036	3,300	22,577

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Contra Costa – Alameda County Line		19,402	88,193	91,544	91,968	65,899	357,007
I 580 W	Interstate 580 W	3,731	17,825	19,161	19,188	14,811	74,716
I 580 E	Interstate 580 E	5,248	16,798	18,558	16,400	12,769	69,772
I 80 W	Interstate 80 W	5,687	25,462	19,735	16,157	10,672	77,713
I 80 E	Interstate 80 E	2,723	11,689	16,160	20,082	19,812	70,466
N	San Pablo Avenue NB	412	3,119	4,214	5,488	1,962	15,196
S	San Pablo Avenue SB	537	4,848	4,710	4,273	1,948	16,316
N	Ashbury Avenue NB	48	367	527	477	258	1,677
S	Ashbury Avenue SB	42	553	610	506	169	1,880
N	Colusa Ave NB	255	1,321	2,007	2,234	1,152	6,968
S	Colusa Ave SB	318	1,960	2,102	1,936	717	7,033
N	Arlington Ave NB	122	933	1,264	1,852	799	4,970
S	Arlington Ave SB	188	1,787	1,516	1,456	422	5,368
N	Grizzly Peak Blvd NB	42	1,215	497	903	139	2,795
S	Grizzly Peak Blvd SB	50	318	483	1,018	270	2,138
N	All other roads	456	2,805	3,933	4,872	2,082	14,148
S	All other roads	324	3,185	3,228	2,633	980	10,349

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Alameda – Santa Clara County Line		33,273	129,395	109,261	121,467	97,615	491,012
N	Fremont Blvd NB	1,504	2,928	3,497	2,707	3,271	13,906
S	Fremont Blvd SB	1,076	2,046	3,365	2,860	2,808	12,155
I 880 N	Interstate 880 N	6,830	30,320	25,027	27,740	31,811	121,727
I 880 S	Interstate 880 N	11,105	33,193	29,214	27,976	27,504	128,993
N	Milmont Dr NB	16	1,802	1,444	1,073	12	4,347
S	Milmont Dr SB	49	774	1,400	1,426	31	3,680
N	Warm Springs Blvd NB	2,651	7,959	9,103	8,041	7,868	35,623
S	Warm Springs Blvd SB	3,928	7,738	9,231	7,911	7,653	36,461
I 680 NB	Interstate 680 NB	1,275	12,531	5,870	11,396	5,635	36,707
I 680 SB	Interstate 680 SB	1,821	17,880	8,199	16,675	6,217	50,791
N	N Park Victoria Dr NB	775	2,522	2,422	3,329	1,783	10,830
S	N Park Victoria Dr SB	1,146	2,857	3,779	3,141	2,605	13,529
N	Calaveras Rd NB	413	2,855	3,494	3,636	211	10,608
S	Calaveras Rd SB	668	3,786	3,179	3,458	130	11,221
N	Mines Rd NB	3	32	20	79	51	185
S	Mines Rd SB	14	172	17	20	26	249

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
San Joaquin County Line		47,370	85,626	70,963	71,784	96,132	371,875
SR 4 W	State Highway Route 4 W	2,888	4,169	4,898	3,948	3,850	19,753
SR 4 E	State Highway Route 4 E	2,608	3,836	5,158	3,773	6,087	21,463
W	Byron Highway WB	2,310	3,170	3,704	2,555	2,191	13,930
E	Byron Highway EB	1,771	2,901	3,348	2,708	3,623	14,352
W	W Grant Line Rd WB	125	519	844	879	392	2,759
E	W Grant Line Rd EB	56	349	1,062	1,239	506	3,211
I 205 W	Interstate 205 WB	10,812	18,964	12,880	12,436	14,049	69,141
I 205 E	Interstate 205 EB	7,161	13,206	13,518	15,251	23,894	73,030
I 580 W	Interstate 580 WB	8,808	15,456	3,155	5,229	8,118	40,766
I 580 E	Interstate 580 EB	3,800	9,137	3,390	7,543	16,014	39,883
W	Patterson Pass Rd WB	2,996	4,463	5,544	4,603	4,578	22,184
E	Patterson Pass Rd EB	2,762	3,830	6,045	5,283	6,899	24,819
W	Tesla Rd WB	434	1,971	3,391	3,043	3,843	12,683
E	Tesla Rd EB	840	3,655	4,025	3,294	2,086	13,901

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Alameda/Contra Costa East-West Screenline		69,777	206,108	192,635	204,993	185,772	859,285
W	Cummings Skyway WB	35	854	356	810	100	2,154
E	Cummings Skyway EB	29	311	260	360	165	1,125
SR 4 WB	State Highway Route 4 WB	3,425	13,345	10,235	11,254	7,061	45,320
SR 4 EB	State Highway Route 4 EB	2,187	9,478	10,148	12,058	7,317	41,187
W	Alhambra Valley Rd WB	60	985	448	651	341	2,484
E	Alhambra Valley Rd EB	74	654	469	1,029	344	2,570
W	San Pablo Dam Rd WB	998	5,427	4,982	4,625	2,599	18,630
E	San Pablo Dam Rd EB	615	3,613	4,817	5,615	4,097	18,758
SR 24 W	State Highway Route 24 WB	4,945	21,509	13,677	10,463	8,556	59,150
SR 24 E	State Highway Route 24 EB	2,062	9,177	13,688	19,289	19,111	63,327
W	Canyon Rd WB	1,855	3,616	4,637	3,688	3,117	16,913
E	Canyon Rd EB	1,254	3,413	4,300	3,556	4,672	17,194
W	Crow Canyon Rd WB	2,139	3,859	4,883	3,926	5,401	20,208
E	Crow Canyon Rd EB	1,263	3,988	4,976	4,345	5,920	20,493
W	Jensen Rd WB	-	-	-	-	-	-
E	Jensen Rd EB	-	-	-	-	-	-
W	E Castro Valley Blvd WB	1,248	5,159	4,470	4,776	2,912	18,566
E	E Castro Valley Blvd EB	1,326	4,233	5,932	6,219	5,909	23,619
I 580 W	Interstate 580 WB	6,219	27,022	14,710	15,016	12,252	75,218
I 580 E	Interstate 580 EB	2,757	14,365	15,876	23,168	20,167	76,333

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
W	Dublin Canyon Rd WB	2,314	4,560	5,050	4,211	4,309	20,443
E	Dublin Canyon Rd EB	1,878	3,910	4,901	4,051	6,312	21,053
SR 84 W	State Highway Route 84 WB	9,383	13,462	13,769	12,808	9,094	58,516
SR 84 E	State Highway Route 84 EB	5,161	9,139	11,634	10,677	16,418	53,028
I 680 W	Interstate 680 WB	12,371	27,259	19,130	19,274	14,799	92,833
I 680 E	Interstate 680 EB	6,178	16,772	19,288	23,126	24,798	90,161

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Alameda/Contra Costa North-South Screenline		43,063	206,390	209,816	221,447	147,480	828,197
N	Wicks Blvd NB	114	1,543	2,682	3,554	425	8,318
S	Wicks Blvd SB	135	1,250	1,803	1,367	284	4,838
I 880 N	Interstate 880 NB	6,193	30,695	24,138	26,740	17,114	104,879
I 880 S	Interstate 880 SB	4,566	24,108	22,057	27,241	19,295	97,265
N	Washington Ave NB	750	4,134	5,071	4,707	2,849	17,511
S	Washington Ave SB	609	2,668	3,085	2,509	3,132	12,004
N	Hesperian Blvd NB	1,182	4,534	6,611	6,419	4,464	23,211
S	Hesperian Blvd SB	1,820	9,863	10,444	10,601	5,513	38,241
N	E 14th St NB	648	4,082	4,001	3,665	2,049	14,446
S	E 14th St SB	259	3,229	3,333	4,370	1,349	12,539
I 580 W	Interstate 580 WB	1,878	12,512	8,662	9,421	6,199	38,672
I 580 E	Interstate 580 EB	4,185	14,006	13,637	16,419	15,717	63,965
N	Redwood Rd NB	209	1,507	833	1,171	393	4,113
S	Redwood Rd SB	267	861	1,021	1,521	998	4,667
N	Crow Cayon Rd NB	1,207	3,653	4,658	4,130	5,693	19,340
S	Crow Cayon Rd SB	2,072	4,552	4,901	4,141	5,306	20,972
N	San Ramon Rd NB	1,733	8,500	12,745	11,962	6,448	41,387
S	San Ramon Rd SB	2,830	8,607	9,167	7,255	4,914	32,773

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
I 680 W	Interstate 680 WB	2,580	18,267	22,011	24,753	17,890	85,501
I 680 E	Interstate 680 EB	2,978	15,606	14,184	13,908	7,857	54,533
N	Village Pkwy NB	414	2,092	3,086	2,985	1,773	10,350
S	Village Pkwy SB	202	1,943	2,344	2,512	679	7,680
N	Dougherty Rd NB	473	3,498	4,492	5,347	3,311	17,121
S	Dougherty Rd SB	1,084	6,901	5,774	5,231	2,490	21,479
N	Tassajara Rd NB	288	1,754	3,496	3,726	2,364	11,628
S	Tassajara Rd SB	466	2,890	2,736	2,082	850	9,024
N	Collier Canyon Rd NB	7	146	86	151	76	466
S	Collier Canyon Rd SB	6	109	73	187	60	435
N	N Livermore Ave NB	59	1,159	2,075	3,255	369	6,917
S	N Livermore Ave SB	159	3,068	2,730	2,253	208	8,417
N	Vasco Rd NB	1,185	3,351	3,774	3,801	5,228	17,339
S	Vasco Rd SB	2,508	5,300	4,109	4,065	2,183	18,165

Table 47: Forecasted Bridge and Screenline Volumes by Time Period – Forecast Year 2050 – AlaCC Model Compared to PBA 2050

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
San Francisco Bay Screenline		11,610	25,833	26,754	32,605	10,766	107,568
I 580 W	Richmond-San Rafael Bridge (CC to Marin)	535	4,886	(2,865)	2,407	(585)	4,378
I 580 E	Richmond-San Rafael Bridge (Marin to CC)	1,360	4,282	586	3,963	1,862	12,052
I 80 W	Bay Bridge (Alameda to San Francisco)	5,801	6,734	17,137	7,712	7,201	44,584
I 80 E	Bay Bridge (San Francisco to Alameda)	2,194	6,141	13,856	12,918	7,509	42,617
SR 92 W	San Mateo Bridge (Alameda to San Mateo)	45	1,104	(2,048)	(90)	(3,521)	(4,511)
SR 92 E	San Mateo Bridge (San Mateo to Alameda)	420	(1,589)	(1,108)	1,092	(991)	(2,176)
SR 84 W	Dumbarton Bridge (Alameda to San Mateo)	664	4,481	18	3,354	(846)	7,671
SR 84 E	Dumbarton Bridge (San Mateo to Alameda)	591	(205)	1,178	1,250	138	2,952
Contra Costa – Solano County Line		6,533	9,471	7,814	10,975	5,878	40,671
I 80 W	Carquinez Bridge (Contra Costa to Solano)	1,228	3,517	(7,173)	(9,499)	(6,250)	(18,176)
I 80 E	Carquinez Bridge (Solano to Contra Costa)	(3,097)	(11,689)	(8,202)	(881)	1,917	(21,952)
I 680 W	Benicia Bridge (Contra Costa to Solano)	2,310	3,406	5,553	3,603	5,136	20,008
I 680 E	Benicia Bridge (Solano to Contra Costa)	3,853	9,114	8,446	9,823	2,455	33,692
SR 160 W	Antioch Bridge (Contra Costa to Solano)	1,919	3,458	4,768	3,943	1,425	15,513
SR 160 E	Antioch Bridge (Solano to Contra Costa)	319	1,665	4,421	3,985	1,195	11,586

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Contra Costa – Alameda County Line		4,181	(2,377)	(4,700)	(2,924)	(4,131)	(9,950)
I 580 W	Interstate 580 W	1,766	4,829	2,430	2,507	1,414	12,946
I 580 E	Interstate 580 E	2,329	(1,420)	302	(34)	3,173	4,349
I 80 W	Interstate 80 W	537	(1,986)	(838)	(1,524)	(810)	(4,621)
I 80 E	Interstate 80 E	(137)	(2,985)	(4,330)	(4,734)	(900)	(13,086)
N	San Pablo Avenue NB	46	425	79	1,561	(1,847)	265
S	San Pablo Avenue SB	(197)	710	130	336	(824)	155
N	Ashbury Avenue NB	35	113	151	(36)	114	377
S	Ashbury Avenue SB	31	48	128	38	24	269
N	Colusa Ave NB	222	1,107	1,650	1,291	868	5,137
S	Colusa Ave SB	280	804	752	1,229	396	3,461
N	Arlington Ave NB	(216)	(1,220)	(1,967)	(1,277)	(2,771)	(7,451)
S	Arlington Ave SB	(359)	(1,593)	(2,061)	(1,572)	(1,888)	(7,474)
N	Grizzly Peak Blvd NB	(89)	60	(767)	(576)	(551)	(1,924)
S	Grizzly Peak Blvd SB	(66)	(1,267)	(359)	(131)	(528)	(2,352)
N	All other roads	263	927	1,537	2,165	559	5,451
S	All other roads	114	471	1,193	424	(355)	1,846

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Alameda – Santa Clara County Line		4,273	(19,460)	(6,655)	(17,489)	1,425	(37,905)
N	Fremont Blvd NB	457	(3,667)	(778)	(1,888)	(71)	(5,948)
S	Fremont Blvd SB	197	(2,939)	(817)	(3,685)	(1,200)	(8,444)
I 880 N	Interstate 880 N	(1,032)	(4,173)	(6,506)	(6,251)	(75)	(18,038)
I 880 S	Interstate 880 N	14	(6,298)	(7,092)	(6,231)	(1,407)	(21,013)
N	Milmont Dr NB	(889)	(1,343)	(2,169)	(2,302)	(3,668)	(10,371)
S	Milmont Dr SB	(1,086)	(1,961)	(2,269)	(1,874)	(3,185)	(10,375)
N	Warm Springs Blvd NB	1,419	(497)	585	(377)	1,297	2,428
S	Warm Springs Blvd SB	2,464	(915)	240	(635)	1,978	3,132
I 680 NB	Interstate 680 NB	(89)	1,460	387	(1,865)	1,440	1,333
I 680 SB	Interstate 680 SB	(126)	(2,938)	1,451	1,549	2,114	2,049
N	N Park Victoria Dr NB	752	(705)	1,441	(284)	1,425	2,628
S	N Park Victoria Dr SB	1,095	(2,329)	2,162	(838)	2,360	2,451
N	Calaveras Rd NB	413	2,855	3,494	3,636	211	10,608
S	Calaveras Rd SB	668	3,786	3,179	3,458	130	11,221
N	Mines Rd NB	3	32	20	79	51	185
S	Mines Rd SB	14	172	17	20	26	249

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
San Joaquin County Line		26,472	41,363	20,023	21,689	67,642	177,189
SR 4 W	State Highway Route 4 W	2,038	2,660	3,363	2,617	3,140	13,818
SR 4 E	State Highway Route 4 E	2,166	2,735	3,565	2,058	4,998	15,523
W	Byron Highway WB	1,410	1,576	2,088	1,157	1,448	7,679
E	Byron Highway EB	1,310	1,746	1,672	899	2,474	8,102
W	W Grant Line Rd WB	125	519	844	879	392	2,759
E	W Grant Line Rd EB	56	349	1,062	1,239	506	3,211
I 205 W	Interstate 205 WB	10,812	18,964	12,880	12,436	14,049	69,141
I 205 E	Interstate 205 EB	7,161	13,206	13,518	15,251	23,894	73,030
I 580 W	Interstate 580 WB	(3,492)	(8,070)	(18,611)	(13,382)	(1,301)	(44,856)
I 580 E	Interstate 580 EB	(2,145)	(6,241)	(19,364)	(17,688)	634	(44,805)
W	Patterson Pass Rd WB	2,996	4,463	5,544	4,603	4,578	22,184
E	Patterson Pass Rd EB	2,762	3,830	6,045	5,283	6,899	24,819
W	Tesla Rd WB	434	1,971	3,391	3,043	3,843	12,683
E	Tesla Rd EB	840	3,655	4,025	3,294	2,086	13,901

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Alameda/Contra Costa East-West Screenline		25,101	17,998	34,033	12,035	40,771	129,938
W	Cummings Skyway WB	35	854	356	810	100	2,154
E	Cummings Skyway EB	29	311	260	360	165	1,125
SR 4 WB	State Highway Route 4 WB	2,333	5,015	4,809	4,593	2,790	19,540
SR 4 EB	State Highway Route 4 EB	1,368	2,654	4,891	4,566	3,071	16,549
W	Alhambra Valley Rd WB	2	283	(215)	(94)	(70)	(95)
E	Alhambra Valley Rd EB	5	18	(130)	227	(198)	(78)
W	San Pablo Dam Rd WB	726	1,352	2,690	1,958	1,395	8,120
E	San Pablo Dam Rd EB	450	1,953	2,711	1,801	2,003	8,919
SR 24 W	State Highway Route 24 WB	2,174	183	3,281	1,460	3,093	10,191
SR 24 E	State Highway Route 24 EB	536	(383)	1,800	1,133	2,311	5,397
W	Canyon Rd WB	352	(1,323)	94	(182)	(472)	(1,531)
E	Canyon Rd EB	593	(299)	58	(822)	(1,742)	(2,213)
W	Crow Canyon Rd WB	1,034	74	361	(641)	(1,260)	(432)
E	Crow Canyon Rd EB	158	203	454	(222)	(741)	(147)
W	Jensen Rd WB	-	-	-	-	-	-
E	Jensen Rd EB	-	-	-	-	-	-
W	E Castro Valley Blvd WB	877	965	(221)	465	261	2,348
E	E Castro Valley Blvd EB	1,021	1,303	1,563	1,215	2,973	8,075
I 580 W	Interstate 580 WB	1,945	5,709	4,027	2,952	6,049	20,681
I 580 W	Interstate 580 EB	776	2,373	4,510	2,481	9,579	19,719

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
W	Dublin Canyon Rd WB	2,314	4,560	5,050	4,211	4,309	20,443
E	Dublin Canyon Rd EB	1,878	3,910	4,901	4,051	6,312	21,053
SR 84 W	State Highway Route 84 WB	4,162	4,265	4,385	4,816	1,371	18,999
SR 84 E	State Highway Route 84 EB	1,639	(2,399)	(1,275)	(4,467)	2,091	(4,412)
I 680 W	Interstate 680 WB	534	(6,364)	(5,023)	(5,284)	(1,059)	(17,196)
I 680 E	Interstate 680 EB	159	(7,217)	(5,303)	(13,350)	(1,561)	(27,273)

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
Alameda/Contra Costa North-South Screenline		3,539	(15,887)	(14,740)	(21,690)	(19,820)	(68,597)
N	Wicks Blvd NB	19	521	(17)	246	(481)	288
S	Wicks Blvd SB	25	(826)	(415)	(1,433)	(484)	(3,134)
I 880 N	Interstate 880 NB	1,602	3,811	912	2,673	3,989	12,986
I 880 S	Interstate 880 SB	1,110	2,109	448	1,242	478	5,385
N	Washington Ave NB	(245)	(3,959)	(4,346)	(3,236)	(2,227)	(14,013)
S	Washington Ave SB	205	(3,627)	(5,460)	(5,580)	(1,857)	(16,318)
N	Hesperian Blvd NB	234	(4,200)	(5,151)	(5,179)	(3,235)	(17,530)
S	Hesperian Blvd SB	509	(1,304)	(2,658)	(596)	(2,959)	(7,008)
N	E 14th St NB	(489)	5	(1,023)	(290)	(3,756)	(5,552)
S	E 14th St SB	(474)	(585)	(1,565)	322	(4,097)	(6,400)
I 580 W	Interstate 580 WB	339	4,519	1,567	(66)	1,863	8,222
I 580 E	Interstate 580 EB	2,529	3,380	3,719	4,065	6,235	19,929
N	Redwood Rd NB	34	332	(107)	56	(1,059)	(744)
S	Redwood Rd SB	(98)	(519)	105	342	129	(42)
N	Crow Cayon Rd NB	110	(119)	417	(352)	(146)	(91)
S	Crow Cayon Rd SB	227	(214)	722	249	766	1,750
N	San Ramon Rd NB	1,075	2,940	5,606	2,734	1,548	13,902
S	San Ramon Rd SB	2,311	840	4,438	1,635	2,809	12,033

		Time Period					
Route Number/ Direction	Link Description	Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
I 680 W	Interstate 680 WB	(1,201)	(3,867)	784	(4,259)	195	(8,348)
I 680 E	Interstate 680 EB	(1,828)	(10,742)	(6,921)	(8,886)	(4,793)	(33,170)
N	Village Pkwy NB	414	2,092	3,086	2,985	1,773	10,350
S	Village Pkwy SB	202	1,943	2,344	2,512	679	7,680
N	Dougherty Rd NB	169	2,260	3,144	4,403	2,671	12,647
S	Dougherty Rd SB	949	6,006	4,049	3,108	668	14,779
N	Tassajara Rd NB	(195)	(2,505)	(4,360)	(4,640)	(3,811)	(15,511)
S	Tassajara Rd SB	(689)	(5,291)	(5,593)	(4,845)	(2,785)	(19,203)
N	Collier Canyon Rd NB	7	146	83	131	75	442
S	Collier Canyon Rd SB	4	40	73	185	60	362
N	N Livermore Ave NB	(81)	12	857	1,495	(727)	1,556
S	N Livermore Ave SB	(19)	1,279	1,395	852	(565)	2,941
N	Vasco Rd NB	(1,110)	(5,060)	(5,609)	(6,843)	(6,675)	(25,297)
S	Vasco Rd SB	(2,103)	(5,306)	(5,261)	(4,718)	(4,101)	(21,489)

Product 18: Region-level, time-period-specific comparison of estimated transit boardings by operator and technology, separately, to MTC's estimates for the horizon year.

Region-level time-period-specific transit boardings by operator and technology are reported for the 2050 forecast year for PBA 2050 (Table 48), the AlaCC model (Table 49) and a comparison between the AlaCC model and PBA 2050 (Table 50).

The AlaCC model estimates of transit boardings are 17 percent lower for the Bay Area compared to PBA 2050. The AlaCC model is within 1 percent of BART ridership, 21 percent of AC Transit ridership, 11 percent of total Muni ridership (bus plus light rail) and 17 percent of Caltrain ridership. The AlaCC model estimates lower ferry ridership compared to PBA 2050. Three factors contribute to the difference in transit ridership estimates:

- The drop in ridership observed in the region between 2015 and 2019, which is reflected in the AlaCC model base year validation but not reflected in the TM1.5.2 base year validation. As discussed in the Model Overview section, the Bay Area lost transit riders in this period despite growing in population.
- Missing PBA 2050 transit investments in the AlaCC model compared to the TM1.5.2 PBA 2050 scenario. Only major regionwide transit investments and selected local transit improvements are reflected in the AlaCC 2050 scenario.
- Higher 2050 auto ownership estimated by the AlaCC model compared to TM1.5.2. This may be due to artificial differences in auto accessibility created by the denser zone system and/or network used in the AlaCC model. Average auto accessibility decreases proportionally more from base to horizon year in TM1.5.2 than in the AlaCC model.

Table 48: Forecasted Transit Boardings by Operator and Technology - 2050 Forecast Year – PBA 2050

Operator	Technology	Time Period					
		Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
AC Transit	Bus	6,300	138,400	87,500	145,200	67,900	445,300
BART	Heavy Rail	39,800	305,100	96,800	234,400	192,000	868,100
Caltrain	Commuter Rail	13,200	129,200	29,200	106,800	70,700	349,100
Golden Gate Transit	Bus	1,200	18,100	6,300	16,700	5,500	47,800
	Ferry	600	10,000	2,300	7,700	3,800	24,400
SamTrans	Bus	2,600	52,700	26,300	54,100	19,800	155,500
SF Muni	Bus	17,800	292,800	199,200	330,900	174,100	1,014,800
	Light Rail	4,500	88,100	60,100	98,400	64,200	315,300
VTA	Bus	2,300	97,600	58,300	102,600	24,500	285,300
	Light Rail	8,200	90,700	50,700	93,200	54,700	297,500
Other	Bus	3,400	76,600	27,200	72,000	20,400	199,600
	Ferry	0	9,900	1,600	7,100	4,400	23,000
	Light Rail	0	13,900	5,900	13,900	0	33,700
	Commuter Rail	1,800	28,200	5,200	24,000	10,100	69,300
All	All	61,900	1,046,200	559,800	1,072,600	520,100	4,128,700

Table 49: Forecasted Transit Boardings by Operator and Technology - 2050 Forecast Year – AlaCC Model

Operator	Technology	Time Period					
		Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
AC Transit	Bus	7,266	106,611	74,725	112,717	51,601	352,920
BART	Heavy Rail	40,698	268,101	145,608	217,600	198,004	870,011
Caltrain	Commuter Rail	18,051	100,330	27,718	81,611	61,045	288,755
Golden Gate Transit	Bus	2,183	16,691	5,201	14,111	6,405	44,591
	Ferry	258	3,500	418	2,169	1,327	7,672
SamTrans	Bus	3,214	43,989	18,577	39,749	13,551	119,080
SF Muni	Bus	14,540	236,995	165,542	242,260	149,937	809,274
	Light Rail	11,004	121,244	75,781	134,348	30,563	372,940
VTA	Bus	3,645	61,473	36,848	63,902	17,001	182,869
	Light Rail	4,934	52,479	28,001	52,511	21,538	159,463
Other	Bus	5,251	47,416	24,173	47,185	15,005	139,030
	Ferry	0	3,647	450	2,049	1,449	7,595
	Light Rail	n/a	n/a	n/a	n/a	n/a	0
	Commuter Rail	3,947	29,564	4,939	26,776	9,883	75,109
All	All	114,991	1,092,040	607,981	1,036,988	577,309	3,429,309

Table 50: Forecasted Transit Boardings by Operator and Technology - 2050 Forecast Year – AlaCC Model Compared to PBA 2050

Operator	Technology	Time Period					
		Early AM (3-6 AM)	AM Peak (6-10 AM)	Midday (10 AM-3 PM)	PM Peak (3-7 PM)	Evening (7 PM-3 AM)	All Time Periods
AC Transit	Bus	15%	-23%	-15%	-22%	-24%	-21%
BART	Heavy Rail	2%	-12%	50%	-7%	3%	0%
Caltrain	Commuter Rail	37%	-22%	-5%	-24%	-14%	-17%
Golden Gate Transit	Bus	82%	-8%	-17%	-16%	16%	-7%
	Ferry	-57%	-65%	-82%	-72%	-65%	-69%
SamTrans	Bus	24%	-17%	-29%	-27%	-32%	-23%
SF Muni	Bus	-18%	-19%	-17%	-27%	-14%	-20%
	Light Rail	145%	38%	26%	37%	-52%	18%
VTA	Bus	58%	-37%	-37%	-38%	-31%	-36%
	Light Rail	-40%	-42%	-45%	-44%	-61%	-46%
Other	Bus	54%	-38%	-11%	-34%	-26%	-30%
	Ferry	n/a	-63%	-72%	-71%	-67%	-67%
	Light Rail	n/a	-100%	-100%	-100%	n/a	-100%
	Commuter Rail	119%	5%	-5%	12%	-2%	8%
All	All	13%	-19%	-7%	-21%	-19%	-17%

Appendix A

This section lists all the PBA projects represented with project cards in the AlaCC model. The projects are listed in order of Opening Year and RTP Identification code. Many PBA projects consist of multiple elements, only some of which can be represented in a network model. The descriptions below are the complete PBA project descriptions rather than the AlaCC project card description. The project cards listed below are applied to the base 2017/2018 network; as such, a few of these projects represent 2020 base year existing conditions.

Exhibit 12: Plan Bay Area Projects Included in AlaCC 2050 Scenario

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
Alameda		Tempo BRT	Tempo is a bus rapid transit (BRT) service in Oakland and San Leandro in California. It is operated by AC Transit as Line 1T. The route has dedicated lanes and center-boarding stations along much of the corridor, prepaid fares, signal preemption, and all-door boarding. The northern terminus of the line is at the Uptown Transit Center, located at the 19th Street/Oakland BART station. The line continues down Broadway in mixed traffic, passing 12th Street/Oakland City Center BART station before the southbound and northbound routes split at 11th and 12th Streets, respectively. Both directions simultaneously meet at and run on Lake Merritt Boulevard before splitting again to 12th Street and International Boulevard. Southbound buses join International Boulevard at 14th Avenue and begin median running in an exclusive bus lane. Services continue to a station near Fruitvale BART and onward to San Leandro, where operation in mixed traffic resumes and the line continues down Davis Street to terminate at San Leandro BART.	N/A	N/A	2016
Marin	MTC	Richmond-San Rafael Bridge Access Improvements	Contra Costa and Marin Counties: On I-580/Richmond-San Rafael Bridge: The Richmond San Rafael (RSR) Bridge Access Improvement Project will convert the existing shoulders on the RSR Bridge to accommodate bicycle and pedestrian access on the upper bridge deck (westbound), and a new automobile travel lane on the lower deck (eastbound). Bicycle and pedestrian access on the upper deck of the RSR Bridge would be provided by installing a barrier to separate bicyclists and pedestrians from motorists. The total length of this project is approximately 6 miles. All proposed improvements are anticipated to be within existing highway and local street rights-of-way except for easements required for utility relocations and for a small length of path connecting to the RSR bridge. The project consists of three major components that are interrelated. Element 1 is an Eastbound I-580 travel lane across the RSR Bridge between Marin County and Contra Costa County. Element 2 is a	MRN150009	21-T06-020	2020

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
			bicycle/pedestrian path in Contra Costa County. Element 3 is the bicycle/pedestrian path on the RSR Bridge and the associated path connections to bridge.			
Santa Clara	VTA	BART - Fremont to Berryesa Extension	Extend BART service from Fremont to Berryesa station. Operations commenced in June 2020.	N/A	N/A	2020
Alameda		I-880 SB Express Lanes, Hegenberger Rd to Dixon Landing Rd	HOV lane conversion.	N/A	N/A	2021
Alameda		I-880 NB Express Lanes, Dixon Landing Rd to Lewelling Blvd	HOV lane conversion.	N/A	N/A	2021
Alameda	BART	Freemont to Berryesa Extension	Extend BART service to Berryesa station.	N/A	N/A	2022
Alameda	ACTC	I-880 North Safety Improvements	Oakland: I-880 between 23rd Ave to 29th Ave: Reconfigure Interchange, including new ramps.	ALA050019	21-T06-024	2025
Marin	TAM	US 101 HOV Lanes - Marin-Sonoma Narrows (Marin)	Marin and Sonoma Counties: From SR 37 in Novato to Old Redwood Highway in Petaluma; Convert expressway to freeway and widen to 6 lanes for HOV lanes.	MRN050034	21-T06-026	2025
Alameda	ACTC	Rte 84 Widening, south of Ruby Hill Dr to I-680	Alameda County: On State Route 84 from south of Ruby Hill Drive to I-680: Upgrade from 2-lane conventional highway to 4-lane expressway, make operational improvements to SR84/I-680 I/C and extend SB express lane about two miles to the north. Caltrans is the implementing agency for the Con phase.	ALA150001	21-T06-037	2025
Alameda	MTC	Bay Bridge Forward: Alameda I-580 WB HOV Lane Ext	Alameda County: On I-580 westbound approach to the San Francisco- Oakland Bay Bridge toll plaza from the SR 24/I-980 interchange to I- 80: Convert one general purpose lane to an HOV lane.	ALA190018	21-T06-049	2025
Various		Service Frequency Boost | AC Transit	This program includes funding to implement improvements to AC Transit's existing local bus service. Improvements include frequency upgrades (5-10 minute peak headways along routes 72/72M/72R, 18, 51A/B, 6, 20/21, 57, 40/40L, 97, 99, Tempo BRT, NL, F-local and F-Transbay) and local/rapid service on some routes		21-T10-065	2025
San Francisco	SFMTA	Van Ness Avenue Bus Rapid Transit	San Francisco: On Van Ness Avenue from Mission to Lombard: Design and implement a BRT project. Project is phased. Project also references RTP IDs 240745 and 240471. The Van Ness BRT corridor is 1.96 miles (3.15 km) long, running north–south between Lombard Street and Mission Street. Van Ness Avenue has six travel lanes on this section – two general-purpose lanes in each direction, plus center-running red bus lanes. Nine stations are located along the corridor. A	SF-070005	21-T10-081	2025

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
			landscaped center median is located between the bus lanes except at stations. The northbound bus lane exits at Filbert Street, with a marked transition for buses to move to the curb lane. Services using the BRT lane and stations. https://en.wikipedia.org/wiki/Van_Ness_Bus_Rapid_Transit.			
San Francisco	SFMTA	Light Rail | Service Expansion to Chinatown ("Central Subway")	This program includes funding to extend Muni's existing T-line to Chinatown through the Central Subway. Improvements include light rail shuttles between Chinatown and Mission Bay (via the Mission Bay Loop) during peak periods and frequency upgrades (7-minute peak headways, 4-5 mins with shuttle).		21-T10-083	2025
Contra Costa	SJRC	Oakley Station Platform	Oakley: North of Main Street between 2nd St and O'Hara Ave: Construct a new train station platform for the Amtrak San Joaquin inter-city rail service. Constructs a station track siding with two turnouts, within the existing railroad right-of-way. Includes shelters, lighting, signage, ADA-compliant pedestrian sidewalks and other associated improvements.	CC-190002	21-T11-115	2025
Alameda	ACTC	I-680 NB HOV/HOT Lane	Route I-680: from South of Auto Mall Parkway to State Route 84 in Alameda County: Construct NB HOV/HOT Lane.	ALA130034	21-T12-116	2025
San Mateo	CCAG	US101 Managed Lanes: Santa Clara Co- S of Grand Ave	San Mateo County: On US 101 between 2 miles south of the Santa Clara County Line (P.M. 50.6 in SCL) and 0.3 mi south of Grand Avenue Interchange (SM 21.8): Install Express Lanes. Utilize existing auxiliary lanes where possible and restore auxiliary lanes where needed for operations. SMCTA is co-sponsoring project.	SM-150017	21-T12-116	2025
Santa Clara	VTA	Santa Clara County - US 101 Express Lanes	Santa Clara County: Implement roadway pricing on US 101 carpool lane. Convert existing US 101 HOV lanes in both directions from Cochrane Avenue to the San Mateo County Line to Express Lanes, including the US 101/SR 85 HOV direct connectors in Mountain View, add a second Express Lane in both directions from Cochrane Road to SR 85 in San Jose and from Blossom Hill Road to Fair Oaks Avenue, convert the second HOV lanes in both directions from Shoreline Boulevard to south of Oregon Expressway to Express Lanes, and add auxiliary lanes at various locations. Construct new Express Lanes from Cochrane Ave to East Dunne Ave in Morgan Hill. The Silicon Valley Express Lanes Phase 3 project includes the conversion of single carpool lanes to express lanes on US101 from near SR-237 to SR-85 in Mountain View and the conversion of double carpool lanes to double Express Lanes on US101 from the US101/SR-85 interchange in Mountain View to near the San Mateo County line in Palo Alto. Earmark funds will be used to design Phase 5 of the	SCL110002	21-T12-116	2025

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
			project-limits US101 PM 45.9 to PM 38.3. Deliver milestones reflect phase 5 of project.			
Solano	MTC	Solano I-80 Managed Lanes	Solano County: I-80 from Red Top Rd to I-505: Convert existing HOV to Managed Lane; I-80 from Air Base Parkway to I-505: Construct new Managed Lanes. Project also references RTP IDs 17-10-0059	SOL110001	21-T12-116	2025
Various		Express Bus | Service Expansion | ReX (Basic) | Blue Line (San Francisco to San Jose)	This program includes funding to implement new express bus service along US-101, SR-85 and I-280 (on express lanes where available) between San Francisco (Salesforce Transit Center) and San Jose (Diridon Station). Improvements include high-frequency service (10-minute peak headways) and station area amenities like upgraded local bus stops, taxi/TNC loading zones, and improved bicycle/pedestrian infrastructure.		21-T12-126	2025
Santa Clara		Express Lanes | SCL 101 Phase 5 Express Lanes: SR 237 to near I-880	New dual express lanes			2025
Alameda		I-880 Express Lanes, SR-237 to US-101	HOV lane conversion			2027
Alameda		Express Lanes | ALA 580 Express Lanes: Bay Bridge to I-238	GP lane conversion			2028
Alameda		Express Lanes | ALA 580 Express Lanes: I-238 to I-680	GP lane conversion			2028
Alameda		Express Lanes | ALA 580 Express Lanes: Greenville Rd Alameda/San Joaquin Co. Line	GP lane conversion			2028
San Joaquin		I-205 Managed Lanes	I-205 Managed Lanes. Widen from 6 to 8 lanes, Alameda County Line to I-5.			2028
San Joaquin		I-5 HOV Lanes, I-205 to Louise Ave	I-5 HOV Mossdale Rd. Add HOVL from I-205 to Louise Ave, both directions.			2028
Contra Costa	CCTA	Reconstruct I-80/San Pablo Dam Rd Interchange	San Pablo: Reconstruct the I-80/San Pablo Dam Rd (SPDR) interchange. Includes relocating WB El Portal on-ramp to the full I/C northwards, providing access to McBryde Ave through a new connector road from SPDR I/C, and replacing Riverside Ave pedestrian Overcrossing. Constructed in phases.	CC-070035	21-T06-013	2030
Contra Costa	MTC	RSR Forward: ORT and I-580 WB HOV Lane	On westbound I-580 approaching the Richmond-San Rafael (RSR) Bridge beginning at the I-580 / Regatta Avenue Interchange, provide safety and operational improvements by converting one of the three existing general-purpose lanes	CC-210010	21-T06-020	2030

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
			to a high-occupancy vehicle (HOV) lane and replacing the existing tolling structure with overhead toll gantries east of the existing toll booths.			
Contra Costa	CCTA	I-680/SR 4 Interchange (I/C) Reconstruction - Phase 1, 2a, 4	Phase 1: Construct a two-lane flyover direct connector from NB 680 to WB SR4 and remove the existing NB 680 to WB SR4 loop, construct auxiliary lanes, a slip ramp and install a ramp metering facility. Phase 2A: extend the SB 680 collector-distributor ramp and install a ramp metering facility for the WB SR4 on-ramp. Phase 4: Construct SB I-680 to EB SR 4 connector.	CC-010023	21-T06-022	2030
Alameda	ACTC	I-880/Whipple Rd Industrial Pkwy SW I/C Imps	Union City/Hayward: at I-880/Whipple Rd Interchange, implement interchange improvements including widening and reconfiguration of northbound diagonal off-ramp, northbound off-ramp, northbound diagonal on-ramp, northbound on-ramp widening of Industrial Parkway Southwest, local street intersection improvements, and construction of ped/bike improvements.	ALA170005	21-T06-024	2030
Alameda	ACTC	Oakland/Alameda Access Project	Oakland and Alameda: Between Fallon Street and Washington Street: Reconfigure interchanges and intersections to improve connections between I-880, the Posey and Webster tubes and the downtown Oakland area. Removal, reconstruction and reconfiguration of ramps with I-880 and I-980 including a new horseshoe connector between Posey Tube and I-880, removal of NB I-880/Broadway off-ramp viaduct, construction of a new through 6th Street connecting Oak Road to Broadway, reconstruction of Westbound I-980/Jackson Street off-ramp, widening of the NB 880 Oak St off-ramp, construction of new sidewalks, bicycle lanes and bike paths, intersection improvements, and local street modifications in downtown Oakland, China Town, Jack London Square, and within City of Alameda.	ALA070009	21-T06-024	2030
Alameda	Hayward	I-880 Auxiliary lanes at Industrial Parkway	Hayward: I-880 NB between Industrial Pkwy and Alameda Creek; I- 808 SB between Industrial Pkwy and Whipple Rd: Construct auxiliary lanes.	ALA090020	21-T06-024	2030
Alameda	Hayward	I-880 I/C Improvements (Winton Ave and A St)	Hayward: I-880/A St. I/C: Reconstruct interchange, add bike lanes, modify signals and reconfigure intersections to improve truck-turning maneuvers. The interchange reconstruction will provide Caltrans with additional width on the I-880 mainline to accommodate auxiliary lanes in each direction between the I-880/Winton Avenue and I-880/A Street interchanges. This project has potential/conditional funding through Local Area Transportation Improvement Program (LATIP).	ALA170046	21-T06-024	2030

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
Alameda	Hayward	I-880/Industrial Parkway West Interchange	Hayward: At I-880/Industrial Parkway West: Reconstruct the interchange including replacement of overcrossing structure and a new 2-lane northbound off-ramp by realigning a section of Ward Creek. Realign northbound diagonal on-ramp, widen and realign southbound off-ramp, construct a southbound loop on-ramp to provide a HOV bypass. Create an auxiliary lane along northbound I-880 between Industrial Parkway West Interchange and Whipple Road/Industrial Parkway Southwest Interchange by restriping existing lanes and shoulders. This project includes widening of local streets to provide dedicated bikeways and sidewalks, signalization modifications and construction of a multiuse bike and pedestrian path over the new overcrossing structure.	ALA110002	21-T06-024	2030
San Mateo		Corridor & Interchange Improvements | US-101			21-T06-027	2030
Contra Costa	CCTA	SR-4 Operational Improvements - Initial Phases	State Route 4 Operational Improvements - Eastbound: (a) Extend a lane from the lane drop at Port Chicago Interchange to the Willow Pass Rd off-ramp and end as a mandatory exit lane. (b)Construct a new general-purpose lane between the Willow Pass Rd off-ramp and the Willow Pass Rd on-ramp. The new general-purpose lane would eliminate the mandatory exit at Willow Pass Rd off-ramp from (a) and connect to the existing auxiliary lane between Willow Pass Rd on-ramp & San Marco Blvd off ramp. Construct a second exit lane at the EB SR4 off-ramp to San Marco Blvd to accommodate existing and future peak hour traffic volumes. (c) Construct auxiliary lane from the San Marco Blvd loop on ramp to the existing deceleration lane at Bailey Rd off-ramp. (d) Construct an auxiliary lane between the Port Chicago Highway on-ramp and the Willow Pass Road off-ramp. Westbound: Construct a lane from Willow Pass Rd on-ramp connecting to the existing added lane, east of the Port Chicago Highway off-ramp and a second exit lane at Port Chicago Highway off-ramp. Modify one of the mandatory exit lanes to SR242 to an optional exit lane, allowing three lanes exit to SR242 and three lanes to continue on WB SR4.	CC-170018	21-T06-031	2030
Regional/ Multi-County	MTC	SR 37 Interim Project - Sears Point to Mare Island	Solano and Sonoma Counties: SR-37 between the Sears Point/SR 121, and Mare Island: Implement a high occupancy vehicle (HOV) lane, implement tolling. This project will improve traffic flow and peak travel times and increase vehicle occupancy. This project provides an incentive to increase multiple occupant vehicle use during peak periods. Currently there is no incentive for a bus route on SR 37 because of the substantial delays and there are no current	VAR210004	21-T06-035	2030

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
			transit routes using SR 37. The Napa Bus Feasibility Study identified a demand for bus service through the corridor, and this project would allow transit operators to implement bus service on SR 37. Other State funds are SB170 funds. Other Federal funds are NHPP.			
Alameda	Hayward	Rt 92/Clawiter/Whitesell Interchange Improvements	Hayward: Rt 92/Clawiter Rd: Upgrade existing Clawiter interchange. Add ramps and overcrossing for Whitesell St. extension. Signalize ramp intersections.	ALA090016	21-T06-041	2030
Alameda	MTC	BBF: I-80 WB Bus Only Lane Extension	Alameda County: On I-80 westbound between SFOBB Toll Plaza and Powell Street interchange: Construct a bus-only or HOV lane.	ALA210028	21-T06-049	2030
Alameda	ACTC	7th Street Grade Separation West	Oakland: Within the Port: Implement roadway and rail improvements including realigning and grade separating the intersection of 7th Street and Maritime St and constructing a rail spur underneath connecting the Joint Intermodal Terminal and the Oakland Harbor Intermodal Terminal yard, also reconstruct and widen the existing multi-use path.	ALA170086	21-T07-055	2030
Alameda	Dublin	Dublin Blvd. - North Canyons Pkwy Extension	Alameda County, Dublin and Livermore: Dublin Blvd-North Canyons Parkway from Fallon Rd to Croak Rd: Construct six lane extension Dublin Blvd-North Canyons Parkway from Croak Rd to Doolan Rd: Construct four lane extension. The new extended street is planned to have bike lanes, off-street Class I trail, sidewalks, landscaping curb and gutter, traffic signals, a raised median, bus stops, and all street utilities. This project will consider the provision of dedicated transit lanes and/or queue jump lanes in addition to the mixed flow travel lanes for higher level of transit service with 10 to 20 minute headways during appropriate peak demand periods. This project will also require enhanced multimodal connectivity to various land uses along its stretch and at its terminus, including creating connectivity to 5 PDAs in two cities.	ALA150003	21-T07-056	2030
Alameda	Dublin	Tassajara Road Widening	Dublin: Tassajara Road between North Dublin Ranch Drive and Quarry Lane School Road: Widen and improve Tassajara Road to a four-lane arterial standard, with buffered bike lanes, sidewalks, landscaped median, stormwater treatment areas, and other associated street improvements including pavement replacement, grinding/overlay, and cross slope correction operations, new and revised striping. The widening project will increase the capacity of Tassajara Road and accommodate future traffic in the next 10 to 15 years generated by several approved developments in eastern Dublin and in Contra Costa County.	ALA210026	21-T07-056	2030

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
Contra Costa	Brentwood	Brentwood Boulevard Widening - North (Phase I)	Brentwood: Brentwood Boulevard from Havenwood Avenue to Homecoming Way: Phase I-Widen from 2 to 4 lanes, with two lanes in each direction with two bike lanes, curbs, gutters, medians, sidewalks, street lights and landscaping on each side of the roadway, including a new parallel bridge over Marsh Creek, traffic signal modification at Brentwood Boulevard / Grant Street, and moving overhead power lines, telephone lines and cable lines underground. CIP 336-3162.	CC-070011	21-T07-056	2030
Contra Costa	Brentwood	Brentwood Boulevard Widening - North (Phase II)	Brentwood: Brentwood Blvd. between Homecoming Way and Lone Tree Way: Widen existing roadway from 2 to 4 lanes for 2600 linear feet including curb, gutter, sidewalk, bike lanes, streetlights and landscaping.	CC-170015	21-T07-056	2030
Contra Costa	CC County	Camino Tassajara Realignment, S of Windemere Pkwy	Contra Costa County: Camino Tassajara between Windemere Parkway and the City of Dublin: Realign curves and widen Camino Tassajara to four lanes. The project will be coordinated with the City of Dublin to tie into improvements along Tassajara Road. The project will widen the roadway to provide two travel lanes and Class 2 bike lane in both directions.	CC-170016	21-T07-056	2030
Contra Costa	San Ramon	Crow Canyon Road (Alcosta to Indian Rice) Widening	San Ramon: Crow Canyon Rd from Alcosta Blvd to Indian Rice Rd: Widen to three lanes in each direction. Work will be completed in two phases. Phase 1 limits: Alcosta to St. George. Phase 2 limits: St. George to Indian Rice Road.	CC-190001	21-T07-056	2030
Contra Costa		AC Transit 23rd St BRT	This program includes funding to implement new BRT service along 23rd St from Hercules to Contra Costa College, Richmond BART and the Richmond Ferry. Improvements include high-frequency service (10-minute peak headways), queue jumps, transit signal priority, new vehicles, improved stops and possible bus-only lanes.		21-T10-076	2030
		AC Transit San Pablo Ave BRT	This program includes funding to implement BRT improvements to existing bus service along San Pablo Ave from 20th St to Richmond Pkwy Transit Center. Improvements include frequency upgrades (5-minute peak headways), improved stop infrastructure, merging of local/rapid stops, dedicated lanes and transit signal priority. Dedicated bus/bike lanes ->The project will convert one travel lane in each direction to a bus-only lane and convert the on-street parking lanes to protected bike lanes (from 16th St in Oakland to Heinz Ave in Berkeley). https://www.alamedactc.org/programs-projects/multimodal-arterial-roads/sanpabloave	ALA230008	21-T10-077	2030
San Francisco	SFMTA	Geary Bus Rapid Transit	San Francisco: Along the Geary corridor between 34th Avenue and Market Street: Design and implement transit	SF-070004	21-T10-079	2030

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
			performance and safety improvements. Phase 1 (Geary Rapid Project) extends from Stanyan St to Market St and opened in 2021. Phase 2 (Geary Blvd Improvement) extends from 34th Avenue to Stanyan St. The Geary Rapid Project decreased the number of travel lanes on Geary Expressway from four to two general-purpose lanes and includes a dedicated bus-only lane in each direction. Phase 2 will add one dedicated bus lane in each direction, no change in the number of GPLs. https://www.sfmta.com/projects/geary-rapid-project.			
San Francisco	SFMTA	SFMTA - Core Capacity Program	SFMTA: Along the K, J and M-Line Corridors: Design and implement high priority route improvements from the Muni Forward Program. Project includes a combination of transit signal priority, transit-only lanes, stop consolidation, and complementary facility and pedestrian improvements. Included in the award are a set of targeted improvements to two key rail corridors: the J and M-Lines and design for the K-line.	SF-190012	21-T10-084	2030
Santa Clara	VTA	Eastridge to BART Regional Connector	San Jose: At the Eastridge Transit Center: Phase I (completed) - pedestrian and environmental improvements, bus improvements, expansion of transit center and relocation of ring road Capitol Expressway light Rail line 2.3 miles from the existing Alum Rock Transit Center to a rebuilt Eastridge Transit Center: Phase II - Provide light rail extension in the East Valley. Project will also include new station facilities, power, signal and way.	SCL050009	21-T10-087	2030
San Francisco	SF County TA	SF Downtown Congestion Pricing	San Francisco: In the downtown area: Implement a demonstration value pricing (tolls and incentives) program	SF-130017	21-T10-091	2030
Regional/ Multi-County	WETA	Ferry Service - Berkeley	WETA: Berkeley: Provide ferry service from Berkeley to San Francisco.	MTC050027	21-T11-096	2030
San Francisco	Port of SF	Mission Bay Ferry Terminal	San Francisco: At the eastern terminus of 16th St: Construct new ferry landing to service San Francisco Mission Bay and Central Waterfront as a part of the Bay area ferry transit system. Project includes RTP	SF-170001	21-T11-097	2030
San Mateo	Redwood City	Redwood City Ferry Service	SF Bay Area: Between Redwood City and San Francisco/Oakland: Environmental clearance and design of ferry transit service	SM-110002	21-T11-098	2030
Alameda	Fremont	Irvington BART Station	Fremont: Along the BART corridor in the Irvington District: Construct a new BART station.	ALA230004	21-T11-104	2030
Alameda	BART	Bay Fair Connection	The Bay Fair Connection project will add one or more additional tracks and one or more additional passenger platforms. Work also includes station modernization,	ALA170044	21-T11-106	2030

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
			modification to switches, tracks, crossovers, train control, signaling, traction power, etc. to the Bay Fair Station in order to accomplish the following goals:- Build necessary infrastructure for achievement of "BART Metro" service plan to better serve the Core Areas * Trains must be able to be brought into service at Bay Fair (through a staging area pocket track) * Trains must be able to be decoupled at Bay Fair Station (short trains, turn backs)- Allow for a seamless and convenient connection between the Tri-Valley and Silicon Valley (e.g. one-seat ride or timed transfer)- Configure station for maximum system performance and operational flexibility in all directions over the long-term- Modernize station, improve the customer experience provide expanded facilities for crew			
Regional/ Multi-County	BART	BART Transbay Core Capacity Improvements	BART: Systemwide: Implement a multi-pronged effort to address capacity issues in the Transbay corridor and is in coordination with the BART Metro Program project. The project elements are: Communication-based train control (CBTC) system to safely enable closer headways and allow BART to operate more frequent service (12-minute frequencies) Expansion of the rail car fleet by 306 vehicles to add cars to existing trains and operate more frequent trains Added traction power substations to allow more frequent service. Project also references 17-10-0016.	REG170017	21-T11-106	2030
Santa Clara	VTA	BART - Berryessa to San Jose Extension	San Jose: Six miles from Berryessa Station in north San Jose to Santa Clara: Extend BART by constructing 4 new stations, a tunnel through downtown San Jose, and a new maintenance / storage yard in Santa Clara. The project constructs new track and dedicated guideway, power systems, signal systems, and purchases new vehicles. The project also includes upgrades to the existing BART system, that are required to extend operations to San Jose/Santa Clara. Other State funds are TIRCP.	BRT030001	21-T11-109	2030
San Francisco	TBJPA	Transbay Terminal/Caltrain Downtown Ext: Ph. 2	From Fourth/Townsend to Salesforce Transit Center: Extend Caltrain /High Speed Rail to Downtown San Francisco (DTX) Extend Caltrain rail service from 4th St/Townsend St in San Francisco to Salesforce Transit Center in downtown San Francisco, including two new stations: Phase 2 of the Transbay Transit Center program is the extension of the Caltrain commuter rail service from its current San Francisco terminus at Fourth and Townsend Streets to a new underground terminus beneath the Salesforce Transit Center building.	SF-050002	21-T11-110	2030

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
Regional/ Multi-County	SMART	SMART Rail and Pathway (Phase 2)	Marin and Sonoma Counties: Sonoma County Airport Station to Windsor: Extend rail and pathway (includes freight rail); Petaluma North at Corona Rd: Construct infill station (includes freight rail gauntlet tracks); McInnis to Smith Ranch in San Rafael, Hanna Ranch Rd. to S. Rowland Blvd. in Novato, Lakeville to Payran in Petaluma, Southpoint in Petaluma to Penngrove at Main St, Rohnert Park at Golf Course to Southwest Santa Rosa at Bellevue, Southwest Santa Rosa to Santa Rosa SMART Station (Joe Rodota Trail to 3rd St), San Miguel Rd. to Airport Blvd. in Santa Rosa: Construct multi-use pathway. Project also references RTP IDs 21-T11-201 and 21-T08-060. Other Federal funds are FRA PTC funds.	VAR210005	21-T11-113	2030
Alameda		Rail | Service Expansion | San Joaquin County-Dublin/ Pleasanton ("Valley Link")	This program includes funding to implement new rail service between San Joaquin Valley and the Dublin/Pleasanton BART station, including three new stations within Alameda County and three-car trains (12-minute peak headways).		21-T11-114	2030
Contra Costa	Hercules	Hercules Intercity Rail Station	Hercules: At future train station: Relocate the Kinder Morgan pipeline, Shell pipeline, fiber optic line, construct the 3rd track for the new station, construct the new station building, multi-use trail, retaining walls, and parking structure.	CC-030002	21-T11-115	2030
Alameda	ACTC	I-680 Express Lanes from SR84 to Alcosta Boulevard	Alameda and Contra Costa Counties: SB I-680 from SR-84 to north of Alcosta Blvd: express lane improvements (Phase 1); NB and SB I-680 from SR-84 to north of Alcosta Blvd: Widen for express lanes (Phase 2). Project limits in Alameda County are PM10.6 to PM21.9 and in Contra Costa County are PM0.0 to PM1.1.	ALA170009	21-T12-116	2030
Contra Costa	CCTA	I-680 NB Express Lane Completion	CC County: I680 NB from Livorna to SR-242: Widen to extend managed Lane from SR-242 to Benicia-Martinez Bridge: Convert HOV to Express Lane; from N Main to Treat: Operational improvements at various locations along I680: install limited access buffers and mitigation projects. The PBA 2050 project ID is 21-T12-116.	CC-170017	21-T12-116	2030
Regional/ Multi-County	BAIFA	ALA/CC-80 and Bay Bridge Approach Express Lanes	Alameda/Contra Costa counties: On I-80 from the Carquinez Bridge to Powell and the Bay Bridge Approaches (I-80, I-580, I-880 and Toll Plaza): Convert HOV lanes to express lanes. Work includes but is not limited to installation of gantries, tolling/traffic monitoring equipment and systems (hardware/software), signage, electrical, communications and fiber, lighting, and restriping, as well as police observation areas for enforcement. Project also references RTP ID 17-10-0045.	VAR170003	21-T12-116	2030
San Francisco	SF County TA	HOV/HOT Lanes on U.S.101 and I-280 in SF	San Francisco: On US 101 from SF/SM County line to I-280 interchange and on I-280 from US 101 interchange to 6th	SF-130008	21-T12-116	2030

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
			Street offramp: Convert a mixed traffic lane in each direction to HOV to enhance carpool and transit operations during peak periods in order to complement HOV lanes through San Mateo county and/or as part of a potential/demonstration congestion charging program in SF and project develop for converting to HOT lanes.			
San Mateo	CCAG	US-101 Managed Lanes North of I-380	San Mateo County: On US-101 from I-380 to logical termini near SF/SM County line: New managed lane construction in each direction. SMCTA is co-sponsoring the project.	SM-190009	21-T12-116	2030
Santa Clara	VTA	SR 85 Express Lanes	Santa Clara County: On SR 85 carpool lane from US 101 in San Jose to US 101 in Mountain View including the US 101/SR 85 HOV direct connectors and approaches: Implement roadway pricing. Convert the existing HOV lanes on SR 85 in both directions from US 101 in South San Jose to US 101 in Mountain View to Express Lanes, including the US 101/SR 85 HOV direct connectors in South San Jose, add a second Express Lane in both directions between SR 87 and I-280, and add an auxiliary lane on SR 85 NB from existing South De Anza Boulevard on-ramp to Stevens Creek Boulevard off-ramp. The Silicon Valley Express Lanes Phase 3 project includes the conversion of carpool lanes to express lanes on SR-85 from SR237/Grant Road to the US 101/SR-85 Interchange in Mountain View including the existing US101/SR-85 carpool lane-to-carpool lane direct connector ramps. Deliver milestones reflect phase 4 of project.	SCL090030	21-T12-116	2030
San Mateo	SamTrans	SamTrans Express Bus Service	San Mateo, San Francisco and Santa Clara Counties: On the US-101 Corridor: Implement a network of four express bus routes	SM-190003	21-T12-119	2030
Contra Costa	CCTA	I-680 Part Time Transit Lane	In Contra Costa County: On I-680 between Ygnacio Valley Rd and Alcosta Blvd: Increase bus service efficiency by implementing bus operations on shoulder (BOS) PBA2050 ID: 21-T12-122. New route running at 20-minute headways between Martinez Amtrak and East Dublin-Pleasanton BART, using I-680 HOV/Express Lanes (north of SR-24), and PTTLs south of SR-24, and including stops at Walnut Creek BART and the Bollinger Canyon Road Park & Ride lot. The PTTL will only be in use for buses in the northbound direction of 680 south of SR-24.	CC-170061	21-T12-122	2030
San Francisco	SFMTA	Express Bus | Service Expansion | SFMTA | US-101 & I-280	This program includes funding to implement improvements to existing express bus service along US-101 and I-280 (on express lanes where available), including frequency upgrades (10-minute peak headways on routes 8BX and 14X).		21-T12-123	2030

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
		Express Bus | Service Expansion | ReX (Premium) | Green Line (Vallejo to SFO Airport)	This program includes funding to implement new express bus service along I-80, I-280 and US-101 (on express lanes where available) between Vallejo and San Francisco International Airport. Improvements include high- frequency service (10-minute peak headways); capital improvements such as in-line bus stations on freeways and arterials; and station area amenities like upgraded local bus stops, taxi/TNC loading zones, and improved bicycle/pedestrian infrastructure.		21-T12-128	2030
Contra Costa		Express Lanes | SR-4 Express Lanes: Hillcrest Ave to I-680/SR4 Interchange	HOV lane conversion			2030
Santa Clara		Express Lanes | SR-87 Express Lanes: US-101 to SR-85	HOV lane conversion			2030
San Joaquin		SR-120 Widening	Widen from 4 to 6 lanes, I-5 to Main St.			2030
Santa Clara		Express Lanes | SCL SR 85 Future Phases: SR-237 to SR-87	New dual express lanes			2031
San Joaquin		SR-99 Widening	Widen from 6 to 8 lanes, SR-120 to Stanislaus County Line.			2032
Alameda		I-680 (NB) Express Lanes: Calaveras Blvd (SR-237) to US-101	GP lane conversion		21-T12-116	2033
Various		Express Bus | Service Expansion | ReX (Basic) | Red Line (Oakland to Redwood City)	This program includes funding to implement new express bus service along I-580, I-238, I-880, SR-84 and US-101 (on express lanes where available) between Downtown Oakland (19th St BART Station) and Redwood City (Caltrain Station). Improvements include high-frequency service (10-minute peak headways) and station area amenities like upgraded local bus stops, taxi/TNC loading zones, and improved bicycle/pedestrian infrastructure.		21-T12-127	2035
Santa Clara		Express Lanes | SCL US 101 Future Phases: I-880 to East Dunne Ave	New dual express lanes			2035
San Joaquin		I-5 HOV Lanes, Hammer Ln to Eight Mile Rd	I-5 HOV. Widen from 6 to 8 lanes, Hammer Ln to n/o Eight Mile Rd			2036
Santa Clara		Express Lanes | I-280 Express Lanes: Leland Ave to Magdalena Avenue	Combination HOV-Lane Conversion and GP-Lane Conversion			2038
Santa Clara		Express Lanes | I-280 Express Lanes: US-101 to Leland Ave	GP-Lane Conversion			2038

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
San Joaquin		I-5 HOV Lanes, French Camp Rd to Charter Way	I-5 HOV. Widen from 6 to 8 lanes, French Camp Rd to Charter Way.			2038
Solano	STA	I-80/I-680/SR 12 Interchange Improvements	Fairfield: I-80/I-680/Route 12 IC: Improve widen I-80 and I-680 as well as improve the connections from westbound I-80 to I-680 and SR12 (West) directly connect northbound I-680 and SR12 (West) connect the I-80/Red Top Road interchange with Business Center Drive and construct or improve interchanges at SR12 (West)/Red Top Road, I-80/Red Top Road, I-80/Green Valley Road, and I-680//Red Top Road. A third eastbound lane would be added to SR12 (East) from the Chadbourne Road on-ramp to the Webster Street off ramp.	SOL070020	21-T06-015	2040
Alameda	Dublin	I-580 Interchange Imps at Hacienda/Fallon Rd, Ph 2	Dublin: I-580/Fallon Rd I/C: Phase 2 reconstruct overcrossing to provide four-lanes in each direction, reconstruct the southbound to eastbound loop on-ramp, widen the eastbound off-ramp to provide two exit lanes with two left turn and two right tum lanes, widen the eastbound on-ramp, widen the westbound off-ramp to provide two left turn and two right turn lanes, and widen the westbound on-ramp, add new bicycle lanes and sidewalks to close a significant gap in these modes I-580 Hacienda Drive Interchange: Reconstruct overcrossing to provide additional northbound lane, widen the eastbound off-ramp to include a third left-turn lane, modify the westbound loop on-ramp, and widen the westbound off-ramp to include a third left-turn lane, and add new bicycle lanes and sidewalks to close a significant gap in these modes.	ALA170045	21-T06-019	2040
Santa Clara		Corridor & Interchange Improvements | US-101	This program includes funding to implement interchange improvements at SR-25, SR-237, Blossom Hill Rd, Buena Vista Ave, Ellis St, Mabury Rd/Taylor St, Mathilda Ave, Moffett Blvd, Montague Expressway, Old Oakland Rd, Shoreline Blvd, Trimble Rd/De La Cruz Blvd/Central Expressway, Zanker Rd/Skyport Dr/Fourth St, and between San Antonio Rd and Charleston Rd/Rengstorff Ave; and ramp metering improvements in Morgan Hill and Gilroy.	21-T06-028	2040	Santa Clara
Sonoma	Son Co TA	US 101 Marin/Sonoma Narrows (Sonoma)	Marin and Sonoma Counties (Sonoma County Portion): From SR37 in Novato to Old Redwood Highway in Petaluma: convert expressway to freeway; Between Lakeville Highway and East Washington Street:	SON070004	21-T06-029	2040
Santa Clara		Corridor & Interchange Improvements | SR-237	Adds 2 miles of freeway lanes in each direction on SR 237		21-T06-043	2040
Contra Costa	Concord	SR 242 / Clayton Road Interchange Improvements	Concord: At the SR242/Clayton Rd Interchange: Construct NB on-ramp and SB off-ramp. On ramp will access NB SR242. SB SR242 Off-ramp will intersect Franquette Way near the	CC-070024	21-T06-045	2040

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
			Clayton west Rd intersection. current phase- PSR phase to evaluate interchange and local road improvement alternatives. CCTA Website: http://www.ccta.net/projects/project/97			
Contra Costa	CC County	Byron Highway - Vasco Road Connection	Contra Costa County: between Byron Highway and Vasco Road: Construct an east-west connection road (1 lane per direction).	CC-070081	21-T06-047	2040
Alameda	Union City	East-West Connector: Decoto and Quarry Lakes Pkwy	Union City and Fremont: Decoto Rd from I-880 to SR-238 (Mission Blvd): Widen roadway and implement complete streets improvements. Decoto Complete Streets is a transit priority corridor with separate Class I system for bikes and pedestrians Quarry Lakes Pkwy alignment between Paseo Padre Pkwy and SR-238: Construct new, 4-lane multimodal corridor. QLP is a new four-lanes, multimodal corridor with buffer bike lanes and separated Class I trail for bikes and pedestrian with landscaped areas, street trees, and utilities to support the 100 acres Union City BART Station PDA. Project will be constructed in usable phases to support new housing developments, access to Union City BART Station and complete trail connections. PBA2050 ID is 21-T07056. Other State funds are SR-84 LATIP funds.	ALA978004	21-T07-056	2040
Contra Costa	CCTA	SR 4 Integrated Corridor Management	Contra Costa County: Along SR 4 between I-80 in Hercules to the SR 4/SR 160 Interchange in the City of Antioch: Implement Integrated Corridor Management along corridor.	CC-150013	21-T07-057	2040
Santa Clara	VTA	LRT Extension to Vasona Junction and Double Track	Campbell and San Jose: From the existing Winchester Station to a new Vasona Junction Station, near Route 85: Extend the light rail line and upgrade existing single track to double track at various location between Downtown San Jose and Winchester Station. Extend station platforms to accommodate 3-car consists.	SCL090040	21-T10-089	2040
San Francisco	SF County TA	Treasure Island Congestion Pricing Program	San Francisco: Treasure Island: Implement Congestion Pricing Program. project is phased	SF-110049	21-T10-092	2040
San Mateo	Caltrain	Peninsula Corridor Electrification Expansion	Caltrain: Electric Multiple Unit (EMU) fleet: Expand fleet through procurement of an additional 40 vehicles. This will build on the initial procurement of 96 EMUs through the Peninsula Corridor Electrification Project, which is fully funded and underway. This includes minor modifications to lengthen some station platforms to accommodate 8-car EMU's as well as wayside bike improvements.	SM-190002	21-T11-107	2040
Alameda	ACTC	I-580/680 Interchange HOV/HOT Widening	Alameda County: On I-580 between Hacienda Dr. and San Ramon/Foothill Road and on I-680 between Stoneridge Dr. and Amado: Widen to add one HOV/HOT lane for WB 580 to SB 680 and NB 680 to EB 580 movements at connector and to	ALA170008	21-T12-116	2040

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
			Tassajara Road to improve capacity, operations, and safety at the interchange, primarily in the westbound direction approaching the interchange.			
Alameda	ACTC	I-880 NB HOV/HOT: North of Hacienda to Hegenberger	Alameda County: I-880 in the northbound direction from north of Hacienda Ave to Hegenberger Road: Widen to provide one HOV/express lane.	ALA170010	21-T12-116	2040
San Joaquin		I-5 HOV Lanes, Louise Ave to French Camp Rd	I-5 HOV. Widen from 6 to 8 lanes, Louise Ave to French Camp Rd.			2040
Alameda		I-680 Express Lanes (NB): SR-262/Automall Pkwy to SR-237/Calaveras Blvd	New express lane construction		21-T12-116	2042
Solano		I-80 Express Lanes: SR-37 – Carquinez Bridge	New express lane construction		21-T12-116	2042
Solano		I-80 Express Lanes: Red Top Rd to SR-37	GP lane conversion		21-T12-116	2042
Alameda		I-880 Express Lanes, Hegenberger Rd to Bay Bridge	GP lane conversion			2042
Alameda	Fremont	State Route 262 (Mission Blvd) Improvements	Fremont: Mission Blvd/I-680 IC: Implement interchange improvements at I-680, new freeway lanes between I-680 and I-880 that create grade separations at Mohave Drive and Warm Springs Boulevard. Reconstruct local access with one-way frontage roads between Warm Springs Boulevard and I-680.	ALA170001	21-T06-046	2050
Various		Ferry | Service Frequency Boost | WETA	This program includes funding to implement improvements to existing ferry service between the San Francisco Ferry Building and Alameda/Oakland, Harbor Bay, Vallejo, Richmond and South San Francisco, including frequency upgrades (15-30 minute peak headways).		21-T11-095	2025-2040
Various		AC Transit Transbay Network: Capital Improvements + Service Increase	This program includes funding to implement improvements to existing express bus service along I-80, I-580 and I- 880 (on express lanes where available). Improvements include frequency upgrades (15-minute peak headways on routes F, O, P, J, V and L) and planning for express bus expansion throughout the inner East Bay.		21-T12-120	2025-2040
Various		Rapid Bus | AC Transit | Modernization	This program includes funding to implement rapid transit improvements to existing bus service. Improvements include new rapid bus service; improved bus stops and stations; new/improved transit signal priority (including on street and on-bus equipment); transit priority infrastructure; dedicated bus lanes; queue jumps; and frequency upgrades (5-12		21-T10-073	2025-2040 (phasing)

County	Sponsor	Project Name	Project Description	TIP ID	RTP ID	Opening Year
			minute peak headways on routes 18, 20/21, 40, 57, 97 and NL).			
Various	BART	Rail | Service Expansion | Oakland-San Francisco ("Link21")	This program includes funding to implement Link21, providing new transbay rail service between San Francisco and Oakland, including new stations in the East Bay and San Francisco (10 trains per hour per direction in peak).		21-T11-112	2036 - 2050
Solano		Benicia-Martinez Bridge HOV Bypass Lane			21-T12-116	

Appendix B

Exhibit 13: 2050 Household Projection Differences, PBA vs AlaCC, East Alameda County (SD 15)

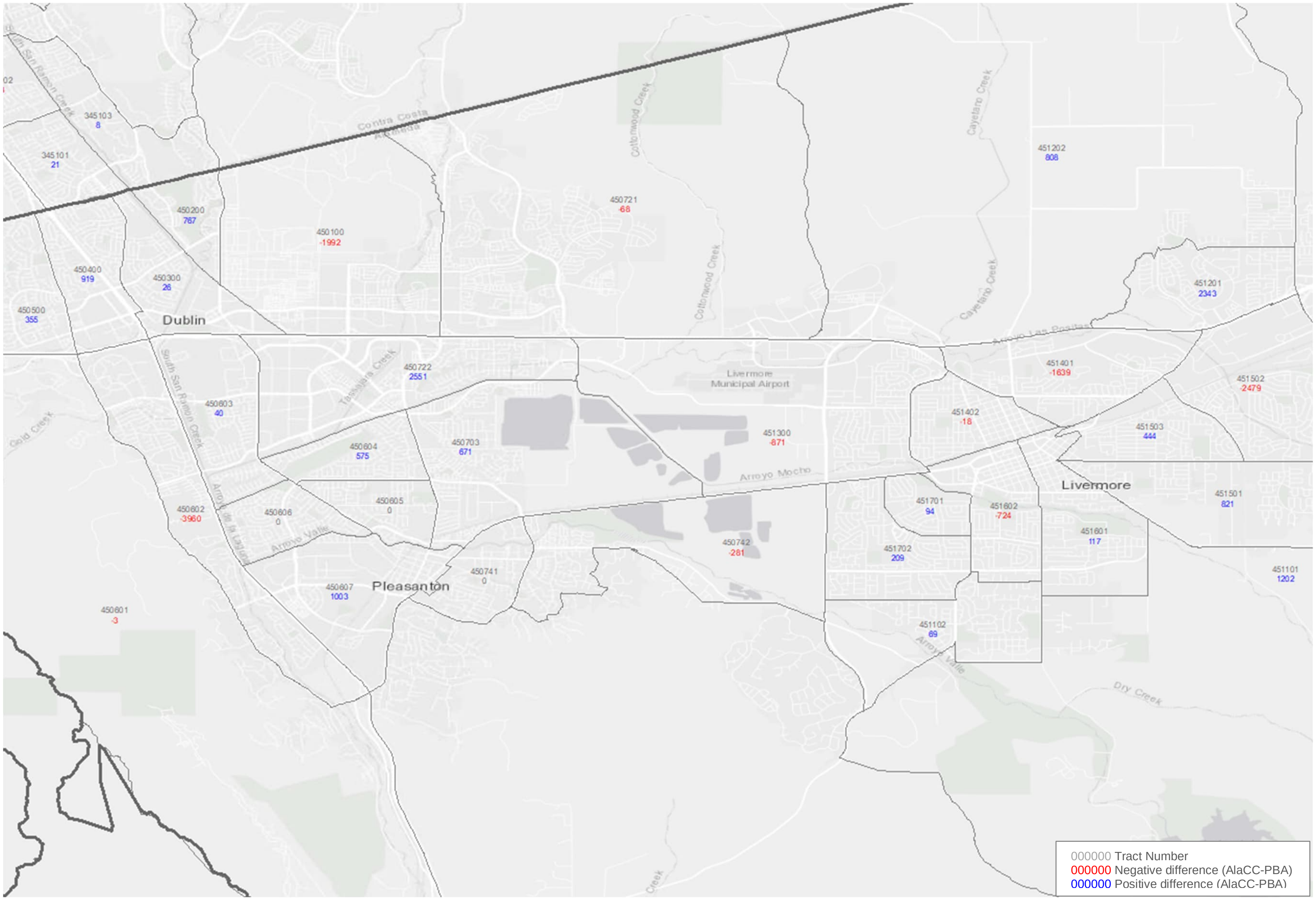


Exhibit 14: 2050 Household Projection Differences, PBA vs AlaCC, South Alameda County (SD 16)

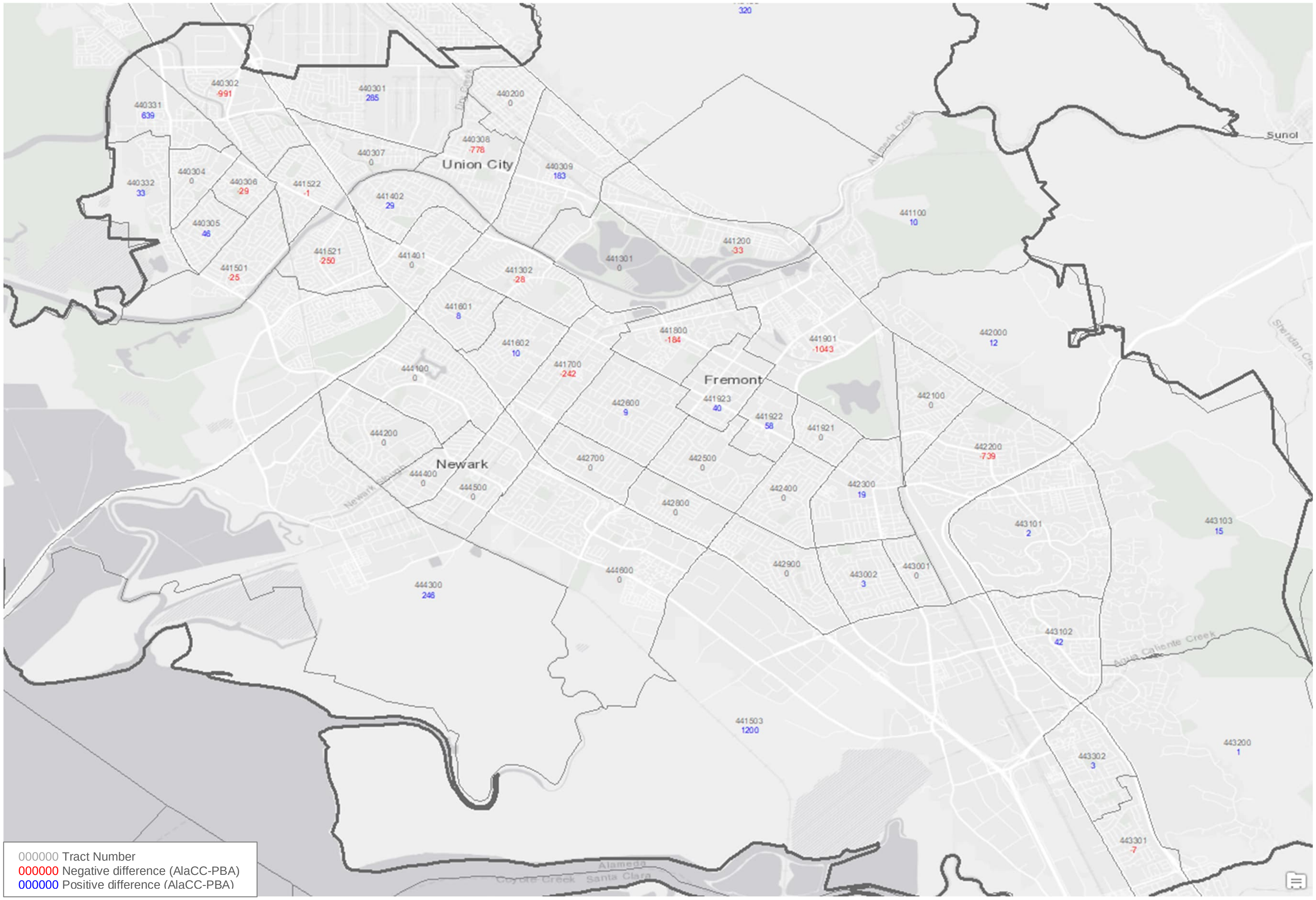


Exhibit 15: 2050 Household Projection Differences, PBA vs AlaCC, Central Alameda County (SD 17)

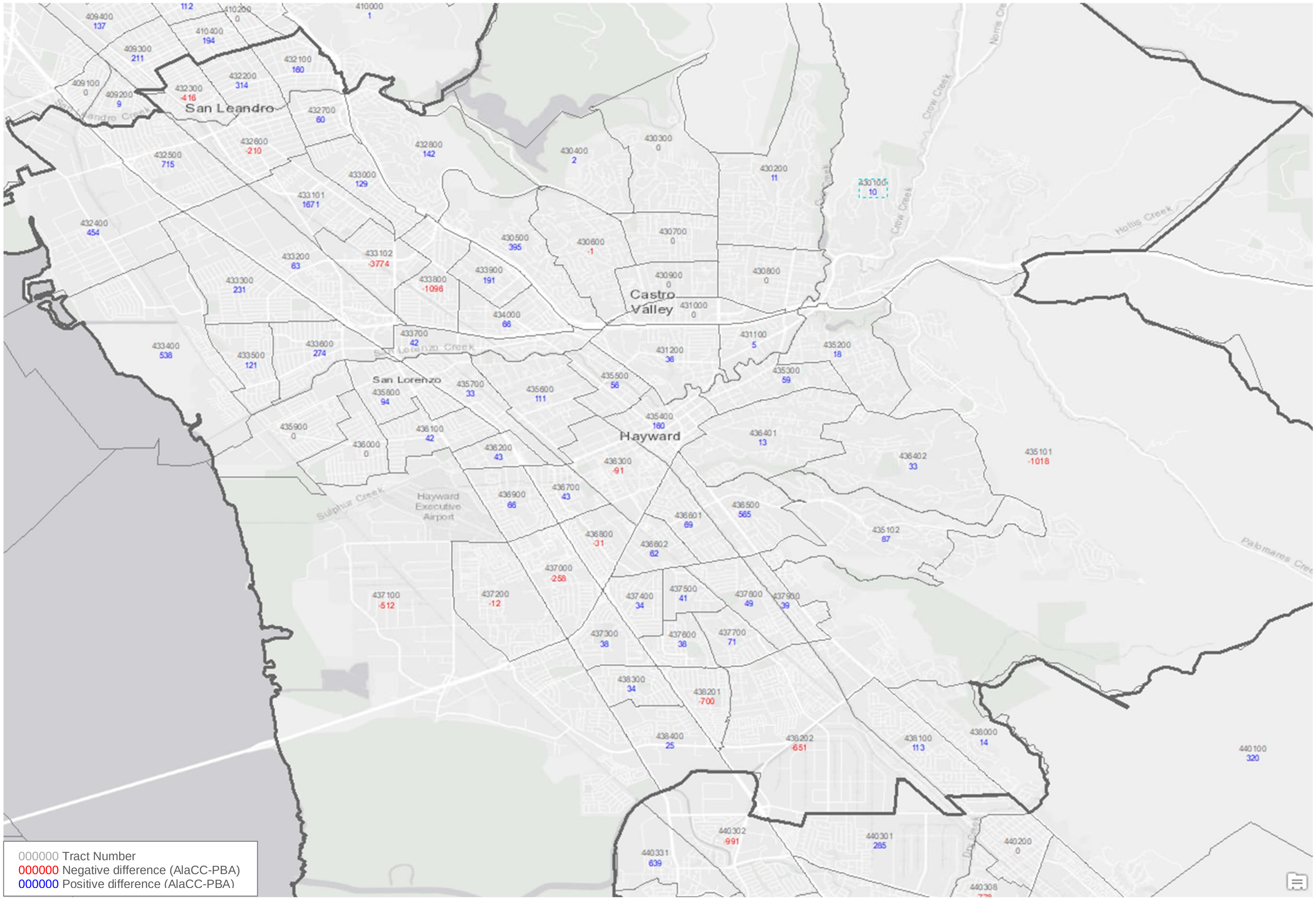


Exhibit 16: 2050 Household Projection Differences, PBA vs AlaCC, North Alameda County (SD 18)

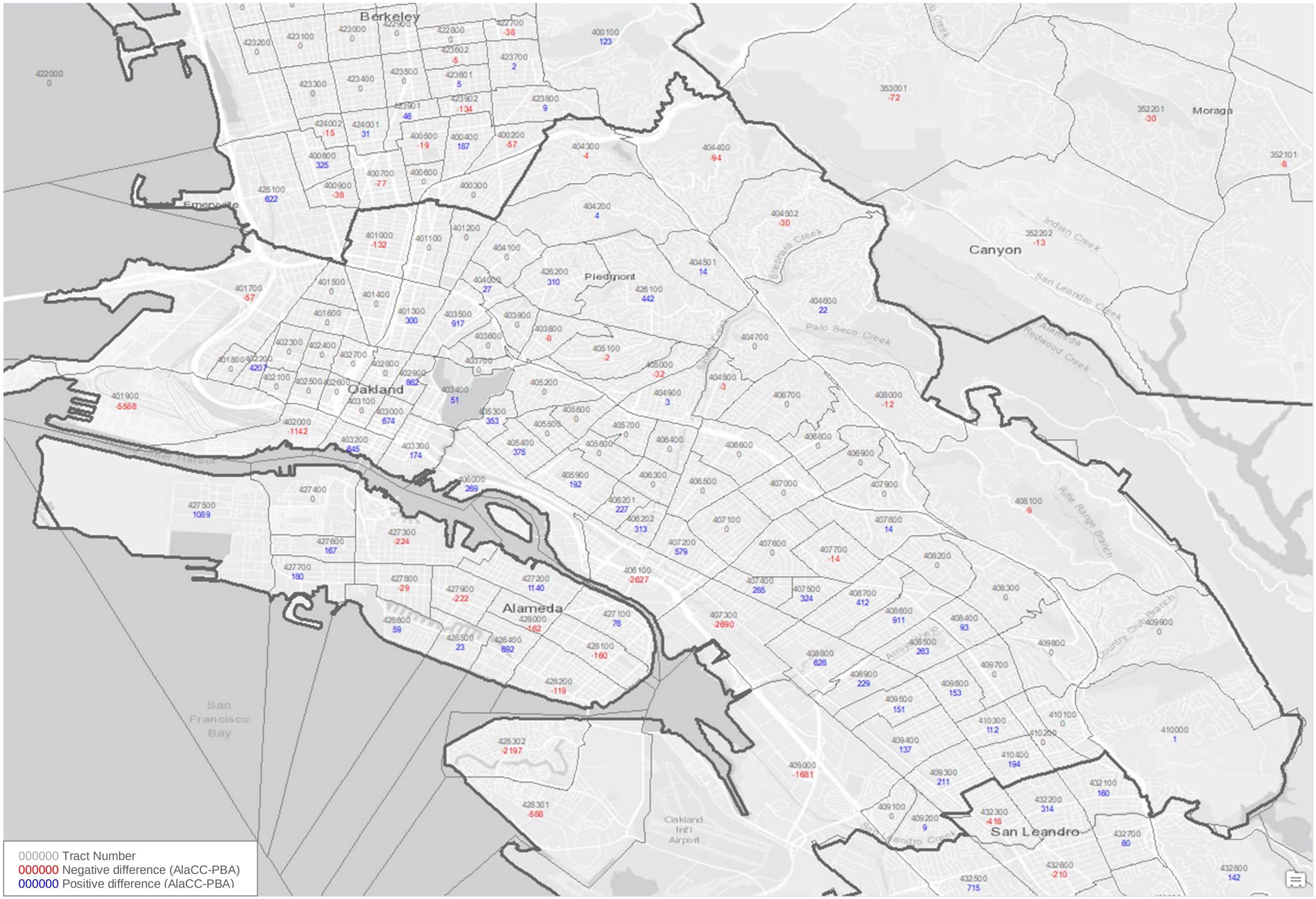


Exhibit 17: 2050 Household Projection Differences, PBA vs AlaCC, Northwest Alameda County (SD 19)

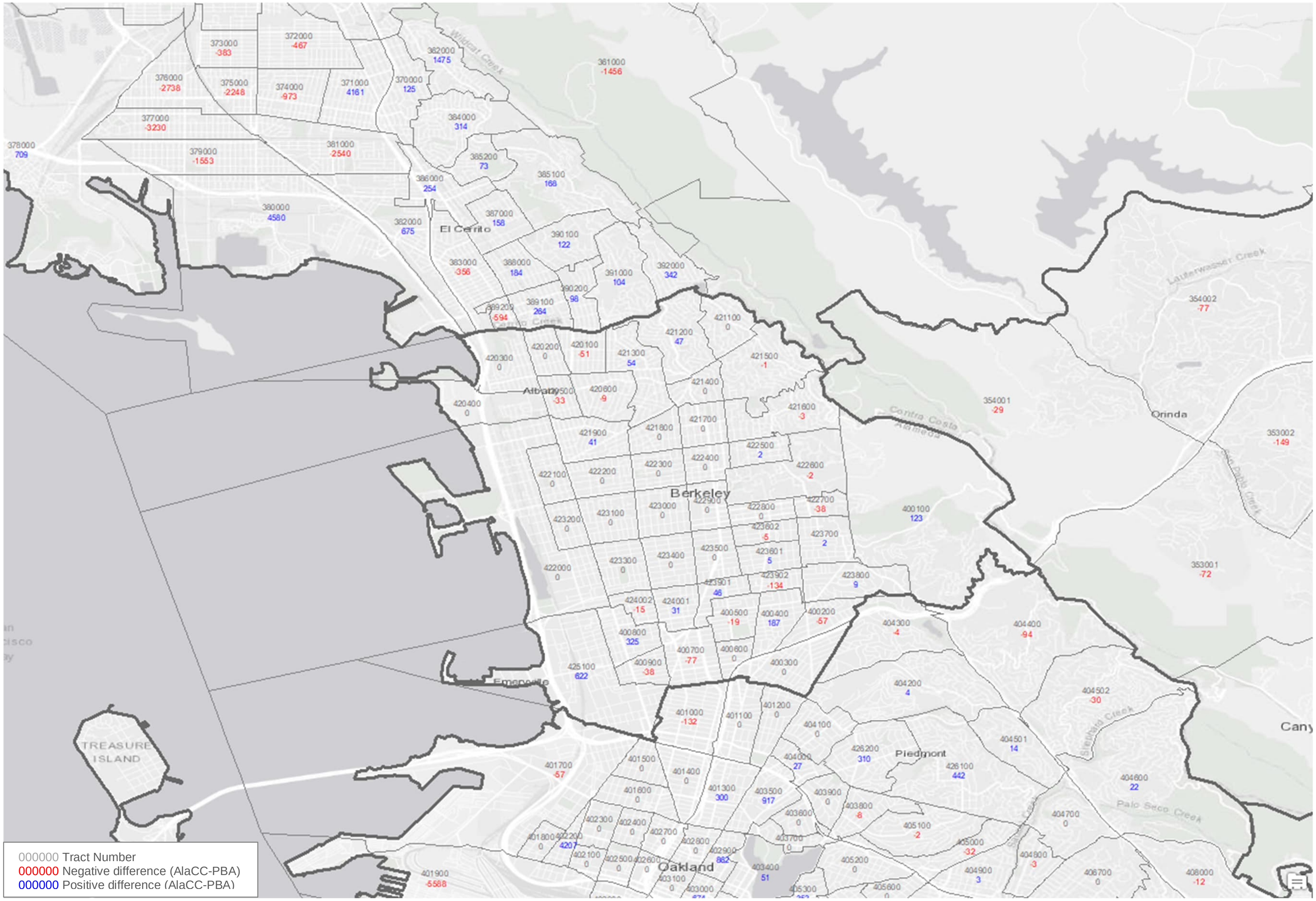


Exhibit 18: 2050 Household Projection Differences, PBA vs AlaCC, West Contra Costa County (SD 20)

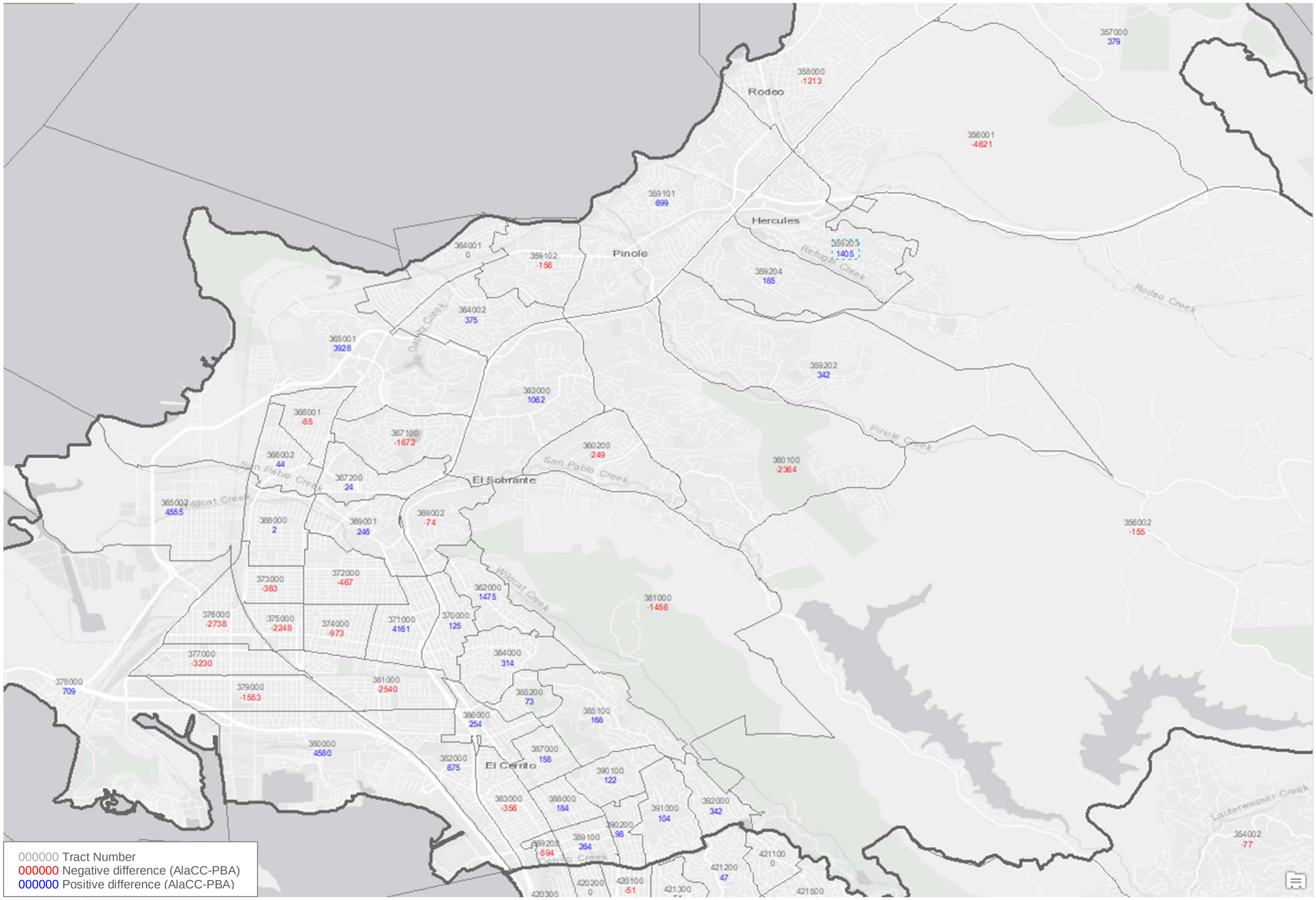


Exhibit 19: 2050 Household Projection Differences, PBA vs AlaCC, North Contra Costa County (SD 21)

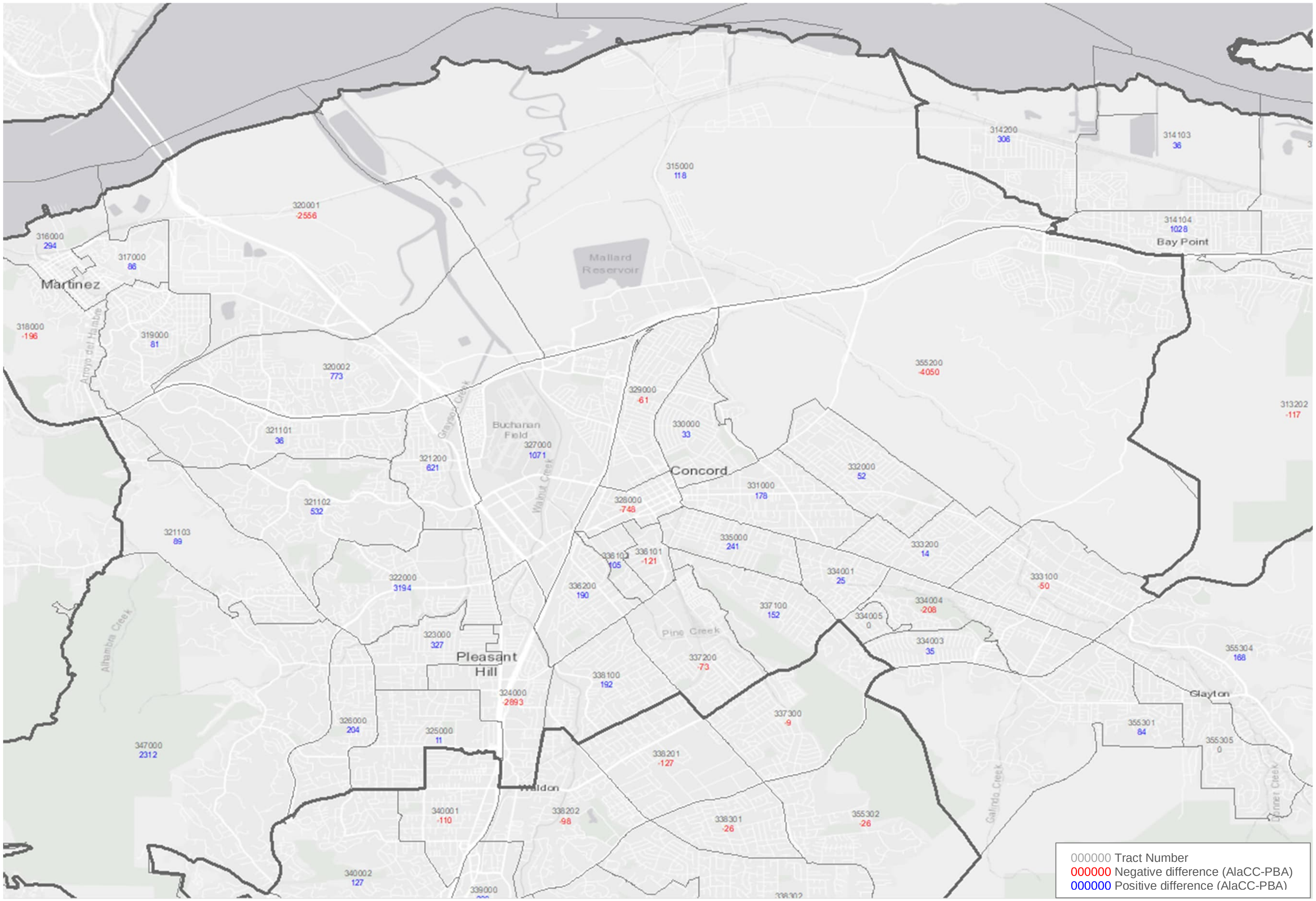


Exhibit 20: 2050 Household Projection Differences, PBA vs AlaCC, Central Contra Costa County (SD 22)

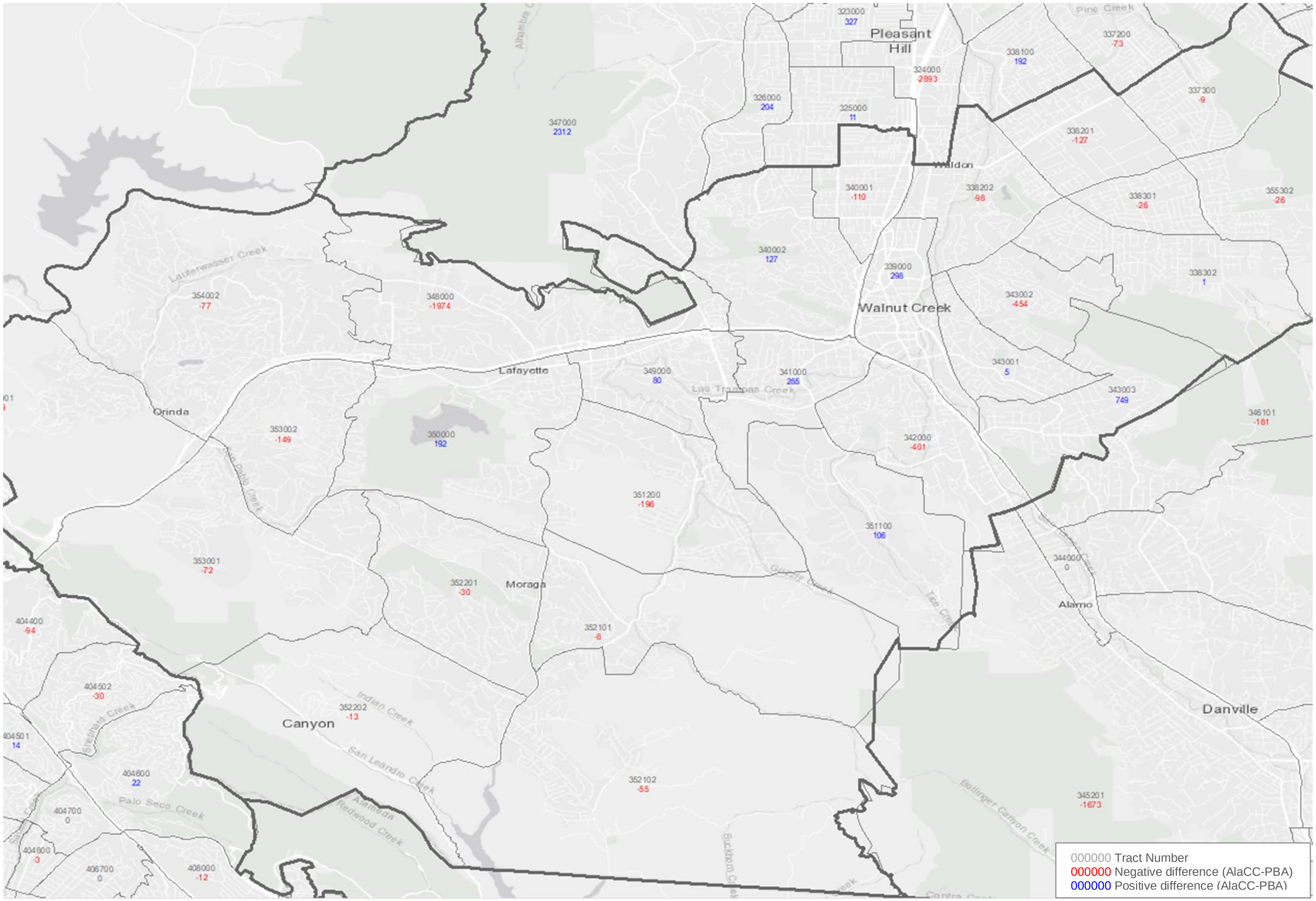


Exhibit 21: 2050 Household Projection Differences, PBA vs AlaCC, South Contra Costa County (SD 23)

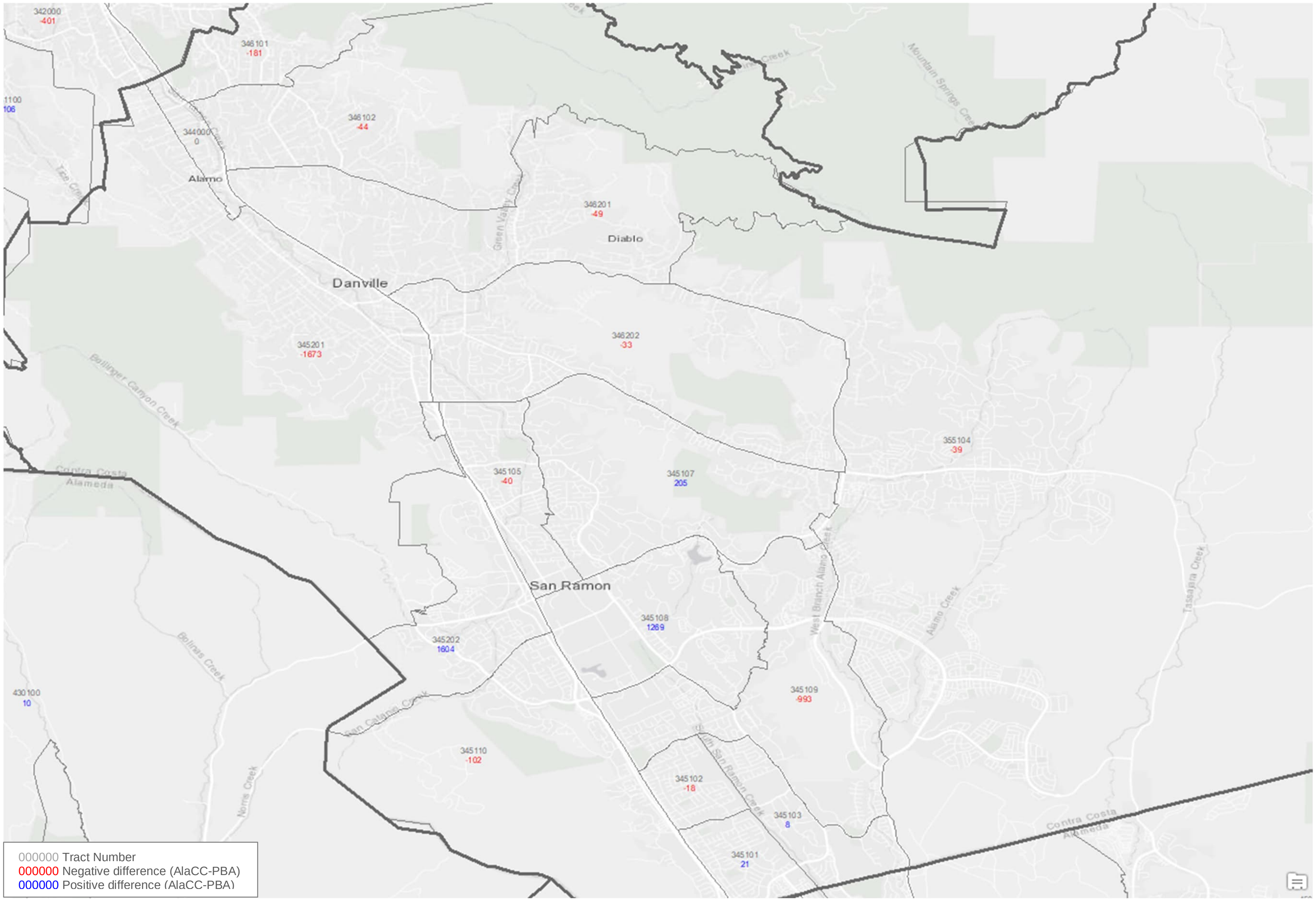


Exhibit 22: 2050 Household Projection Differences, PBA vs AlaCC, East Contra Costa County, Pittsburg-Antioch (SD 23)

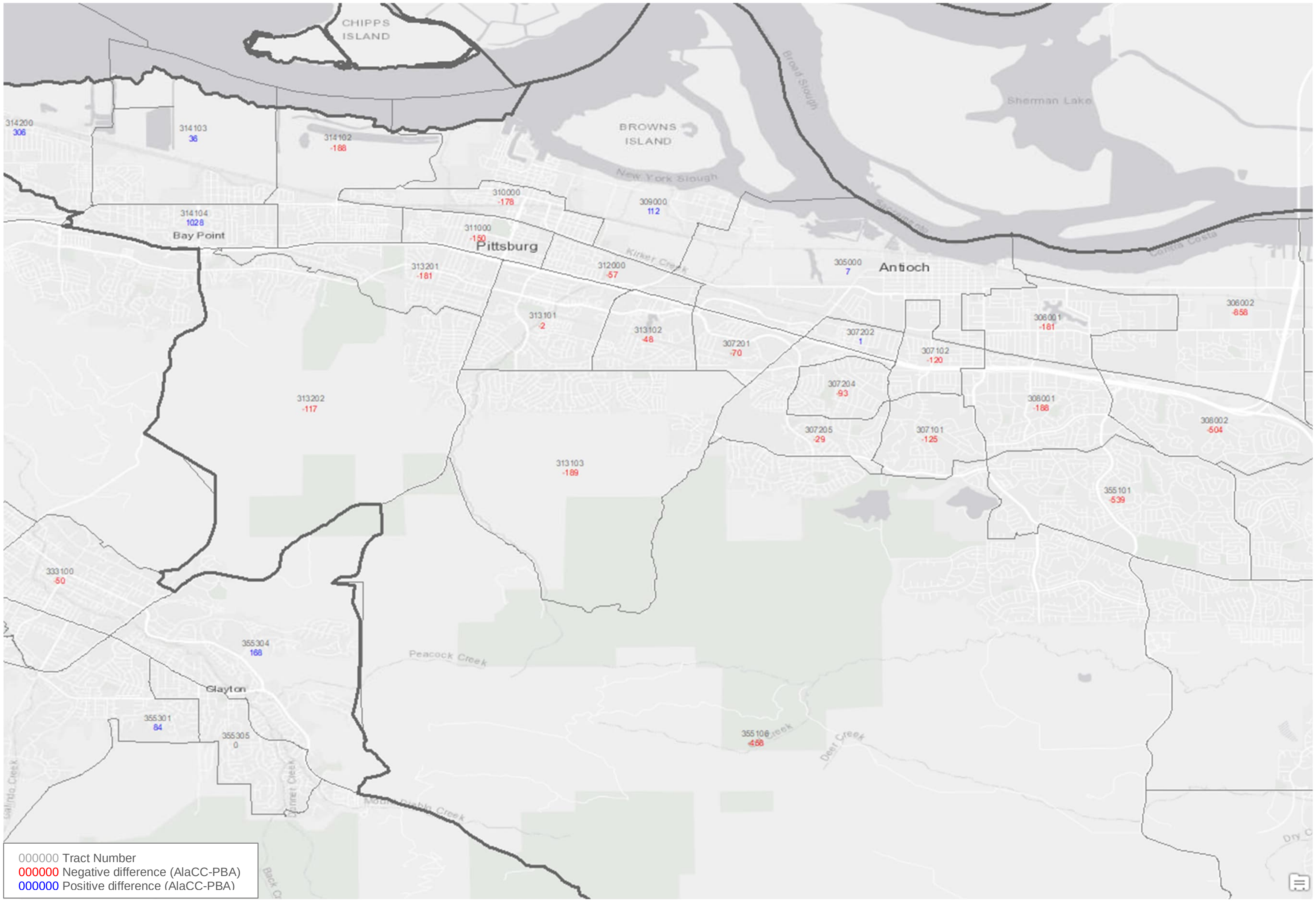


Exhibit 23: 2050 Household Projection Differences, PBA vs AlaCC, East Contra Costa County, Oakley-Brentwood (SD 23)

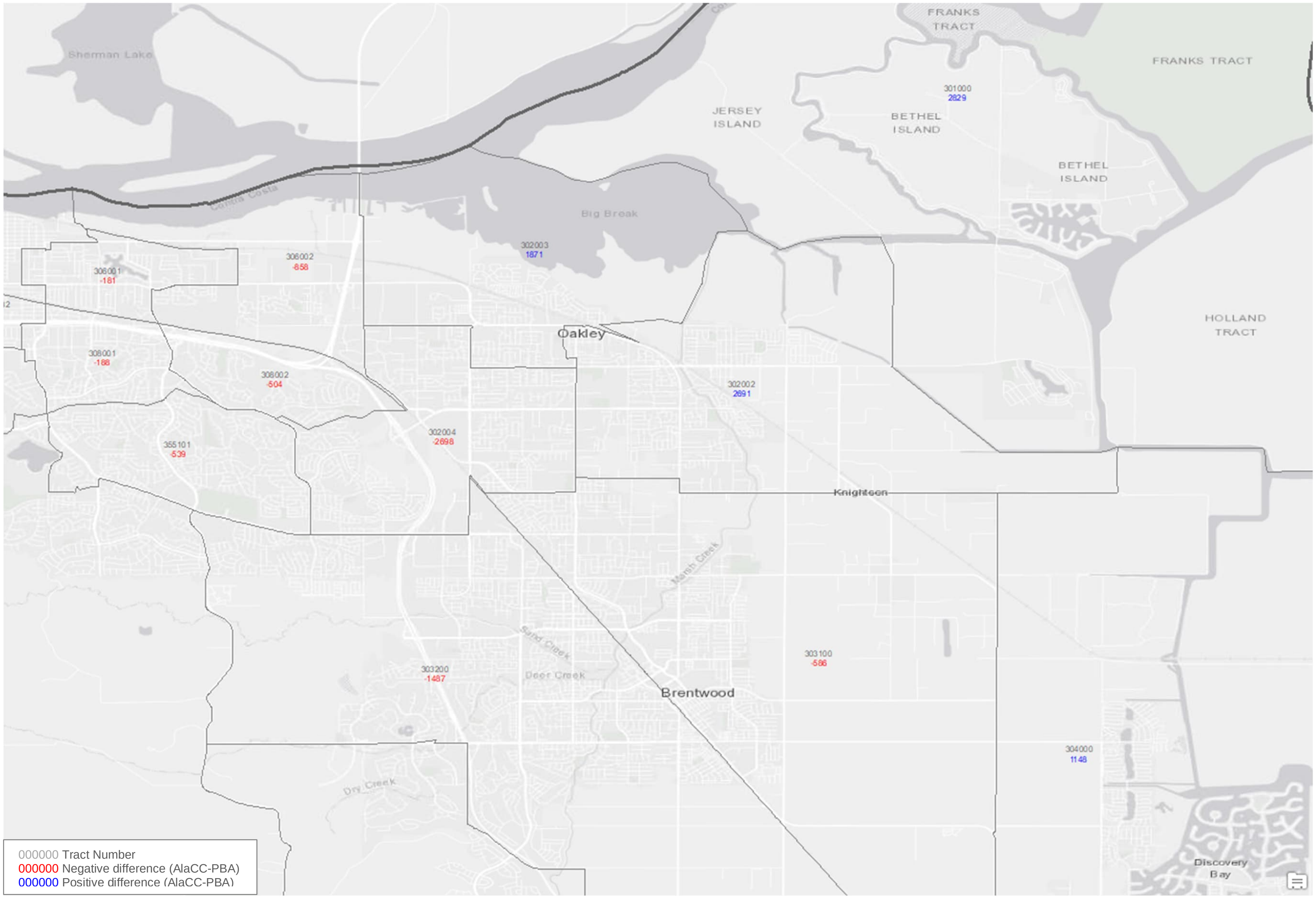


Exhibit 24: 2050 Employment Projection Differences, PBA vs AlaCC, East Alameda County (SD 15)

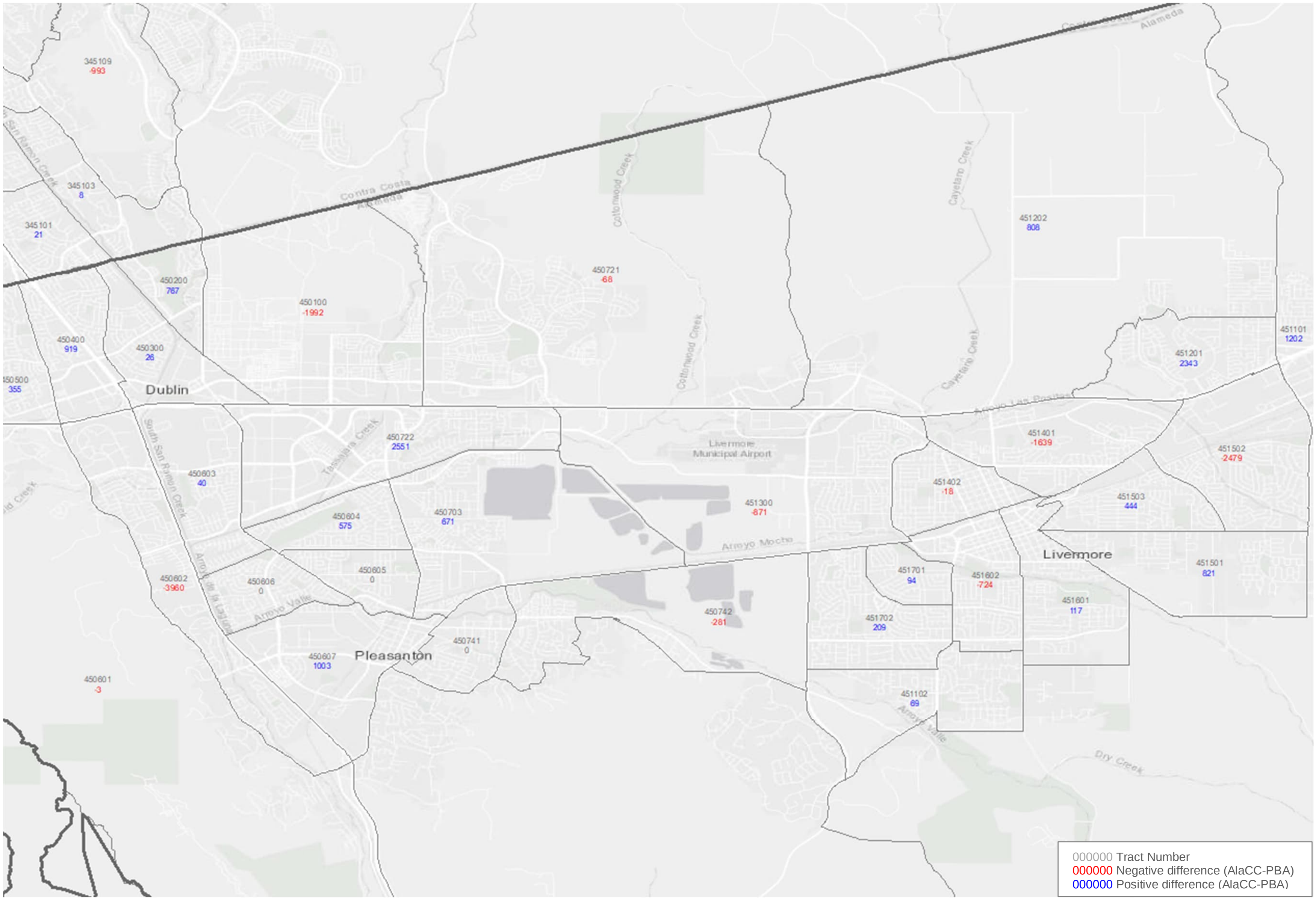


Exhibit 25: 2050 Employment Projection Differences, PBA vs AlaCC, South Alameda County (SD 16)

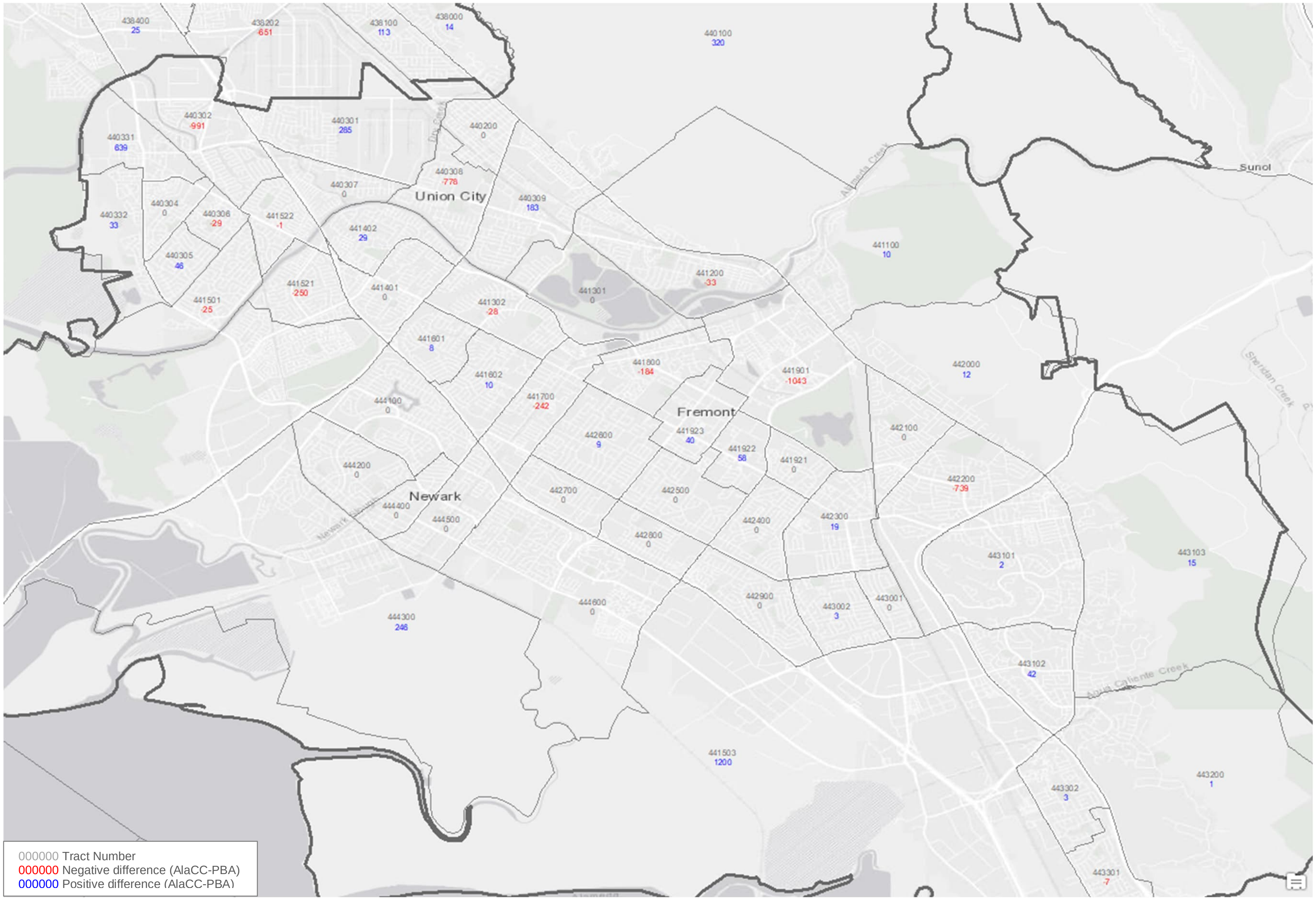


Exhibit 26: 2050 Employment Projection Differences, PBA vs AlaCC, Central Alameda County (SD 17)

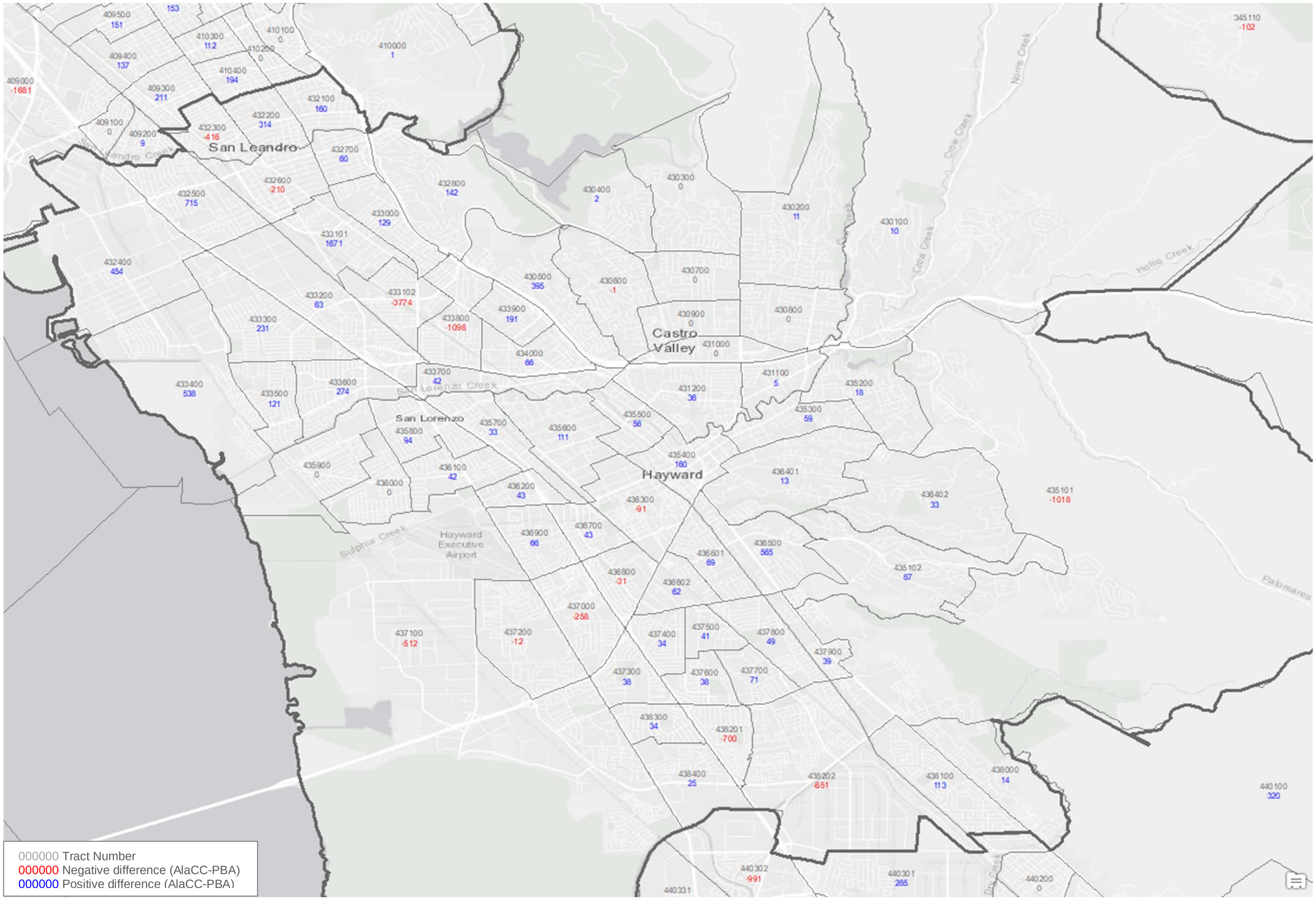


Exhibit 27: 2050 Employment Projection Differences, PBA vs AlaCC, North Alameda County (SD 18)

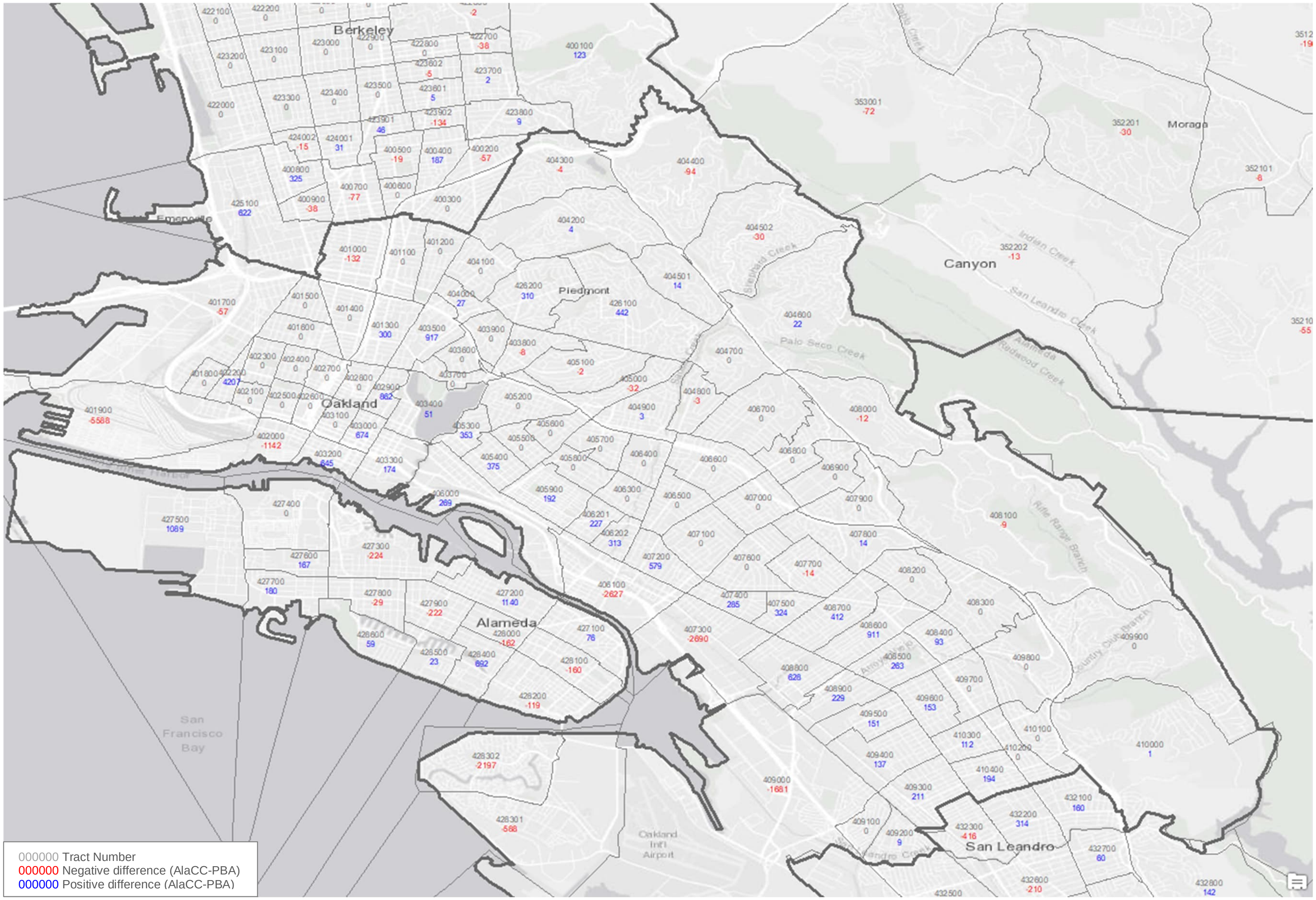


Exhibit 28: 2050 Employment Projection Differences, PBA vs AlaCC, Northwest Alameda County (SD 18)

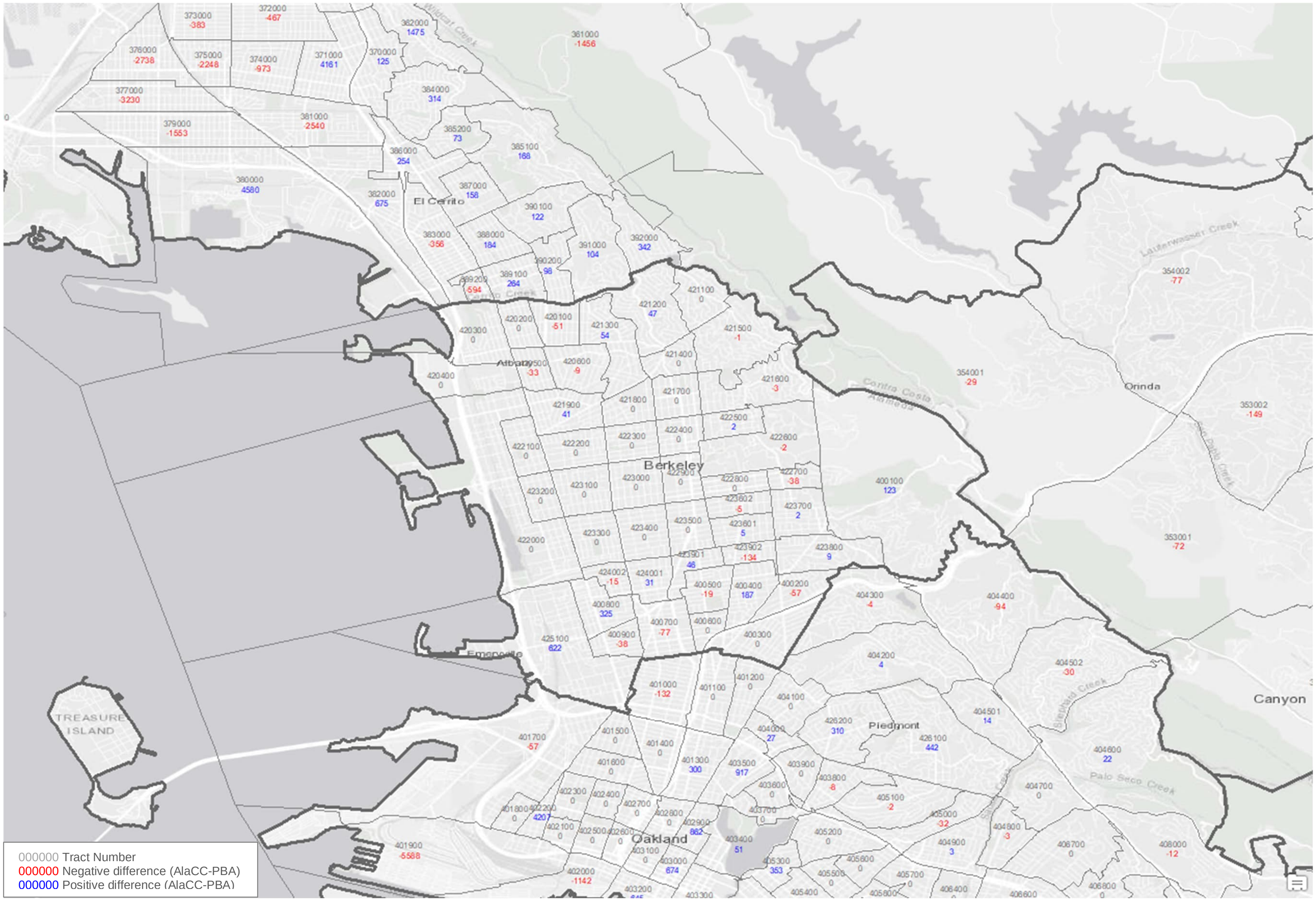


Exhibit 29: 2050 Employment Projection Differences, PBA vs AlaCC, West Contra Costa County (SD 20)

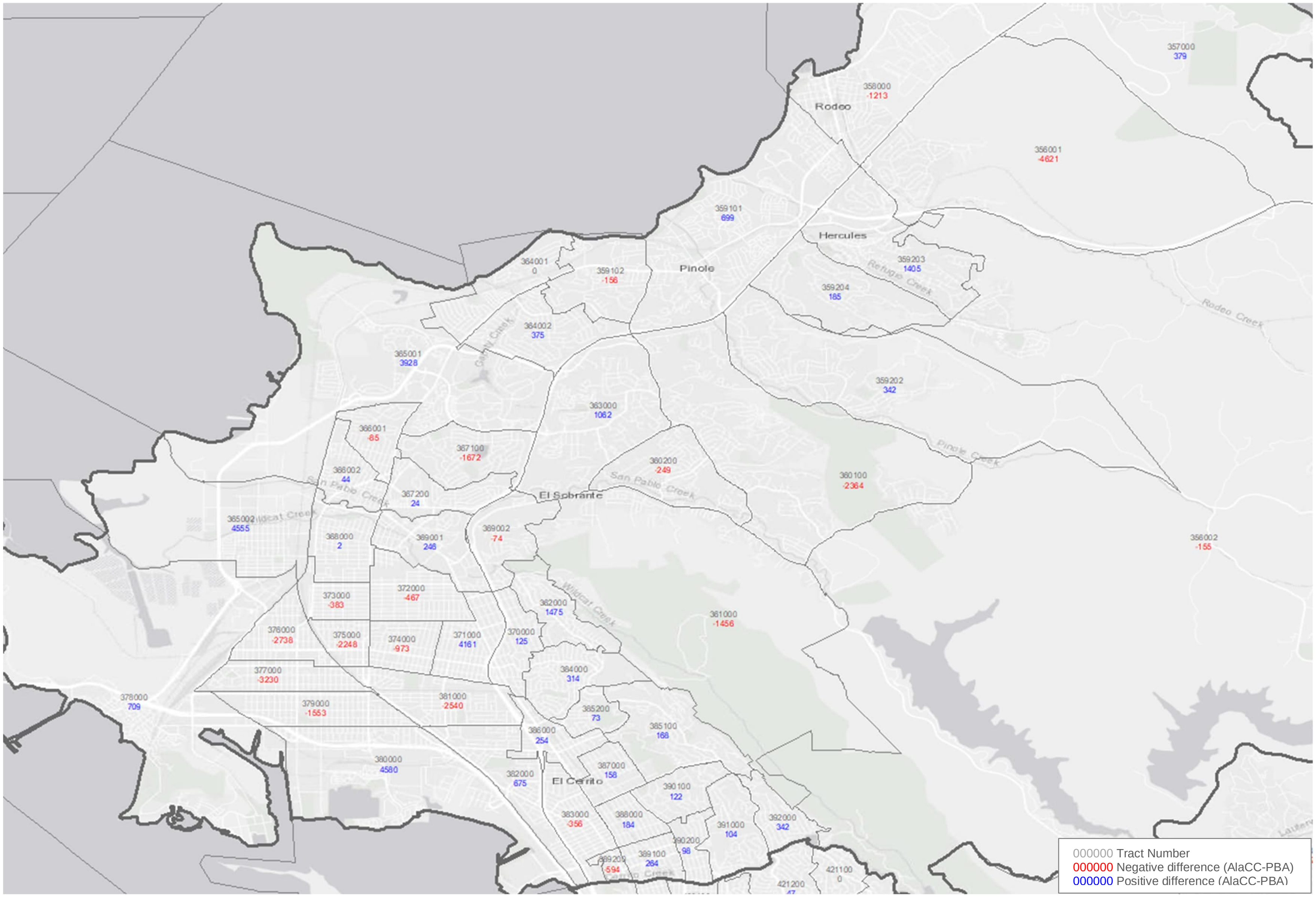


Exhibit 30: 2050 Employment Projection Differences, PBA vs AlaCC, North Contra Costa County (SD 21)

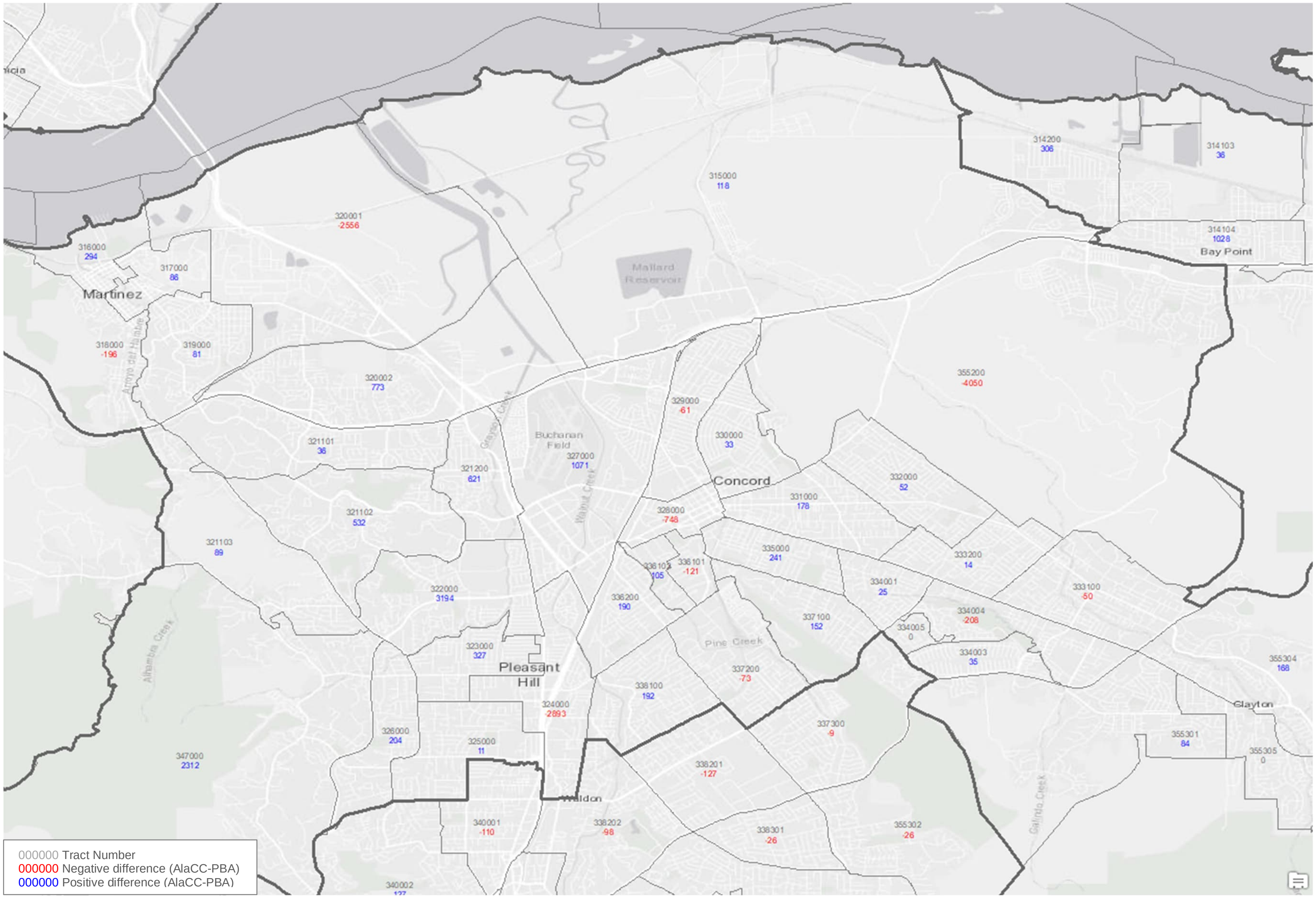


Exhibit 31: 2050 Employment Projection Differences, PBA vs AlaCC, Central Contra Costa County (SD 22)

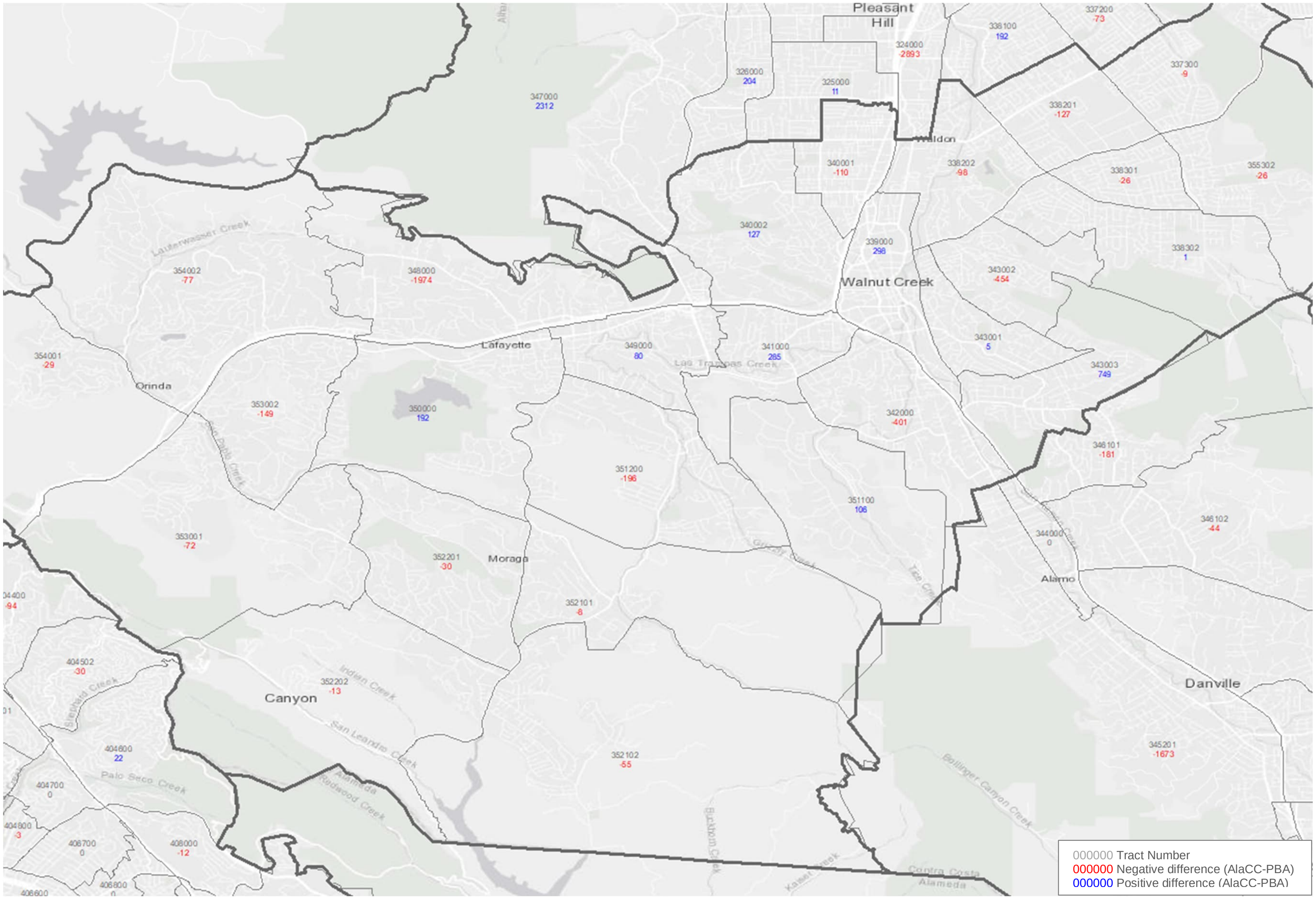


Exhibit 32: 2050 Employment Projection Differences, PBA vs AlaCC, South Contra Costa County (SD 23)

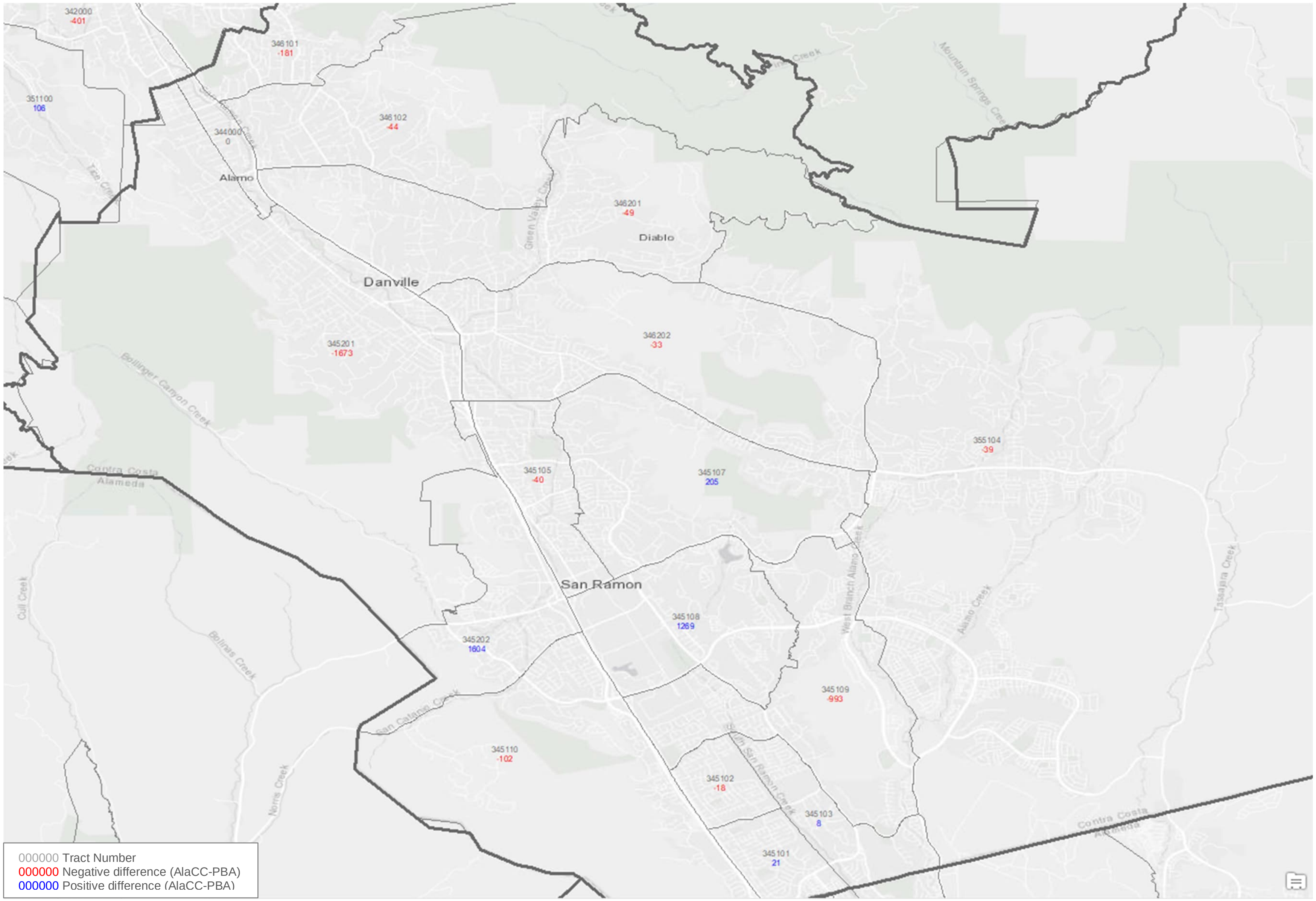


Exhibit 33: 2050 Employment Projection Differences, PBA vs AlaCC, West Contra Costa County, Pittsburg-Antioch (SD 24)

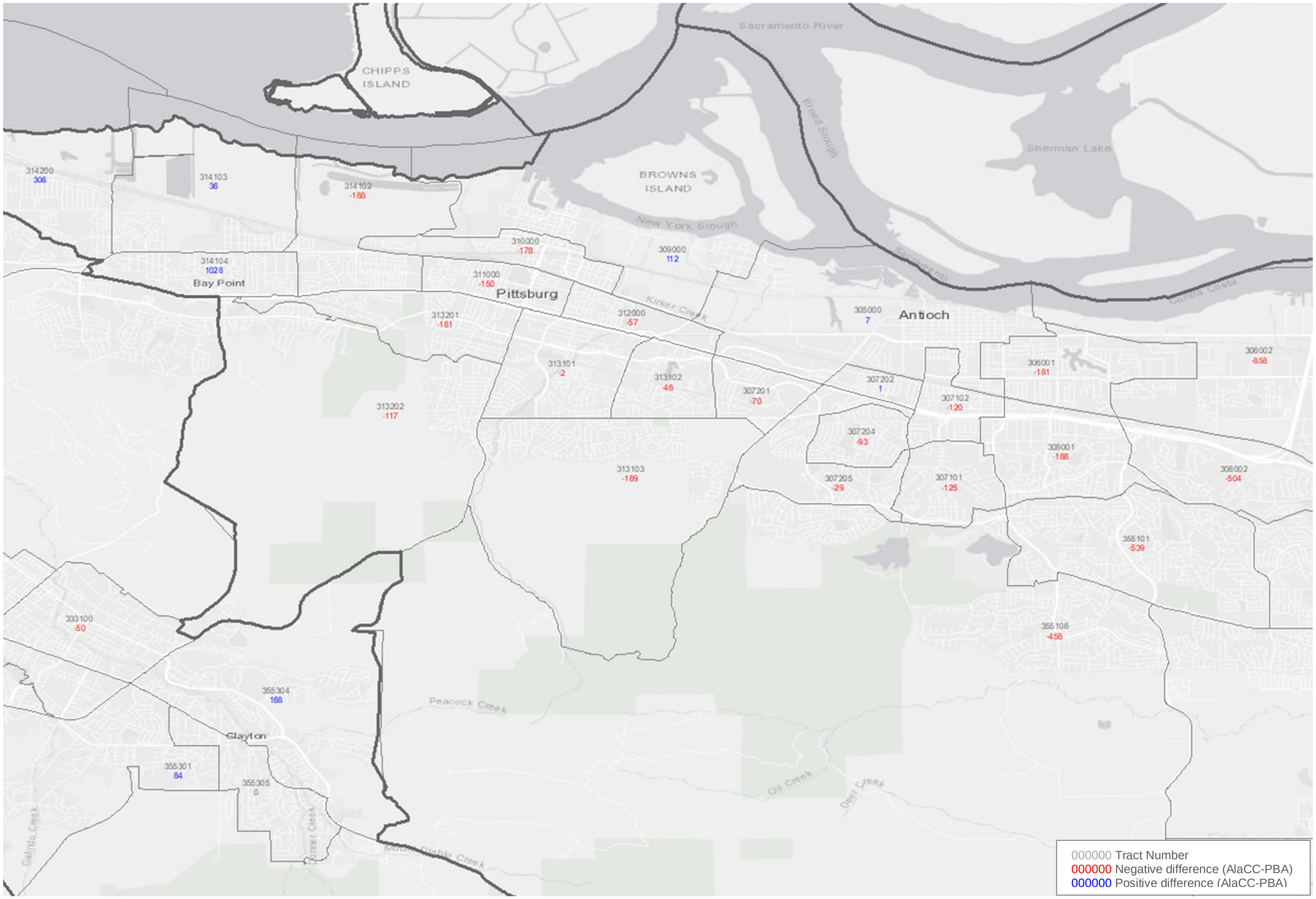


Exhibit 34: 2050 Employment Projection Differences, PBA vs AlaCC, West Contra Costa County, Oakley-Brentwood (SD 24)